



Logic: deduction and induction, Analogy, Numerical relations and reasoning

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	1	1	2	0	0	1	1	1	0	0	0
2 Marks Count	2	3	0	2	0	1	1	1	1	1	0
Total Marks	5	7	2	4	0	3	3	3	2	2	0

Welcome to the "General Aptitude: Analytical Aptitude" chapter, a crucial component of your GATE Computer Science exam preparation. This section is designed to sharpen your logical reasoning, problem-solving abilities, and critical thinking skills, which are not only vital for the exam but also fundamental to a successful career in computer science. Analytical Aptitude questions typically form a significant portion of the 15 marks allocated to the General Aptitude section in GATE, often appearing as Multiple Choice Questions (MCQs), Multiple Select Questions (MSQs), or Numerical Answer Type (NAT) questions. Mastering these topics will not only boost your overall score but also enhance your ability to approach complex computational problems with a structured and logical mindset.

Topic-wise Key Concepts

Age Relation

Definition and Core Idea: Age relation problems involve determining the current or past/future ages of individuals based on given conditions, often expressed as ratios, sums, or differences. The core idea is to translate these verbal conditions into algebraic equations.

• Important Formulas and Results:

1. If the current age of a person is x years, then their age n years ago was $(x - n)$ years.
2. If the current age of a person is x years, then their age n years hence (in the future) will be $(x + n)$ years.
3. If the ratio of ages of two persons A and B is $a : b$, their ages can be represented as ax and bx for some constant x .

• Key Properties and Identities:

- The difference in ages between two people remains constant throughout their lives.
- Ratios of ages change over time, but the underlying age values can be determined by setting up equations.

• Common Pitfalls or Tricky Points:

- Confusing "ago" with "hence" when setting up equations.
- Incorrectly applying ratios (e.g., assuming the constant x in $ax : bx$ is the same for different time periods without adjustment).
- Making calculation errors in solving linear equations.

• Standard Problem-Solving Techniques or Shortcuts:

- Assign variables (e.g., x , y) to the current ages of the individuals involved.
- Formulate linear equations based on the given conditions.
- Solve the system of equations. For ratio problems, use a common multiplier.
- Always check if the calculated ages are logically consistent (e.g., no negative ages).

Analogy

Definition and Core Idea: Analogy questions test your ability to identify the relationship between a given pair of items (words, numbers, letters, or symbols) and then find another pair that shares the exact same relationship. The core idea is pattern recognition and logical extension.

• Important Formulas and Results: No mathematical formulas. The "results" are types of relationships:

1. **Synonym/Antonym:** e.g., GOOD : BAD :: HAPPY : SAD
2. **Part-Whole:** e.g., FINGER : HAND :: TOE : FOOT
3. **Cause-Effect:** e.g., FIRE : ASH :: EVENT : MEMORY
4. **Worker-Tool:** e.g., CARPENTER : SAW :: WRITER : PEN
5. **Product-Raw Material:** e.g., SHOE : LEATHER :: WINE : GRAPES
6. **Quantity-Unit:** e.g., LENGTH : METER :: MASS : KILOGRAM
7. **Gender:** e.g., LION : LIONESS :: NEPHEW : NIECE
8. **Study-Topic:** e.g., BIOLOGY : LIFE :: GEOLOGY : EARTH
9. **Function:** e.g., KNIFE : CUT :: ERASER : ERASE

10. **Degree of Intensity:** e.g., WARM : HOT :: ANNOY : ENRAGE

- **Key Properties and Identities:**

- The relationship must be consistent and precise.
- Consider the order of elements in the pair; it often matters.
- Relationships can be direct or indirect, but should be the strongest possible link.

- **Common Pitfalls or Tricky Points:**

- Choosing an option with a weak or secondary relationship.
- Overthinking and finding obscure connections.
- Not considering all options before making a choice.
- Ignoring the direction of the relationship (e.g., tool-worker vs. worker-tool).

- **Standard Problem-Solving Techniques or Shortcuts:**

- Clearly identify the relationship between the first pair. Express it as a sentence (e.g., "A is a type of B," "A causes B").
- Apply the exact same relationship to the options provided.
- If multiple options seem plausible, look for the most specific and direct relationship.
- Consider different categories of relationships (functional, structural, categorical, etc.).

Code Words

Definition and Core Idea: Code word problems involve deciphering a hidden rule used to transform one set of words or letters into another. The core idea is to identify the pattern of transformation and apply it to a new word.

- **Important Formulas and Results:** No mathematical formulas. "Results" are common coding patterns:

1. **Letter Shifting:** Each letter is shifted by a fixed number of positions in the alphabet (e.g., A becomes C, B becomes D).
2. **Letter Reversal:** The order of letters in the word is reversed.
3. **Substitution:** Specific letters are replaced by other specific letters or symbols.
4. **Position-based Coding:** Letters are coded based on their position in the word or alphabet.
5. **Mixed Patterns:** A combination of the above, e.g., shifting and reversing.

- **Key Properties and Identities:**

- The coding rule is consistent throughout the given examples.
- The rule can be simple (e.g., +1 shift) or complex (e.g., vowel-consonant specific shifts).

- **Common Pitfalls or Tricky Points:**

- Missing subtle patterns, especially when multiple rules are combined.
- Assuming a simple rule when a more complex one is in play.
- Not considering the alphabet's cyclical nature (e.g., $Z + 1 = A$).

- **Standard Problem-Solving Techniques or Shortcuts:**

- Write down the alphabet and its numerical positions ($A=1, B=2, \dots, Z=26$).
- Compare the original word with its coded form letter by letter.
- Look for common differences in positions, reversals, or direct substitutions.
- Test your identified rule with any other given examples before applying it to the question.

Coding Decoding

Definition and Core Idea: Coding-decoding problems are an extension of code words, often involving numbers, symbols, or more intricate letter manipulations. The core idea remains to identify the underlying coding rule and apply it to decode a message or encode a new one.

- **Important Formulas and Results:** No specific formulas, but common methods:

1. **Alphabet Position Values:** Using $A = 1, B = 2, \dots, Z = 26$ and performing arithmetic operations.
2. **Opposite Letter Coding:** Replacing a letter with its opposite (e.g., A with Z, B with Y).
3. **Symbol Substitution:** Each letter or digit is replaced by a unique symbol.
4. **Matrix/Grid Coding:** Letters are coded based on their position in a predefined grid.
5. **Mixed Operations:** Combining letter shifts with numerical operations or symbol substitutions.

- **Key Properties and Identities:**

- The coding scheme is logical and consistent.
- Often, the same letter/number will have the same code throughout a problem.

- **Common Pitfalls or Tricky Points:**

- Overlooking multiple layers of coding (e.g., shift then reverse).
- Miscalculating numerical positions or applying incorrect arithmetic.
- Assuming a simple pattern when a more complex one exists.

- **Standard Problem-Solving Techniques or Shortcuts:**

- For letter-to-number codes, always write down the alphabet with its numerical positions.
- Look for common letters in different coded words to deduce their codes.
- Identify if the code is based on position, value, type (vowel/consonant), or a combination.
- Systematically test hypotheses about the coding rule.

Direction Sense

Definition and Core Idea: Direction sense problems involve determining the relative position and direction of a person or object after a series of movements. The core idea is to visualize or diagram the movements and apply principles of geometry.

• Important Formulas and Results:

1. **Pythagorean Theorem:** For finding the shortest distance between two points when movements form a right-angled triangle. If a person moves x units in one perpendicular direction and y units in another, the shortest distance d from the starting point is given by $d = \sqrt{x^2 + y^2}$.
2. **Cardinal Directions:** North (N), South (S), East (E), West (W).
3. **Intermediate Directions:** Northeast (NE), Northwest (NW), Southeast (SE), Southwest (SW).

• Key Properties and Identities:

- A right turn means a 90-degree clockwise rotation. A left turn means a 90-degree anti-clockwise rotation.
- "Facing North" and "moving North" are distinct concepts.
- Shadow problems: In the morning, shadow falls to the West. In the evening, shadow falls to the East. At noon, no shadow (or directly below).

• Common Pitfalls or Tricky Points:

- Confusing left/right turns, especially when facing different directions.
- Misinterpreting "from" and "to" in distance calculations.
- Calculation errors in applying the Pythagorean theorem.
- Not distinguishing between "distance travelled" and "shortest distance from start".

• Standard Problem-Solving Techniques or Shortcuts:

- Draw a clear diagram. Use a cross (+) to represent cardinal directions.
- Start from a central point and mark each movement.
- Use coordinates (x, y) to track position, where North is +y, South is -y, East is +x, West is -x.
- For shortest distance, identify if a right-angled triangle is formed.

Family Relationship

Definition and Core Idea: Family relationship problems require you to deduce the relationship between two individuals based on a series of statements describing their familial connections. The core idea is to construct a mental or physical family tree.

• Important Formulas and Results: No mathematical formulas. "Results" are standard family relations:

1. **Parent:** Father/Mother
2. **Sibling:** Brother/Sister
3. **Spouse:** Husband/Wife
4. **Child:** Son/Daughter
5. **Grandparent:** Father's/Mother's Father/Mother
6. **Grandchild:** Son's/Daughter's Son/Daughter
7. **Uncle/Aunt:** Parent's Brother/Sister
8. **Nephew/Niece:** Sibling's Son/Daughter
9. **Cousin:** Uncle's/Aunt's Child
10. **In-laws:** Relations by marriage (e.g., Father-in-law, Sister-in-law).

• Key Properties and Identities:

- Relationships are reciprocal (e.g., if A is B's father, B is A's child).
- Gender is crucial; do not assume gender unless explicitly stated or implied.

• Common Pitfalls or Tricky Points:

- Confusing genders (e.g., assuming "cousin" implies male or female).
- Misinterpreting complex statements with multiple "of"s (e.g., "my father's only son's wife").
- Making assumptions not explicitly stated in the problem.
- Overlooking the possibility of multiple relationships for a single individual.

• Standard Problem-Solving Techniques or Shortcuts:

- Draw a family tree diagram. Use symbols for gender (e.g., + for male, - for female) and relationships (e.g., horizontal line for siblings, vertical line for parent-child, double horizontal line for marriage).
- Break down complex statements into smaller, manageable parts.

- Start from a definite relationship and build outwards.
- Work backwards from the person whose relationship is being asked.

Inequality

Definition and Core Idea: Inequality problems involve comparing quantities using relational operators like greater than ($>$), less than ($<$), greater than or equal to ($\geq$), less than or equal to ($\leq$), and not equal to ($\neq$). The core idea is to deduce valid conclusions from a set of given inequality statements.

• Important Formulas and Results:

1. **Transitivity:** If $a > b$ and $b > c$, then $a > c$. Similarly for $<$, $\geq$, $\leq$.
2. **Addition/Subtraction:** If $a > b$, then $a + c > b + c$ and $a - c > b - c$.
3. **Multiplication/Division by Positive:** If $a > b$ and $c > 0$, then $ac > bc$ and $\frac{a}{c} > \frac{b}{c}$.
4. **Multiplication/Division by Negative:** If $a > b$ and $c < 0$, then $ac < bc$ and $\frac{a}{c} < \frac{b}{c}$. (Note: The inequality sign reverses).
5. **Combining Inequalities:**
 - If $a > b$ and $c > d$, then $a + c > b + d$.
 - If $a > b$ and $c < d$, no definite conclusion can be drawn about $a + c$ vs $b + d$.

• Key Properties and Identities:

- The direction of the inequality sign is crucial.
- An equality sign ($=$) can be treated as both $\geq$ and $\leq$.
- If there is a break in the chain of comparison (e.g., $A > B$ and $C < B$), direct comparison between A and C might not be possible without more information.

• Common Pitfalls or Tricky Points:

- Incorrectly reversing the inequality sign when multiplying or dividing by a negative number.
- Assuming a direct relationship between two variables when the chain of comparison is broken.
- Confusing strict inequalities ($>$, $<$) with non-strict ones ($\geq$, $\leq$).

• Standard Problem-Solving Techniques or Shortcuts:

- Combine the given statements into a single chain of inequalities (e.g., $A > B \geq C < D$).
- Identify common elements to link different statements.
- For conclusions, trace the path between the two variables in question. If a consistent direction of inequality exists, the conclusion is valid.
- If the path involves opposing inequality signs (e.g., $A > B < C$), no definite conclusion can be drawn.

Logical Inference

Definition and Core Idea: Logical inference involves drawing valid conclusions from a set of given premises (statements) using established rules of logic. The core idea is to determine what *must* be true if the premises are true, without introducing external information.

• Important Formulas and Results: No mathematical formulas. "Results" are rules of inference:

1. **Modus Ponens:** If P then Q. P. Therefore, Q. $((P \rightarrow Q) \wedge P) \rightarrow Q$
2. **Modus Tollens:** If P then Q. Not Q. Therefore, Not P. $((P \rightarrow Q) \wedge \neg Q) \rightarrow \neg P$
3. **Hypothetical Syllogism:** If P then Q. If Q then R. Therefore, If P then R. $((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$
4. **Disjunctive Syllogism:** P or Q. Not P. Therefore, Q. $((P \vee Q) \wedge \neg P) \rightarrow Q$
5. **Conjunction:** P. Q. Therefore, P and Q. $(P \wedge Q) \rightarrow (P \wedge Q)$
6. **Simplification:** P and Q. Therefore, P. $(P \wedge Q) \rightarrow P$

• Key Properties and Identities:

- A conclusion is logically valid if it necessarily follows from the premises, regardless of whether the premises themselves are true in reality.
- Validity is about the structure of the argument, not the truth of its content.

• Common Pitfalls or Tricky Points:

- Making assumptions not explicitly stated in the premises.
- Confusing correlation with causation.
- Committing logical fallacies (e.g., affirming the consequent, denying the antecedent).
- Drawing conclusions that are merely possible, not necessary.

• Standard Problem-Solving Techniques or Shortcuts:

- Represent statements symbolically (e.g., P for "It is raining," Q for "The ground is wet").
- Use Venn diagrams for categorical propositions (All A are B, Some A are B, No A are B).
- Test each conclusion against the premises to see if it *must* be true.

- Eliminate options that introduce new information or are only possibly true.

Logical Reasoning

Definition and Core Idea: Logical reasoning is a broad category encompassing various types of puzzles and problems that require deductive or inductive thinking to arrive at a solution. It tests your ability to analyze information, identify patterns, and draw sound conclusions. The core idea is systematic analysis and deduction.

- **Important Formulas and Results:** No specific mathematical formulas. "Results" are principles of reasoning:
 1. **Deductive Reasoning:** Moving from general principles to specific conclusions (guaranteed if premises are true).
 2. **Inductive Reasoning:** Moving from specific observations to general conclusions (probabilistic).
 3. **Abductive Reasoning:** Inferring the simplest or most likely explanation for an observation.
- **Key Properties and Identities:**
 - Consistency: The solution must not contradict any given statement.
 - Completeness: All conditions must be satisfied.
 - Uniqueness: Often, there is only one correct solution (though some problems may have multiple).
- **Common Pitfalls or Tricky Points:**
 - Jumping to conclusions without considering all possibilities.
 - Misinterpreting complex statements or conditions.
 - Failing to account for all constraints.
 - Getting overwhelmed by the amount of information.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Break down the problem into smaller, manageable parts.
 - Use tables, grids, or diagrams to organize information and track deductions.
 - Start with the most definite statements or constraints.
 - Use elimination: rule out options that contradict any condition.
 - Test hypotheses: try a possible solution and see if it fits all conditions.

Number Relations

Definition and Core Idea: Number relation problems involve identifying the underlying pattern or relationship between numbers in a given sequence, set, or pair. The core idea is to discover the rule that governs the numbers and apply it to find missing terms or solve related questions.

- **Important Formulas and Results:**
 1. **Arithmetic Progression (AP):**
 - n -th term: $a_n = a_1 + (n - 1)d$, where a_1 is the first term and d is the common difference.
 - Sum of n terms: $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$ or $S_n = \frac{n}{2}(a_1 + a_n)$.
 2. **Geometric Progression (GP):**
 - n -th term: $a_n = a_1 r^{n-1}$, where a_1 is the first term and r is the common ratio.
 - Sum of n terms: $S_n = \frac{a_1(r^n - 1)}{r - 1}$ (for $r \neq 1$).
 3. **Harmonic Progression (HP):** A sequence is an HP if the reciprocals of its terms form an AP.
 4. **Special Number Series:**
 - Squares: $1, 4, 9, 16, \dots, n^2$
 - Cubes: $1, 8, 27, 64, \dots, n^3$
 - Prime Numbers: $2, 3, 5, 7, 11, \dots$
 - Fibonacci Sequence: $F_n = F_{n-1} + F_{n-2}$ (e.g., $0, 1, 1, 2, 3, 5, \dots$)
- **Key Properties and Identities:**
 - Patterns can involve addition, subtraction, multiplication, division, squares, cubes, prime numbers, or combinations.
 - Sometimes patterns involve alternating operations or multiple interleaved series.
 - Digit sums or specific properties of digits can also form patterns.
- **Common Pitfalls or Tricky Points:**
 - Assuming a simple AP or GP when the pattern is more complex (e.g., double difference series).
 - Missing alternating patterns or interleaved series.
 - Calculation errors, especially with larger numbers or multiple operations.
 - Not considering all types of number properties (primes, squares, cubes).
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Calculate differences between consecutive terms. If not constant, calculate differences of differences (second-order differences).
 - Calculate ratios between consecutive terms.

- Check for squares, cubes, or prime numbers.
- Look for alternating patterns or two interleaved series.
- Consider operations on digits (sum of digits, product of digits).
- Test your identified pattern with all given terms before predicting the next.

Odd One

Definition and Core Idea: "Odd One Out" problems require you to identify the item (word, number, letter, or figure) that does not belong to a group, based on a shared characteristic among the others. The core idea is classification and identifying the exception.

- **Important Formulas and Results:** No mathematical formulas. "Results" are common classification criteria:
 1. **Category:** All items belong to a specific category except one (e.g., fruits, vegetables, animals).
 2. **Property:** All items share a common property except one (e.g., all are prime numbers, all are even).
 3. **Function:** All items serve a similar function except one (e.g., tools for cutting, modes of transport).
 4. **Structure/Pattern:** All items follow a specific structural or sequential pattern except one (e.g., letter series, number series).
 5. **Relationship:** All items have a specific relationship with each other except one.
- **Key Properties and Identities:**
 - The shared characteristic must be consistent and evident for the majority of the items.
 - The "odd one" will lack this specific characteristic.
- **Common Pitfalls or Tricky Points:**
 - Finding a superficial or weak connection for the "odd one" instead of a strong common property for the rest.
 - Not considering all possible patterns or categories.
 - Overlooking a more obvious pattern in favor of a complex one.
 - Subjectivity: sometimes multiple patterns might exist, but usually one is intended.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - Analyze each item individually to identify its properties.
 - Look for a common theme, category, or pattern that applies to at least three (or more, if applicable) of the items.
 - The item that does not fit this common theme is the odd one out.
 - Consider different types of relationships: semantic, numerical, alphabetical, functional.

Passage Reading

Definition and Core Idea: Passage reading (or reading comprehension) problems involve reading a given text and answering questions based solely on the information provided in that passage. The core idea is to understand the main idea, identify supporting details, and draw logical inferences from the text.

- **Important Formulas and Results:** No mathematical formulas. "Results" are reading comprehension strategies:
 1. **Main Idea:** The central theme or argument of the passage.
 2. **Supporting Details:** Facts, examples, or explanations that elaborate on the main idea.
 3. **Inference:** A conclusion logically derived from the text, but not explicitly stated.
 4. **Author's Tone:** The author's attitude towards the subject matter (e.g., critical, objective, enthusiastic).
- **Key Properties and Identities:**
 - All answers must be directly supported by the passage.
 - Avoid bringing in outside knowledge or personal opinions.
 - Distinguish between what is explicitly stated and what can be logically inferred.
- **Common Pitfalls or Tricky Points:**
 - Misinterpreting details or the main idea.
 - Making assumptions not supported by the text.
 - Falling for distractors that are partially true or true in general but not in the context of the passage.
 - Time pressure leading to superficial reading.
- **Standard Problem-Solving Techniques or Shortcuts:**
 - **Read the questions first:** This helps you know what to look for while reading the passage.
 - **Skim the passage for the main idea:** Read the first and last sentences of each paragraph.
 - **Read the passage carefully:** Pay attention to keywords, transition words, and specific details.
 - **Locate answers:** For specific detail questions, find the relevant sentence(s) in the passage.
 - **Infer carefully:** For inference questions, ensure your conclusion is a necessary consequence of the text.
 - **Eliminate options:** Rule out options that are clearly incorrect, not mentioned, or contradict the passage.

Round Table Arrangement

Definition and Core Idea: Round table arrangement problems involve arranging people or objects around a circular table or in a circular formation based on given conditions. The core idea is to understand relative positioning in a circle and apply principles of circular permutations.

• **Important Formulas and Results:**

1. **Circular Permutations (Distinct items):** For n distinct items arranged in a circle, the number of arrangements is $(n - 1)!$. This is because in a circle, the starting point is arbitrary, so we fix one person's position and arrange the remaining $(n - 1)$ people linearly.
2. **Circular Permutations (Symmetric items):** If clockwise and anti-clockwise arrangements are considered the same (e.g., arranging beads in a necklace), the number of arrangements is $\frac{(n-1)!}{2}$.

• **Key Properties and Identities:**

- In a circular arrangement, "left" and "right" are relative to the person's facing direction.
- "Opposite" means directly across the table.
- Fixing one person's position eliminates rotational symmetry and converts the problem into a linear arrangement for the remaining people.

• **Common Pitfalls or Tricky Points:**

- Confusing left/right turns in a circular setup.
- Misinterpreting "between" (e.g., "A is between B and C" means B-A-C or C-A-B).
- Not fixing a reference point, leading to overcounting arrangements.
- Assuming a specific number of seats when not explicitly stated.

• **Standard Problem-Solving Techniques or Shortcuts:**

- Draw a circle and mark the number of seats.
- Always start by fixing one person's position (e.g., at the "top" of the circle) to establish a reference.
- Place individuals with definite positions or relationships first (e.g., "A is opposite B").
- Use symbols or abbreviations for people.
- Deduce relative positions step-by-step, eliminating possibilities.
- For "left" and "right", imagine yourself sitting in that person's position.

Seating Arrangement

Definition and Core Idea: Seating arrangement problems involve arranging people or objects in a linear row, multiple rows, or other geometric shapes (excluding circles, which are covered separately) based on given conditions. The core idea is to deduce the exact positions by satisfying all constraints.

• **Important Formulas and Results:**

1. **Linear Permutations:** For n distinct items arranged in a line, the number of arrangements is $n!$.
2. **Adjacent Items:** If certain items must sit together, treat them as a single block and arrange the block and other items, then arrange items within the block.

• **Key Properties and Identities:**

- In a linear arrangement, "left" and "right" are absolute (from the perspective of an observer facing the arrangement).
- If people are facing each other in two rows, their left/right directions will be opposite.

• **Common Pitfalls or Tricky Points:**

- Misinterpreting "immediate left/right" versus "left/right" (which can mean anywhere to the left/right).
- Confusing "between" (e.g., "A is between B and C" means B-A-C, not necessarily immediately).
- Incorrectly handling "facing each other" in multi-row arrangements.
- Not considering all possible scenarios before finalizing an arrangement.

• **Standard Problem-Solving Techniques or Shortcuts:**

- Draw a diagram representing the seating arrangement (e.g., a line for a single row, two parallel lines for two rows).
- Start with definite information (e.g., "A is at one end").
- Place individuals with clear relationships first (e.g., "B is immediately to the left of C").
- Use symbols or abbreviations.
- For complex conditions, create multiple possible scenarios and eliminate those that contradict other statements.
- Pay close attention to words like "only," "not," "exactly," which impose strict constraints.

Sequence Series

Definition and Core Idea: Sequence and series problems involve identifying the pattern in a given set of numbers or letters arranged in a specific order and then finding the missing term, the next term, or a specific term in the sequence. The core idea is to discover the rule that generates the sequence.

- **Important Formulas and Results:**

1. **Arithmetic Progression (AP):**

- n -th term: $a_n = a_1 + (n - 1)d$
- Sum of n terms: $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$ or $S_n = \frac{n}{2}(a_1 + a_n)$

2. **Geometric Progression (GP):**

- n -th term: $a_n = a_1 r^{n-1}$
- Sum of n terms: $S_n = \frac{a_1(r^n - 1)}{r - 1}$ (for $r \neq 1$)

3. **Harmonic Progression (HP):** A sequence whose reciprocals form an AP.

4. **Fibonacci Sequence:** $F_n = F_{n-1} + F_{n-2}$ (starting with $F_0 = 0, F_1 = 1$).

5. **Other Common Patterns:**

- Squares: 1, 4, 9, 16, ...
- Cubes: 1, 8, 27, 64, ...
- Prime numbers: 2, 3, 5, 7, 11, ...
- Double difference series (differences between terms form an AP/GP).
- Alternating series (two patterns interleaved).
- Series based on operations on digits (sum of digits, product of digits).

- **Key Properties and Identities:**

- Patterns can involve arithmetic operations, powers, roots, prime numbers, or combinations.
- Letter series often use alphabetical positions (A=1, B=2, etc.) and apply numerical patterns.
- Sometimes the pattern is a combination of two or more simple patterns.

- **Common Pitfalls or Tricky Points:**

- Assuming a simple pattern (AP/GP) when a more complex one (e.g., double difference, alternating) is present.
- Calculation errors, especially with larger numbers or multiple steps.
- Not considering all possible types of numerical relationships (powers, primes, etc.).
- In letter series, forgetting the cyclical nature of the alphabet (Z to A).

- **Standard Problem-Solving Techniques or Shortcuts:**

- Calculate the differences between consecutive terms. If not constant, calculate the differences of these differences.
- Calculate the ratios between consecutive terms.
- Check if terms are squares, cubes, or related to prime numbers.
- Look for alternating patterns or two interleaved series.
- For letter series, convert letters to their numerical positions and look for numerical patterns.
- Test your identified pattern with all given terms to ensure consistency.

Statements Follow

Definition and Core Idea: "Statements Follow" problems (often syllogisms) present a set of statements (premises) and ask you to determine which of the given conclusions logically and necessarily follow from these statements. The core idea is to deduce conclusions strictly based on the provided information, without using external knowledge.

- **Important Formulas and Results:** No mathematical formulas. "Results" are principles of syllogistic reasoning:

1. **Categorical Propositions:**

- All A are B (Universal Affirmative)
- No A are B (Universal Negative)
- Some A are B (Particular Affirmative)
- Some A are not B (Particular Negative)

2. **Rules for Validity:** A conclusion is valid if it necessarily follows from the premises.

3. **Venn Diagrams:** A visual tool to represent categorical propositions and test the validity of conclusions.

- **Key Properties and Identities:**

- The truth of the premises is assumed for the purpose of testing validity.
- A conclusion is valid if it is impossible for the premises to be true and the conclusion false.
- "Some" implies "at least one" and does not exclude "all."

- **Common Pitfalls or Tricky Points:**

- Making assumptions based on real-world knowledge that are not stated in the premises.
- Confusing "some" with "all" or "most."
- Drawing conclusions that are merely possible, not necessary.
- Misinterpreting the negation of a statement (e.g., the negation of "All A are B" is "Some A are not B," not "No A are B").

- **Standard Problem-Solving Techniques or Shortcuts:**

- **Use Venn Diagrams:** Draw overlapping circles for each category mentioned in the statements. Shade or mark regions to represent the information.

- **Combine Statements:** Look for a common term between two statements to link them.
- **Test Each Conclusion:** Check if the conclusion is necessarily true based on your diagram or combined statements. If there's any possibility for the conclusion to be false while premises are true, it's invalid.
- **Avoid External Knowledge:** Stick strictly to the information given in the statements.
- **Focus on Necessity:** The conclusion must *necessarily* follow, not just possibly.

Quick Formula Reference

- **Age Relation:**
 - Age n years ago: $x - n$
 - Age n years hence: $x + n$
 - Age difference is constant.
- **Direction Sense:**
 - Shortest distance (Pythagorean): $d = \sqrt{x^2 + y^2}$
 - Right turn = 90° clockwise; Left turn = 90° anti-clockwise.
- **Inequality:**
 - Transitivity: $a > b, b > c \implies a > c$
 - Multiply/Divide by negative: Reverse inequality sign ($a > b, c < 0 \implies ac < bc$)
- **Logical Inference:**
 - Modus Ponens: $((P \rightarrow Q) \wedge P) \rightarrow Q$
 - Modus Tollens: $((P \rightarrow Q) \wedge \neg Q) \rightarrow \neg P$
 - Hypothetical Syllogism: $((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$
- **Number Relations & Sequence Series:**
 - **AP:** $a_n = a_1 + (n - 1)d, S_n = \frac{n}{2}(2a_1 + (n - 1)d)$
 - **GP:** $a_n = a_1 r^{n-1}, S_n = \frac{a_1(r^n - 1)}{r - 1}$
 - **HP:** Reciprocals form an AP.
 - **Fibonacci:** $F_n = F_{n-1} + F_{n-2}$
- **Round Table Arrangement:**
 - n distinct items: $(n - 1)!$ arrangements
 - Symmetric items (e.g., necklace): $\frac{(n-1)!}{2}$ arrangements
- **Seating Arrangement:**
 - n distinct items in a line: $n!$ arrangements
 - Treat adjacent items as a block for permutation calculations.
- **General (Analogy, Coding, Odd One, Family, Passage, Statements Follow):**
 - Focus on pattern recognition, logical deduction, and strict adherence to given information.
 - Use diagrams (family tree, Venn diagrams, direction maps) to visualize and organize information.
 - Avoid external assumptions.

Important Tips for GATE

1. **Read Carefully and Comprehensively:** Many errors in Analytical Aptitude stem from misreading the question or statements. Pay close attention to keywords like "only," "all," "some," "not," "immediate," "opposite," and "between."
2. **Visualize and Diagram:** For topics like Direction Sense, Family Relationship, Seating Arrangement, Round Table Arrangement, and Statements Follow, drawing clear diagrams (maps, family trees, tables, Venn diagrams) is invaluable. It helps organize information and prevents errors.
3. **Break Down Complex Problems:** Don't get overwhelmed by lengthy problem descriptions. Break them into smaller, manageable statements. Address definite information first, then deduce relative positions or conditions.
4. **Avoid External Assumptions:** In logical reasoning and inference problems, stick strictly to the information provided in the statements. Do not bring in real-world knowledge or personal opinions unless explicitly required (e.g., general knowledge analogies).
5. **Practice Pattern Recognition:** For Number Relations, Sequence Series, Coding-Decoding, and Analogy, consistent practice helps you quickly identify common patterns (AP, GP, squares, cubes, prime numbers, letter shifts, types of relationships).
6. **Time Management:** Analytical Aptitude questions can be time-consuming. If a problem seems overly complex or you're stuck, mark it and move on. Return to it if time permits. Sometimes, eliminating options can be faster than finding the direct answer.
7. **Check Your Work:** Before finalizing an answer, quickly cross-verify your solution against all given conditions. This is especially crucial for arrangement problems where a single misplaced person can invalidate the entire setup.
8. **Understand Negations:** Be clear about what constitutes a negation of a statement. For instance, the negation of "All A are B" is "Some A are not B," not "No A are B." This is vital for logical inference and statements follow.

questions.

8.1 Age Relation (2)

8.1.1 Age Relation: GATE CSE 2010 | Question: 62



Hari(H), Gita(G), Irfan(I) and Saira(S) are siblings (i.e., brothers and sisters). All were born on 1st January. The age difference between any two successive siblings (that is born one after another) is less than three years. Given the following facts:

- Hari's age + Gita's age > Irfan's age + Saira's age
- The age difference between Gita and Saira is one year. However Gita is not the oldest and Saira is not the youngest.
- There are no twins.

In what order they were born (oldest first)?

- A. HSIG B. SGHI C. IGSB D. IHSG

gatecse-2010 analytical-aptitude logical-reasoning normal age-relation

Answer key

8.1.2 Age Relation: GATE2013 CE: GA-10



Abhishek is elder to Savar. Savar is younger to Anshul. Which of the given conclusions is logically valid and is inferred from the above statements?

- A. Abhishek is elder to Anshul B. Anshul is elder to Abhishek
C. Abhishek and Anshul are of the same age D. No conclusion follows

gate2013-ce logical-reasoning age-relation easy

Answer key

8.2 Analogy (1)

8.2.1 Analogy: GATE CSE 2026 | Set 2 | GA | Question: 6



Water : P : Food : Q

Choose the P and Q combination from the options below to form a meaningful analogy.

- A. P = Thirst; Q = Hunger B. P = Drink; Q = Hunger
C. P = Thirst; Q = Satiated D. P = Wet; Q = Critic

gatecse-2026-set2 analytical-aptitude analogy two-marks

Answer key

8.3 Code Words (2)

Practice Test: Test 1 (5Q)

8.3.1 Code Words: GATE CSE 2015 | Set 3 | GA | Question: 1



If ROAD is written as URDG, then SWAN should be written as:

- A. VXDQ B. VZDQ C. VZDP D. UXDQ

gatecse-2015-set3 analytical-aptitude easy logical-reasoning code-words code-language

Answer key

8.3.2 Code Words: GATE CSE 2016 | Set 1 | GA | Question: 04



If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?

- A. zentaga B. tagafer. C. tagazen. D. relffer.

Answer key

8.4

Coding Decoding (1)

8.4.1 Coding Decoding: GATE CSE 2025 | Set 2 | GA | Question: 7



If IMAGE and FIELD are coded as FHBNJ and EMFJG respectively then, which one among the given options is the most appropriate code for BEACH?

- A. CEADP B. IDBFC C. JGIBC D. IBCEC

gatecse2025-set2 analytical-aptitude coding-decoding two-marks

Answer key

8.5

Direction Sense (5)

Practice Test: Test 1 (14Q)

8.5.1 Direction Sense: GATE CSE 2015 | Set 2 | GA | Question: 7



Four branches of a company are located at M, N, O and P. M is north of N at a distance of 4 km; P is south of O at a distance of 2 km; N is southeast of O by 1 km. What is the distance between M and P in km?

- A. 5.34 B. 6.74 C. 28.5 D. 45.49

gatecse-2015-set2 analytical-aptitude normal direction-sense

Answer key

8.5.2 Direction Sense: GATE CSE 2017 | Set 2 | GA | Question: 3



There are five buildings called V, W, X, Y and Z in a row (not necessarily in that order). V is to the West of W. Z is to the East of X and the West of V. W is to the West of Y. Which is the building in the middle?

- A. V B. W C. X D. Y

gatecse-2017-set2 analytical-aptitude direction-sense normal

Answer key

8.5.3 Direction Sense: GATE CSE 2019 | GA | Question: 10



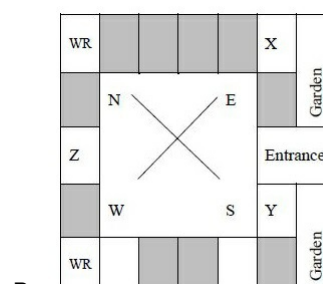
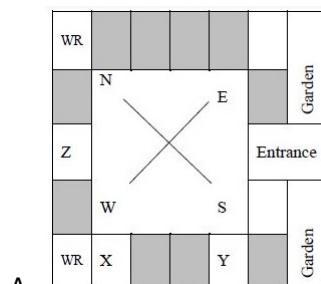
Three of the five students are allocated to a hostel put in special requests to the warden, Given the floor plan of the vacant rooms, select the allocation plan that will accommodate all their requests.

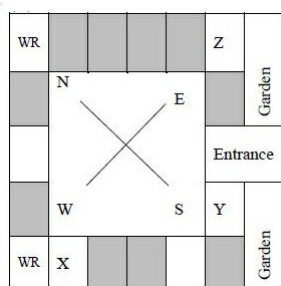
Request by X: Due to pollen allergy, I want to avoid a wing next to the garden.

Request by Y: I want to live as far from the washrooms as possible, since I am very much sensitive to smell.

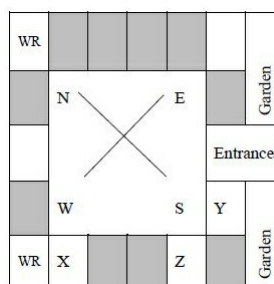
Request by Z: I believe in Vaastu and so I want to stay in South-West wing.

The shaded rooms are already occupied. WR is washroom





C.



D.

gatecse-2019 general-aptitude analytical-aptitude easy direction-sense two-marks

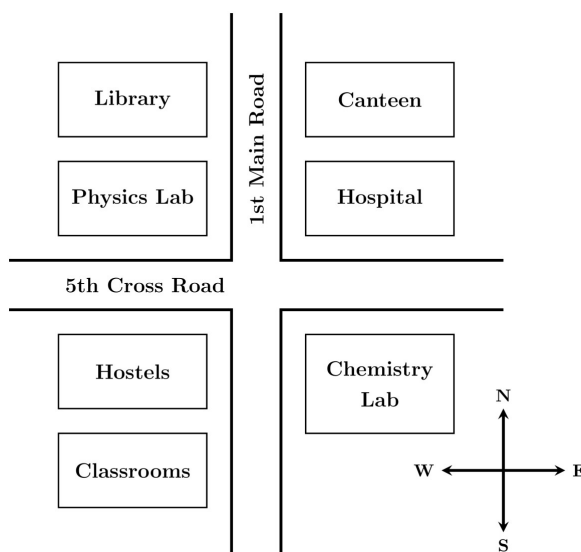
Answer key

8.5.4 Direction Sense: GATE CSE 2025 | Set 1 | GA | Question: 5



According to the map shown in the figure, which one of the following statements is correct?

Note: The figure shown is representative.



- A. The library is located to the northwest of the canteen.
- B. The hospital is located to the east of the chemistry lab.
- C. The chemistry lab is to the southeast of physics lab.
- D. The classrooms and canteen are next to each other.

gatecse2025-set1 analytical-aptitude direction-sense easy one-mark

Answer key

8.5.5 Direction Sense: GATE2014 AG: GA-9



X is 1 km northeast of Y . Y is 1 km southeast of Z . W is 1 km west of Z . P is 1 km south of W . Q is 1 km east of P . What is the distance between X and Q in km?

- A. 1
- B. $\sqrt{2}$
- C. $\sqrt{3}$
- D. 2

gate2014-ag analytical-aptitude direction-sense normal

Answer key

8.6

Family Relationship (1)

Practice Test: Test 1 (9Q)

8.6.1 Family Relationship: GATE CSE 2025 | Set 1 | GA | Question: 4



Consider the relationships among P, Q, R, S, and T :

- P is the brother of Q.
- S is the daughter of Q.
- T is the sister of S.
- R is the mother of Q.

The following statements are made based on the relationships given above.

1. R is the grandmother of S.
2. P is the uncle of S and T.
3. R has only one son.
4. Q has only one daughter.

Which one of the following options is correct?

- A. Both (1) and (2) are true. B. Both (1) and (3) are true.
C. Only (3) is true. D. Only (4) is true.

gatecse2025-set1 analytical-aptitude logical-reasoning family-relationship one-mark

Answer key

8.7

Logical Inference (1)

8.7.1 Logical Inference: GATE CSE 2025 | Set 2 | GA | Question: 6



Based only on the conversation below, identify the logically correct inference:

"Even if I had known that you were in the hospital, I would not have gone there to see you", Ramya told Josephine.

- A. Ramya knew that Josephine was in the hospital.
B. Ramya did not know that Josephine was in the hospital.
C. Ramya and Josephine were once close friends; but now, they are not.
D. Josephine was in the hospital due to an injury to her leg.

gatecse2025-set2 analytical-aptitude logical-inference two-marks

Answer key

8.8

Logical Reasoning (18)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(15Q\)](#) [Test 4 \(15Q\)](#) [Test 5 \(3Q\)](#)

8.8.1 Logical Reasoning: GATE CSE 2010 | Question: 61



If $137 + 276 = 435$ how much is $731 + 672$?

- A. 534 B. 1403 C. 1623 D. 1513

gatecse-2010 analytical-aptitude normal logical-reasoning

Answer key

8.8.2 Logical Reasoning: GATE CSE 2015 | Set 3 | GA | Question: 7



The head of newly formed government desires to appoint five of the six selected members P, Q, R, S, T and U to portfolios of Home, Power, Defense, Telecom, and Finance. U does not want any portfolio if S gets one of the five. R wants either Home or Finance or no portfolio. Q says that if S gets Power or Telecom, then she must get the other one. T insists on a portfolio if P gets one.

Which is the valid distribution of portfolios?

- A. P -Home, Q -Power, R -Defense, S -Telecom, T -Finance
B. R -Home, S -Power, P -Defense, Q -Telecom, T -Finance

- C. *P*-Home, *Q*-Power, *T*-Defense, *S*-Telecom, *U*-Finance
D. *Q*-Home, *U*-Power, *T*-Defense, *R*-Telecom, *P*-Finance

gatecse-2015-set3 analytical-aptitude normal logical-reasoning

Answer key

8.8.3 Logical Reasoning: GATE CSE 2017 | Set 2 | GA | Question: 7



There are three boxes. One contains apples, another contains oranges and the last one contains both apples and oranges. All three are known to be incorrectly labeled. If you are permitted to open just one box and then pull out and inspect only one fruit, which box would you open to determine the contents of all three boxes?

- A. The box labeled 'Apples' B. The box labeled 'Apples and Oranges'
C. The box labeled 'Oranges' D. Cannot be determined

gatecse-2017-set2 analytical-aptitude normal tricky logical-reasoning

Answer key

8.8.4 Logical Reasoning: GATE CSE 2021 | Set 2 | GA | Question: 10



Six students P, Q, R, S, T and U, with distinct heights, compare their heights and make the following observations.

- Observation I: S is taller than R.
- Observation II: Q is the shortest of all.
- Observation III: U is taller than only one student.
- Observation IV: T is taller than S but is not the tallest

The number of students that are taller than R is the same as the number of students shorter than _____.

- A. T B. R C. S D. P

gatecse-2021-set2 analytical-aptitude logical-reasoning two-marks

Answer key

8.8.5 Logical Reasoning: GATE CSE 2023 | GA | Question: 4



A survey for a certain year found that 90% of pregnant women received medical care at least once before giving birth. Of these women, 60% received medical care from doctors, while 40% received medical care from other healthcare providers.

Given this information, which one of the following statements can be inferred with *certainty*?

- A. More than half of the pregnant women received medical care at least once from a doctor.
B. Less than half of the pregnant women received medical care at least once from a doctor.
C. More than half of the pregnant women received medical care at most once from a doctor.
D. Less than half of the pregnant women received medical care at most once from a doctor.

gatecse-2023 analytical-aptitude logical-reasoning one-mark

Answer key

8.8.6 Logical Reasoning: GATE CSE 2026 | Set 1 | GA | Question: 5



'When the teacher is in the room, all students stand silently.'

If the above statement is true, which one of the following statements is **not necessarily** true?

- A. If any student is not standing silently, then the teacher is not in the room.
B. When the teacher is in the room, all students are silent.
C. If all students are standing, then the teacher is in the room.
D. When the teacher is in the room, all students are standing.

gatecse-2026-set1 analytical-aptitude logical-reasoning one-mark

8.8.7 Logical Reasoning: GATE CSE 2026 | Set 1 | GA | Question: 6



Combinatorics deals with problems involving counting. For example, "How many distinct arrangements of N distinct objects in M spaces on a circle are possible?" is a typical problem in combinatorics. This kind of counting is sometimes used in the modeling of several physical phenomena. Often, in such models, the different combinatorial possibilities are assigned probability values. Assigning probabilities enables the computation of the average values of physical quantities.

Consider the following statements:

P : Combinatorics is always invoked in the modeling of physical phenomena.

Q : Modeling some physical phenomena involves assigning probabilities to combinatorial possibilities in order to compute average values of physical quantities.

Based on the passage above, what can be inferred about statements **P** and **Q**?

- | | |
|------------------------------|-----------------------------|
| A. P is False and Q is False | B. P is False and Q is True |
| C. P is True and Q is False | D. P is True and Q is True |

gatecse-2026-set1 analytical-aptitude logical-reasoning combinatory two-marks

8.8.8 Logical Reasoning: GATE CSE 2026 | Set 1 | GA | Question: 9



In the 2020 summer Olympics' Javelin throw finals, Neeraj Chopra exhibited a spectacular performance to win the gold medal. The silver medal was won by Jakub Vadlejch and the bronze medal was won by Vitezlav Vesely. There were six rounds of throws with each athlete having one throw per round. The best of all the throws of each athlete is considered for the medal. Following were the observations about the throws:

- The first and second rounds were dominated by Neeraj Chopra with a gold medal performance in his second throw, while the other two athletes did not have any medal winning throws in these rounds.
- The throws in the last round by both Jakub Vadlejch and Vitezlav Vesely were fouls and were not considered for scoring.
- After four rounds, Vitezlav Vesely was in the second position and could not improve upon his best throw in the succeeding rounds.
- In the fourth round, the throw by Jakub Vadlejch was the best in that round.

In which round did Vitezlav Vesely have his best throw?

- | | | | |
|----------|-----------|----------|----------|
| A. Third | B. Fourth | C. Fifth | D. Sixth |
|----------|-----------|----------|----------|

gatecse-2026-set1 analytical-aptitude logical-reasoning two-marks

8.8.9 Logical Reasoning: GATE CSE 2026 | Set 2 | GA | Question: 5



'When it is raining, peacocks dance.'

Based only on this sentence, which one of the following options is necessarily true?

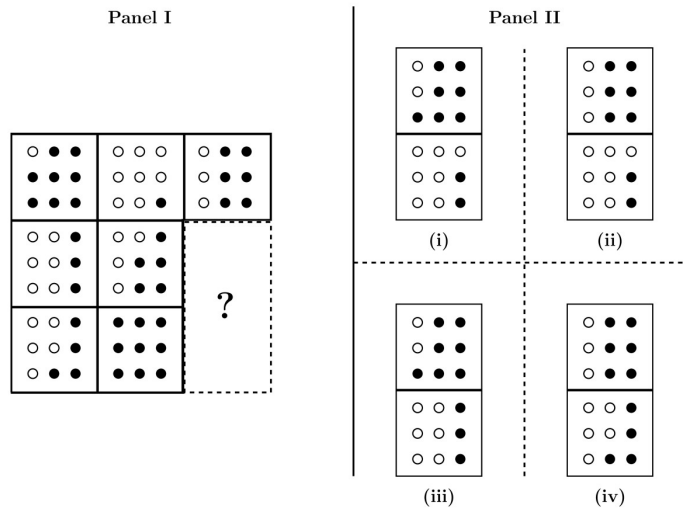
- Peacocks dance only when it is raining.
- When peacocks dance, it is raining.
- When peacocks are not dancing, it is not raining.
- When it is not raining, peacocks do not dance.

gatecse-2026-set2 analytical-aptitude logical-reasoning one-mark

8.8.10 Logical Reasoning: GATE CSE 2026 | Set 2 | GA | Question: 7



Two tiles are missing in Panel I. Which one of the options in Panel II is the appropriate choice for the missing tiles?



- A. (i) B. (ii) C. (iii) D. (iv)

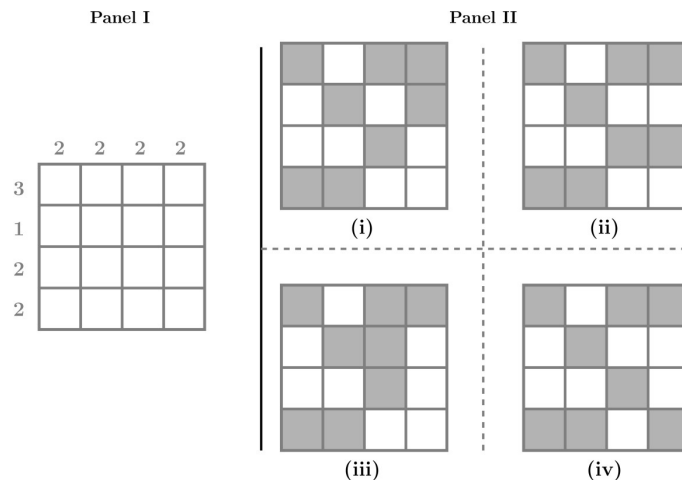
gatecse-2026-set2 analytical-aptitude two-marks logical-reasoning

Answer key

8.8.11 Logical Reasoning: GATE CSE 2026 | Set 2 | GA | Question: 9



The figure in Panel I below is a grid of cells with four rows and four columns. The numbers on the top and on the left represent the number of cells that are to be shaded in that column and row, respectively. Which one of the options shown in Panel II below represents the grid shaded correctly?



- A. (i) B. (ii) C. (iii) D. (iv)

gatecse-2026-set2 analytical-aptitude two-marks logical-reasoning

Answer key

8.8.12 Logical Reasoning: GATE DA 2026 | GA | Question: 5



'If his latest movie had been a commercial success, the actor would have made enough money to sponsor his next movie.'

Based only on the above sentence, which one of the following statements is true?

- A. The actor will certainly sponsor his next movie.
 B. His latest movie was a commercial success.
 C. The actor made enough money from his latest movie.
 D. His latest movie was not commercially successful.

Answer key

8.8.13 Logical Reasoning: GATE DA 2026 | GA | Question: 8



Rishi and Swathi are students of Class 5. Pavan and Tanvi are students of Class 4. Rishi and Pavan are boys. Swathi and Tanvi are girls. The four students played a total of three games of chess. The games were played one after another. A player who lost a game did not participate in any more games. It was observed that:

- I. the first game was the only game where two students of the same class played against each other,
- II. the students of Class 5 won more games than the students of Class 4, and
- III. the boys won two games and the girls won one game.

The student who did not lose any game is ____.

- A. Pavan B. Rishi C. Swathi D. Tanvi

gateda-2026 analytical-aptitude logical-reasoning two-marks

Answer key

8.8.14 Logical Reasoning: GATE2011 GG: GA-8



Three sisters (R , S , and T) received a total of 24 toys during Christmas. The toys were initially divided among them in a certain proportion. Subsequently, R gave some toys to S which doubled the share of S . Then S in turn gave some of her toys to T , which doubled T 's share. Next, some of T 's toys were given to R , which doubled the number of toys that R currently had. As a result of all such exchanges, the three sisters were left with equal number of toys. How many toys did R have originally?

- A. 8 B. 9 C. 11 D. 12

gate2011-gg logical-reasoning analytical-aptitude

Answer key

8.8.15 Logical Reasoning: GATE2011 MN: GA-63



L , M and N are waiting in a queue meant for children to enter the zoo. There are 5 children between L and M , and 8 children between M and N . If there are 3 children ahead of N and 21 children behind L , then what is the minimum number of children in the queue?

- A. 28 B. 27 C. 41 D. 40

analytical-aptitude gate2011-mn logical-reasoning

Answer key

8.8.16 Logical Reasoning: GATE2012 AR: GA-8



Ravi is taller than Arun but shorter than Iqbal. Sam is shorter than Ravi. Mohan is shorter than Arun. Balu is taller than Mohan and Sam. The tallest person can be

- A. Mohan B. Ravi C. Balu D. Arun

gate2012-ar logical-reasoning

Answer key

8.8.17 Logical Reasoning: GATE2012 CY: GA-9



There are eight bags of rice looking alike, seven of which have equal weight and one is slightly heavier. The weighing balance is of unlimited capacity. Using this balance, the minimum number of weighings required to identify the heavier bag is

- A. 2 B. 3 C. 4 D. 8

gate2012-cy analytical-aptitude logical-reasoning

Answer key

Answer key

8.11

Passage Reading (2)

 **Practice Tests:** [Test 1 \(6Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(15Q\)](#) [Test 4 \(15Q\)](#) [Test 5 \(5Q\)](#)

8.11.1 Passage Reading: GATE CSE 2022 | GA | Question: 7



In a recently conducted national entrance test, boys constituted 65% of those who appeared for the test. Girls constituted the remaining candidates and they accounted for 60% of the qualified candidates.

Which one of the following is the correct logical inference based on the information provided in the above passage?

- A. Equal number of boys and girls qualified
- B. Equal number of boys and girls appeared for the test
- C. The number of boys who appeared for the test is less than the number of girls who appeared
- D. The number of boys who qualified the test is less than the number of girls who qualified

gatecse-2022 analytical-aptitude logical-reasoning passage-reading two-marks

Answer key

8.11.2 Passage Reading: GATE CSE 2023 | GA | Question: 6



The country of Zombieland is in distress since more than 75% of its working population is suffering from serious health issues. Studies conducted by competent health experts concluded that a complete lack of physical exercise among its working population was one of the leading causes of their health issues. As one of the measures to address the problem, the Government of Zombieland has decided to provide monetary incentives to those who ride bicycles to work.

Based only on the information provided above, which one of the following statements can be logically inferred with *certainty*?

- A. All the working population of Zombieland will henceforth ride bicycles to work.
- B. Riding bicycles will ensure that all of the working population of Zombieland is free of health issues.
- C. The health experts suggested to the Government of Zombieland to declare riding bicycles as mandatory.
- D. The Government of Zombieland believes that riding bicycles is a form of physical exercise.

gatecse-2023 analytical-aptitude logical-reasoning passage-reading two-marks

Answer key

8.12

Round Table Arrangement (1)

 **Practice Test:** [Test 1 \(9Q\)](#)

8.12.1 Round Table Arrangement: GATE CSE 2017 | Set 1 | GA | Question: 7



Six people are seated around a circular table. There are at least two men and two women. There are at least three right-handed persons. Every woman has a left-handed person to her immediate right. None of the women are right-handed. The number of women at the table is

- A. 2
- B. 3
- C. 4
- D. Cannot be determined

gatecse-2017-set1 analytical-aptitude round-table-arrangement

Answer key

8.13

Seating Arrangement (1)

8.13.1 Seating Arrangement: GATE CSE 2017 | Set 1 | GA | Question: 3



Rahul, Murali, Srinivas and Arul are seated around a square table. Rahul is sitting to the left of Murali. Srinivas is sitting to the right of Arul. Which of the following pairs are seated opposite each other?

A. Rahul and Murali

B. Srinivas and Arul

C. Srinvas and Murali

D. Srinivas and Rahul

gatecse-2017-set1 analytical-aptitude logical-reasoning seating-arrangement

Answer key

8.14

Sequence Series (3)

Practice Tests: Test 1 (15Q) Test 2 (1Q)

8.14.1 Sequence Series: GATE CSE 2020 | GA | Question: 7



If $P = 3$, $R = 27$, $T = 243$, then $Q + S =$ _____

A. 40

B. 80

C. 90

D. 110

gatecse-2020 analytical-aptitude logical-reasoning sequence-series two-marks

Answer key

8.14.2 Sequence Series: GATE CSE 2024 | Set 2 | GA | Question: 5



In the sequence 6, 9, 14, x , 30, 41, a possible value of x is

A. 25

B. 21

C. 18

D. 20

gatecse-2024-set2 analytical-aptitude sequence-series one-mark

Answer key

8.14.3 Sequence Series: GATE2010 MN: GA-7



Given the sequence $A, B, B, C, C, C, D, D, D, D, \dots$ etc., that is one A , two B 's, three C 's, four D 's, five E 's and so on, the 240th letter in the sequence will be :

A. V

B. U

C. T

D. W

general-aptitude logical-reasoning gate2010-mn sequence-series

Answer key

8.15

Statements Follow (7)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (7Q)

8.15.1 Statements Follow: GATE CSE 2016 | Set 1 | GA | Question: 08



Consider the following statements relating to the level of poker play of four players P, Q, R and S .

- I. P always beats Q
- II. R always beats S
- III. S loses to P only sometimes.
- IV. R always loses to Q

Which of the following can be logically inferred from the above statements?

- i. P is likely to beat all the three other players
- ii. S is the absolute worst player in the set

A. (i). only

B. (ii) only

C. (i) and (ii) only'

D. neither (i) nor (ii)

gatecse-2016-set1 analytical-aptitude normal statements-follow

Answer key

8.15.2 Statements Follow: GATE CSE 2016 | Set 2 | GA | Question: 08



All hill-stations have a lake. Ooty has two lakes.

Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

- i. Ooty is not a hill-station.
- ii. No hill-station can have more than one lake.

- A. (i) only.
- B. (ii) only.
- C. Both (i) and (ii)
- D. Neither (i) nor (ii)

gatecse-2016-set2 analytical-aptitude easy statements-follow

Answer key

8.15.3 Statements Follow: GATE CSE 2021 | Set 1 | GA | Question: 9



Given below are two statements 1 and 2, and two conclusions I and II

- Statement 1: All bacteria are microorganisms.
- Statement 2: All pathogens are microorganisms.
- Conclusion I: Some pathogens are bacteria.
- Conclusion II: All pathogens are not bacteria.

Based on the above statements and conclusions, which one of the following options is logically CORRECT?

- A. Only conclusion I is correct
- B. Only conclusion II is correct
- C. Either conclusion I or II is correct
- D. Neither conclusion I nor II is correct

gatecse-2021-set1 analytical-aptitude logical-reasoning statements-follow two-marks

Answer key

8.15.4 Statements Follow: GATE CSE 2022 | GA | Question: 4



Given below are four statements.

- Statement 1 : All students are inquisitive.
- Statement 2 : Some students are inquisitive.
- Statement 3 : No student is inquisitive.
- Statement 4 : Some students are not inquisitive.

From the given four statements, find the two statements that CANNOT BE TRUE simultaneously, assuming that there is at least one student in the class.

- A. Statement 1 and Statement 3
- B. Statement 1 and Statement 2
- C. Statement 2 and Statement 4
- D. Statement 3 and Statement 4

gatecse-2022 analytical-aptitude logical-reasoning statements-follow one-mark

Answer key

8.15.5 Statements Follow: GATE DS&AI 2024 | GA | Question: 6



Thousands of years ago, some people began dairy farming. This coincided with a number of mutations in a particular gene that resulted in these people developing the ability to digest dairy milk.

Based on the given passage, which of the following can be inferred?

- A. All human beings can digest dairy milk.
- B. No human being can digest dairy milk.
- C. Digestion of dairy milk is essential for human beings.
- D. In human beings, digestion of dairy milk resulted from a mutated gene.

gate-ds-ai-2024 analytical-aptitude logical-reasoning statements-follow two-marks

Answer key

8.15.6 Statements Follow: GATE2013 AE: GA-9



- All professors are researchers

- Some scientists are professors

Which of the given conclusions is logically valid and is inferred from the above arguments?

- A. All scientists are researchers
 C. Some researchers are scientists
 B. All professors are scientists
 D. No conclusion follows

gate2013-ae analytical-aptitude logical-reasoning statements-follow

Answer key

8.15.7 Statements Follow: GATE2014 AG: GA-6



In a group of four children, Som is younger to Riaz. Shiv is elder to Ansu. Ansu is youngest in the group. Which of the following statements is/are required to find the eldest child in the group?

Statements

1. Shiv is younger to Riaz.
2. Shiv is elder to Som.

- A. Statement 1 by itself determines the eldest child.
 B. Statement 2 by itself determines the eldest child.
 C. Statements 1 and 2 are both required to determine the eldest child.
 D. Statements 1 and 2 are not sufficient to determine the eldest child.

gate2014-ag analytical-aptitude normal statements-follow

Answer key

Answer Keys

8.1.1	B	8.1.2	D	8.2.1	A	8.3.1	B	8.3.2	C
8.4.1	B	8.5.1	A	8.5.2	A	8.5.3	D	8.5.4	C
8.5.5	C	8.6.1	A	8.7.1	B	8.8.1	C	8.8.2	B
8.8.3	B	8.8.4	C	8.8.5	A	8.8.6	C	8.8.7	B
8.8.8	A	8.8.9	C	8.8.10	A	8.8.11	B	8.8.12	D
8.8.13	D	8.8.14	C	8.8.15	A	8.8.16	C	8.8.17	A
8.8.18	B	8.9.1	A	8.10.1	D	8.10.2	D	8.11.1	D
8.11.2	D	8.12.1	A	8.13.1	C	8.14.1	C	8.14.2	B
8.14.3	A	8.15.1	D	8.15.2	D	8.15.3	C;D	8.15.4	A
8.15.5	D	8.15.6	C	8.15.7	A				



Syllabus: Numerical computation, Numerical estimation, Numerical reasoning and data interpretation

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	2	2	1	2	4	3	1	2	1	2	1
2 Marks Count	2	2	3	3	2	2	2	2	3	2	2
Total Marks	6	6	7	8	8	7	5	6	7	6	5

Welcome to the Quantitative Aptitude section of your GATE Computer Science preparation! This chapter is designed to equip you with the essential mathematical and logical reasoning skills crucial for success in the General Aptitude section of the GATE exam. Quantitative Aptitude tests your ability to solve numerical problems, interpret data, and apply fundamental mathematical concepts efficiently and accurately. Typically, this section carries a significant weightage, often contributing 8-10 marks out of the total 15 marks for General Aptitude, with questions ranging from direct formula application to complex problem-solving and data interpretation. Mastering these topics is not just about scoring in aptitude; it also sharpens your analytical thinking, which is invaluable for the core Computer Science subjects.

Topic-wise Key Concepts

Absolute Value

The absolute value of a real number x , denoted $|x|$, is its non-negative value regardless of its sign. It represents the distance of x from zero on the number line.

- **Definition:** $|x| = x$ if $x \geq 0$, and $|x| = -x$ if $x < 0$.
- **Properties:**
 1. $|x| \geq 0$
 2. $|-x| = |x|$
 3. $|xy| = |x||y|$
 4. $|x/y| = |x|/|y|$ (for $y \neq 0$)
 5. Triangle Inequality: $|x + y| \leq |x| + |y|$
- **Solving Equations/Inequalities:**
 1. $|x| = a \implies x = a$ or $x = -a$ (for $a \geq 0$)
 2. $|x| < a \implies -a < x < a$ (for $a > 0$)
 3. $|x| > a \implies x < -a$ or $x > a$ (for $a \geq 0$)
- **Pitfall:** Forgetting to consider both positive and negative cases when removing the absolute value sign.
- **Technique:** Isolate the absolute value term, then apply the appropriate definition or rule.

Age Relation

Age relation problems involve finding the current or future/past ages of individuals based on given conditions, usually forming linear equations.

- **Core Idea:** Translate word problems into algebraic equations involving variables for ages.
- **Technique:** Assign variables (e.g., x for current age), form equations based on the relationships (e.g., "A is twice as old as B" $\implies A = 2B$), and solve the system of equations.
- **Pitfall:** Misinterpreting phrases like "X years ago" or "X years hence".

Algebra

Algebra deals with symbols and rules for manipulating these symbols. It involves basic operations, factorization, and algebraic identities.

- **Identities:**
 1. $(a + b)^2 = a^2 + 2ab + b^2$
 2. $(a - b)^2 = a^2 - 2ab + b^2$
 3. $a^2 - b^2 = (a - b)(a + b)$
 4. $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$

5. $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
 6. $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
 7. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
 8. $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
 9. If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$
- **Pitfall:** Sign errors, incorrect order of operations (PEMDAS/BODMAS).
 - **Technique:** Simplify expressions using identities, factorize, and solve equations systematically.

Alligation Mixture

Alligation is a rule that helps determine the ratio in which two or more ingredients of different prices must be mixed to produce a mixture of a desired price.

- **Formula:**

$$\frac{\text{Quantity of Cheaper}}{\text{Quantity of Dearer}} = \frac{\text{Price of Dearer} - \text{Mean Price}}{\text{Mean Price} - \text{Price of Cheaper}}$$

- **Technique:** Use the "alligation cross" diagram to visualize and solve problems quickly.
- **Pitfall:** Incorrectly identifying the cheaper, dearer, or mean price.

Area

Area is the measure of the extent of a two-dimensional surface or shape.

- **Formulas:**
 1. **Square:** s^2 (where s is side)
 2. **Rectangle:** $l \times w$ (where l is length, w is width)
 3. **Circle:** πr^2 (where r is radius)
 4. **Triangle:** $\frac{1}{2} \times \text{base} \times \text{height}$ (or Heron's formula: $\sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a + b + c)/2$)
 5. **Parallelogram:** $\text{base} \times \text{height}$
 6. **Trapezoid:** $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$
 7. **Rhombus:** $\frac{1}{2} \times d_1 \times d_2$ (where d_1, d_2 are diagonals)
- **Pitfall:** Using perimeter formulas instead of area, or incorrect units.
- **Technique:** Identify the shape, recall the correct formula, and ensure consistent units.

Arithmetic Series

An arithmetic series is the sum of terms in an arithmetic progression, where the difference between consecutive terms is constant (common difference d).

- **Formulas:**
 1. n -th term: $a_n = a_1 + (n - 1)d$
 2. Sum of n terms: $S_n = \frac{n}{2}(a_1 + a_n)$ or $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$
- **Pitfall:** Off-by-one errors when calculating n , especially when given ranges.
- **Technique:** Clearly identify a_1 , d , and n .

Average

The average (arithmetic mean) is the sum of a set of values divided by the number of values.

- **Formula:** $\text{Average} = \frac{\text{Sum of quantities}}{\text{Number of quantities}}$
- **Weighted Average:** $\text{Weighted Average} = \frac{\sum (w_i x_i)}{\sum w_i}$
- **Technique:** Use the formula directly. For weighted averages, ensure correct weights are applied.
- **Pitfall:** Confusing average with median or mode.

Bar Graph

A bar graph uses rectangular bars to represent data, with the length or height of the bars proportional to the values they

represent.

- **Core Idea:** Visual representation of discrete data for comparison.
- **Technique:** Carefully read labels, scales, and legends. Compare bar heights/lengths to answer questions about quantities, differences, or ratios.
- **Pitfall:** Misinterpreting scales or units on the axes.

Bayes Theorem

Bayes' Theorem describes the probability of an event, based on prior knowledge of conditions that might be related to the event.

- **Formula:** $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- **Key Terms:** $P(A|B)$ is posterior probability, $P(B|A)$ is likelihood, $P(A)$ is prior probability, $P(B)$ is evidence.
- **Pitfall:** Confusing conditional probabilities $P(A|B)$ and $P(B|A)$.
- **Technique:** Break down the problem into individual probabilities and substitute into the formula.

Calendar

Calendar problems involve determining the day of the week for a given date or finding specific dates based on calendar logic.

- **Core Idea:** Based on the concept of "odd days" (remainder when days are divided by 7).
- **Key Concepts:**
 1. Ordinary year = 365 days = 52 weeks + 1 odd day
 2. Leap year = 366 days = 52 weeks + 2 odd days
 3. Leap year rules: Divisible by 4, unless it's a century year not divisible by 400.
 4. Odd days for centuries: 100 years = 5 odd days, 200 years = 3, 300 years = 1, 400 years = 0.
- **Technique:** Calculate total odd days from a reference date (e.g., Jan 1, 0001) to the target date.
- **Pitfall:** Incorrectly identifying leap years or counting odd days.

Cartesian Coordinates

Cartesian coordinates define the position of a point in a plane using two perpendicular axes (x and y).

- **Key Formulas:**
 1. **Distance between (x_1, y_1) and (x_2, y_2) :** $D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 2. **Midpoint of a segment:** $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
 3. **Section Formula (dividing in ratio $m : n$):** $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$
 4. **Area of triangle with vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) :** $\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$
- **Pitfall:** Sign errors in distance or section formula.
- **Technique:** Plot points if helpful, apply formulas carefully.

Circle

A circle is the set of all points in a plane that are equidistant from a central point.

- **Formulas:**
 1. **Circumference:** $C = 2\pi r = \pi d$
 2. **Area:** $A = \pi r^2$
 3. **Equation of a circle (center (h, k) , radius r):** $(x - h)^2 + (y - k)^2 = r^2$
 4. **Area of sector:** $\frac{\theta}{360^\circ} \times \pi r^2$ (where θ is central angle in degrees)
 5. **Length of arc:** $\frac{\theta}{360^\circ} \times 2\pi r$
- **Pitfall:** Confusing radius with diameter, or circumference with area.
- **Technique:** Identify given parameters and apply the correct formula.

Clock Time

Clock time problems involve calculating angles between clock hands, or determining exact times based on relative speeds of hands.

- **Key Rates:**

1. Hour hand: 0.5° per minute
2. Minute hand: 6° per minute
3. Relative speed: 5.5° per minute (minute hand gains on hour hand)

- **Formula for angle between hands at H hours M minutes:** $|30H - \frac{11}{2}M|$

- **Technique:** Calculate the position of each hand relative to 12, then find the difference.

- **Pitfall:** Not considering the hour hand's movement within the hour.

Combinatory (Permutation and Combination)

Combinatory deals with counting arrangements (permutations) and selections (combinations) of objects.

- **Formulas:**

1. **Permutation (arrangement of k items from n):** $P(n, k) = \frac{n!}{(n-k)!}$
2. **Combination (selection of k items from n):** $C(n, k) = \frac{n!}{k!(n-k)!}$
3. $n! = n \times (n-1) \times \dots \times 1$, $0! = 1$

- **Key Principle:** Permutation is about order, combination is not. "Arrangement" implies permutation, "selection" or "group" implies combination.

- **Pitfall:** Confusing permutation with combination.

- **Technique:** Determine if order matters. Use multiplication principle for sequential events, addition principle for mutually exclusive events.

Compound Interest

Compound interest is interest calculated on the initial principal, which also includes all of the accumulated interest from previous periods on a deposit or loan.

- **Formulas:**

1. **Amount (compounded annually):** $A = P\left(1 + \frac{R}{100}\right)^T$
2. **Compound Interest (CI):** $CI = A - P$
3. **Amount (compounded n times a year):** $A = P\left(1 + \frac{R}{100n}\right)^{nT}$
4. **Amount (compounded continuously):** $A = Pe^{RT/100}$

- **Pitfall:** Confusing simple interest with compound interest, or incorrect compounding frequency.

- **Technique:** Identify principal (P), rate (R), time (T), and compounding frequency.

Conditional Probability

Conditional probability is the probability of an event occurring given that another event has already occurred.

- **Formula:** $P(A|B) = \frac{P(A \cap B)}{P(B)}$ (where $P(B) > 0$)

- **Key Idea:** The sample space is reduced to event B.

- **Pitfall:** Misidentifying the intersection event or the conditioning event.

- **Technique:** Clearly define events A and B, find their intersection, and the probability of B.

Cones

A cone is a three-dimensional geometric shape that tapers smoothly from a flat base (usually circular) to a point called the apex or vertex.

- **Formulas:**

1. **Volume:** $V = \frac{1}{3}\pi r^2 h$
2. **Curved Surface Area (CSA):** $CSA = \pi r l$ (where l is slant height, $l = \sqrt{r^2 + h^2}$)
3. **Total Surface Area (TSA):** $TSA = \pi r(l + r)$

- **Pitfall:** Confusing height (h) with slant height (l).

- **Technique:** Use Pythagorean theorem to find l if r and h are given.

Contour Plots

Contour plots (or isoline maps) display three-dimensional data on a two-dimensional plane, where lines connect points of equal value.

- **Core Idea:** Visualize a third variable (e.g., elevation, temperature) over a surface.
- **Technique:** Interpret the density and values of contour lines. Closely spaced lines indicate a steep gradient, widely spaced lines indicate a gentle gradient.
- **Pitfall:** Misinterpreting the values represented by the lines or the gradient.

Cost Market Price

These problems involve concepts of cost price (CP), selling price (SP), market price (MP), profit, loss, and discount.

- **Formulas:**
 1. Profit = SP – CP (if SP > CP)
 2. Loss = CP – SP (if CP > SP)
 3. Profit/Loss % = $\frac{\text{Profit/Loss}}{\text{CP}} \times 100$
 4. Discount = MP – SP
 5. Discount % = $\frac{\text{Discount}}{\text{MP}} \times 100$
 6. SP = CP $\times (1 \pm \frac{\text{Profit/Loss \%}}{100})$
 7. SP = MP $\times (1 - \frac{\text{Discount \%}}{100})$
- **Pitfall:** Calculating profit/loss % on SP instead of CP, or discount % on CP instead of MP.
- **Technique:** Clearly identify CP, SP, MP, and the base for percentage calculations.

Counting

Counting principles involve determining the number of possible outcomes for various events.

- **Principles:**
 1. **Addition Principle:** If event A can occur in m ways and event B in n ways, and they are mutually exclusive, then A or B can occur in $m + n$ ways.
 2. **Multiplication Principle:** If event A can occur in m ways and event B in n ways, then A and B can occur in $m \times n$ ways.
- **Technique:** Break down complex counting problems into simpler, sequential or mutually exclusive steps.
- **Pitfall:** Overcounting or undercounting by misapplying the principles.

Cubes

A cube is a three-dimensional solid object bounded by six square faces, facets or sides, with three meeting at each vertex.

- **Formulas:**
 1. **Volume:** $V = a^3$ (where a is side length)
 2. **Surface Area:** $SA = 6a^2$
 3. **Diagonal of a face:** $a\sqrt{2}$
 4. **Main Diagonal of the cube:** $a\sqrt{3}$
- **Pitfall:** Confusing surface area with volume, or face diagonal with main diagonal.
- **Technique:** Visualize the cube and its dimensions.

Currency Notes

Problems involve calculating the total value of currency notes of different denominations or determining the number of notes of each denomination given a total amount.

- **Core Idea:** Form linear equations based on the value and count of notes.
- **Technique:** Assign variables to the number of notes of each denomination and set up equations.
- **Pitfall:** Calculation errors with large numbers or multiple denominations.

Curves

In quantitative aptitude, "curves" generally refer to the graphs of functions, requiring interpretation of their shape, slope, and points of interest.

- **Core Idea:** Visual representation of relationships between variables.
- **Technique:** Analyze the trend (increasing/decreasing), points of maxima/minima, and intersections.
- **Pitfall:** Misinterpreting the axes or the units.

Data Interpretation

Data interpretation involves extracting, analyzing, and drawing conclusions from various forms of data presentation like tables, charts (bar, pie, line, scatter, radar), and graphs.

- **Core Idea:** Critical analysis of presented data to answer specific questions.
- **Technique:** Read titles, labels, scales, and legends carefully. Perform calculations (percentages, ratios, averages) based on the data.
- **Pitfall:** Rushing calculations, misreading data points, or making assumptions not supported by the data.

Digital Image Processing

While a CS topic, in QA it might appear as basic numerical problems related to image size, resolution, or data storage.

- **Core Idea:** Image represented as a grid of pixels.
- **Key Concepts:**
 1. **Resolution:** Width $\times$ Height (in pixels)
 2. **Color Depth:** Bits per pixel (e.g., 8-bit for 256 colors, 24-bit for true color)
 3. **Image Size (uncompressed):** Resolution $\times$ Color Depth (in bits)
- **Technique:** Apply multiplication principles for calculating total bits/bytes.
- **Pitfall:** Unit conversions (bits to bytes, KB to MB).

Factors

Factors (or divisors) of a number are integers that divide the number evenly without leaving a remainder.

- **Core Idea:** Finding numbers that multiply to give the original number.
- **Properties:**
 1. For $N = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$ (prime factorization):
 2. **Number of factors:** $(a_1 + 1)(a_2 + 1) \dots (a_k + 1)$
 3. **Sum of factors:** $(1 + p_1 + \dots + p_1^{a_1})(1 + p_2 + \dots + p_2^{a_2}) \dots$
- **Technique:** Prime factorization is key to finding number and sum of factors.
- **Pitfall:** Missing factors, especially 1 and the number itself.

Fractions

Fractions represent a part of a whole, expressed as a ratio of two integers (numerator/denominator).

- **Operations:** Addition, subtraction, multiplication, division.
- **Properties:** Equivalent fractions, proper/improper fractions, mixed numbers.
- **Technique:** Find common denominators for addition/subtraction. Invert and multiply for division. Simplify fractions.
- **Pitfall:** Errors in finding common denominators or simplifying.

Functions

A function is a relation between a set of inputs and a set of permissible outputs with the property that each input is related to exactly one output.

- **Core Idea:** Mapping inputs to outputs. $y = f(x)$.
- **Key Concepts:** Domain (set of inputs), Range (set of outputs), types (linear, quadratic, polynomial).
- **Technique:** Evaluate functions at given points, understand their graphs, and solve for unknowns.
- **Pitfall:** Incorrectly applying function rules or misinterpreting domain/range restrictions.

Geometry

Geometry deals with the properties and relations of points, lines, surfaces, solids, and higher dimensional analogs.

- **Core Idea:** Understanding shapes, angles, and spatial relationships.
- **Key Theorems:** Pythagorean theorem, properties of parallel lines and transversals, angle sum property of triangles, similar/congruent triangles.
- **Technique:** Draw diagrams, identify knowns and unknowns, apply relevant theorems and formulas.
- **Pitfall:** Assuming properties not explicitly stated or derivable.

Graph Coloring

Graph coloring assigns colors to elements of a graph subject to certain constraints. In QA, it's usually simplified to finding the minimum number of colors for a small graph.

- **Core Idea:** Assign colors to vertices such that no two adjacent vertices share the same color.
- **Technique:** Start coloring from a vertex with high degree; systematically assign colors, avoiding conflicts with neighbors.
- **Pitfall:** Missing a valid coloring or not finding the minimum.

Inequality

Inequalities are mathematical statements comparing two expressions using symbols like $<$, $>$, $\leq$, $\geq$.

- **Properties:**
 1. Adding/subtracting a number from both sides does not change the inequality direction.
 2. Multiplying/dividing by a positive number does not change the inequality direction.
 3. Multiplying/dividing by a negative number **reverses** the inequality direction.
- **Technique:** Solve like equations, but remember to flip the sign when multiplying/dividing by a negative number. For quadratic inequalities, use sign analysis of factors.
- **Pitfall:** Forgetting to reverse the inequality sign.

LCM HCF

LCM (Least Common Multiple) is the smallest positive integer divisible by each of a given set of integers. HCF (Highest Common Factor) or GCD (Greatest Common Divisor) is the largest positive integer that divides each of a given set of integers.

- **Formulas:**
 1. For two numbers a and b : $\text{LCM}(a, b) \times \text{HCF}(a, b) = a \times b$
 2. To find LCM/HCF: Use prime factorization method.
- **Technique:** Prime factorization is the most reliable method.
- **Pitfall:** Confusing LCM with HCF, especially in word problems.

Line Graph

A line graph displays information as a series of data points connected by straight line segments, showing trends over time or categories.

- **Core Idea:** Visualizing trends and changes over a continuous variable.
- **Technique:** Observe the slope of the lines to understand rates of change, identify peaks and troughs.
- **Pitfall:** Misinterpreting the scale or the meaning of the trend.

Lines

Lines in coordinate geometry are one-dimensional figures with properties like slope, intercepts, and equations.

- **Formulas:**
 1. **Slope (m) of a line through (x_1, y_1) and (x_2, y_2) :** $m = \frac{y_2 - y_1}{x_2 - x_1}$
 2. **Equation of a line (slope-intercept form):** $y = mx + c$ (where c is y-intercept)
 3. **Equation of a line (point-slope form):** $y - y_1 = m(x - x_1)$
 4. **Parallel lines:** $m_1 = m_2$
 5. **Perpendicular lines:** $m_1 m_2 = -1$
- **Pitfall:** Sign errors in slope calculation, confusing parallel with perpendicular conditions.

- **Technique:** Identify slope and a point, then use the appropriate equation form.

Logarithms

Logarithms are the inverse operation to exponentiation. The logarithm of a number x to the base b is the exponent to which b must be raised to produce x .

- **Definition:** $\log_b a = c \iff b^c = a$
- **Properties:**
 1. $\log_b(xy) = \log_b x + \log_b y$
 2. $\log_b(x/y) = \log_b x - \log_b y$
 3. $\log_b x^n = n \log_b x$
 4. $\log_b b = 1, \log_b 1 = 0$
 5. **Change of Base:** $\log_b a = \frac{\log_c a}{\log_c b}$
- **Pitfall:** Misapplying properties, especially for addition/subtraction.
- **Technique:** Convert between logarithmic and exponential forms, use properties to simplify.

Maps

Map problems involve interpreting scale, distances, and directions on maps.

- **Core Idea:** Scale represents the ratio of a distance on the map to the corresponding distance on the ground.
- **Formula:** Map Distance = Actual Distance $\times$ Scale
- **Technique:** Pay close attention to the scale (e.g., 1:100000 or 1 cm = 10 km) and units.
- **Pitfall:** Incorrect unit conversions.

Maxima Minima

Finding the maximum or minimum value of a function or expression.

- **Core Idea:** Identifying the highest or lowest point a function can reach.
- **For Quadratic Function** $f(x) = ax^2 + bx + c$:
 1. Vertex at $x = -b/(2a)$
 2. If $a > 0$, minimum value at vertex.
 3. If $a < 0$, maximum value at vertex.
- **General Technique (Calculus-based, if applicable):** Find derivative $f'(x)$, set to zero to find critical points. Use second derivative test for max/min.
- **Pitfall:** Confusing local maxima/minima with global maxima/minima.

Mensuration

Mensuration is the branch of mathematics that deals with the measurement of geometric figures and their parameters like length, area, and volume. This topic consolidates formulas for various 2D and 3D shapes.

- **Core Idea:** Applying geometric formulas to calculate dimensions, areas, and volumes.
- **Technique:** Refer to specific topics like Area, Volume, Cones, Cubes, Circles, Triangles for detailed formulas.
- **Pitfall:** Incorrectly applying formulas or unit conversions.

Modular Arithmetic

Modular arithmetic is a system of arithmetic for integers, where numbers "wrap around" when reaching a certain value—the modulus.

- **Definition:** $a \equiv b \pmod{m}$ means a and b have the same remainder when divided by m , or m divides $(a - b)$.
- **Properties:**
 1. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.
 2. $(a \times b) \pmod{m} = ((a \pmod{m}) \times (b \pmod{m})) \pmod{m}$
- **Technique:** Use properties to simplify large numbers before finding remainders. Useful for unit digit problems.
- **Pitfall:** Incorrectly handling negative numbers in modulo operations.

Number Representation

Number representation deals with how numbers are written and interpreted, often involving different bases (e.g., decimal, binary).

- **Core Idea:** Converting numbers between different bases.
- **Technique:** For base-10 to base-B, repeatedly divide by B and collect remainders. For base-B to base-10, use positional notation (sum of digits $\times$ base to power of position).
- **Pitfall:** Errors in powers of the base or order of remainders.

Number Series

Number series problems involve identifying the pattern in a sequence of numbers and finding the next term or a missing term.

- **Types:** Arithmetic, geometric, difference series, squares/cubes, prime numbers, alternating patterns, Fibonacci-like.
- **Technique:** Look for common differences, common ratios, differences of differences, squares/cubes, or combinations of operations.
- **Pitfall:** Overlooking subtle patterns or assuming a simple pattern too quickly.

Number System

The number system classifies numbers (natural, whole, integers, rational, irrational, real, complex) and includes concepts like divisibility rules, prime/composite numbers.

- **Divisibility Rules:** (e.g., by 2, 3, 4, 5, 6, 8, 9, 10, 11).
- **Classification:**
 1. Natural Numbers: $\{1, 2, 3, \dots\}$
 2. Whole Numbers: $\{0, 1, 2, 3, \dots\}$
 3. Integers: $\{\dots, -2, -1, 0, 1, 2, \dots\}$
 4. Rational Numbers: $\{p/q \mid p, q \text{ are integers, } q \neq 0\}$
 5. Irrational Numbers: Non-terminating, non-repeating decimals (e.g., $\sqrt{2}, \pi$)
 6. Real Numbers: Rational + Irrational
- **Technique:** Apply divisibility rules, understand properties of number types.
- **Pitfall:** Confusing different sets of numbers (e.g., integers vs. whole numbers).

Number Theory

Number theory is a branch of pure mathematics devoted primarily to the study of integers and integer-valued functions, including primes, divisibility, and modular arithmetic.

- **Core Idea:** Properties of integers. (Overlaps with Factors, Prime Numbers, Modular Arithmetic).
- **Key Concepts:** Prime numbers, composite numbers, factors, multiples, HCF, LCM, divisibility.
- **Technique:** Use prime factorization, apply divisibility rules, understand modular arithmetic.
- **Pitfall:** Misapplying theorems or properties.

Numerical Computation

Numerical computation involves performing calculations accurately and efficiently, often with approximations, rounding, and significant figures.

- **Core Idea:** Precision and accuracy in calculations.
- **Technique:** Estimate answers, use approximations (e.g., $\pi \approx 22/7$ or 3.14), understand rules for significant figures and rounding.
- **Pitfall:** Rounding too early, leading to accumulated errors.

Percentage

Percentage is a way of expressing a number or ratio as a fraction of 100. It is often denoted using the percent sign, "%".

- **Formulas:**
 1. $\text{Value} = \text{Base} \times \frac{\text{Percentage}}{100}$
 2. $\text{Percentage Change} = \frac{\text{Change}}{\text{Original Value}} \times 100$

- 3. Successive Percentage Change: If an item changes by $x\%$ then by $y\%$, net change is $(x + y + \frac{xy}{100})\%$.
- **Pitfall:** Calculating percentage change based on the wrong base value.
- **Technique:** Convert percentages to decimals or fractions for calculations.

Permutation and Combination

Refer to "Combinatory" section for details.

Pie Chart

A pie chart is a circular statistical graphic, which is divided into slices to illustrate numerical proportion.

- **Core Idea:** Visualizing parts of a whole as proportions.
- **Properties:** The sum of all percentages in a pie chart must be 100%, and the sum of all central angles must be 360° .
- **Technique:** Relate percentages to degrees ($1\% = 3.6^\circ$). Calculate proportions and compare sectors.
- **Pitfall:** Misinterpreting the size of sectors or the total quantity.

Polynomials

A polynomial is an expression consisting of variables and coefficients, that involves only the operations of addition, subtraction, multiplication, and non-negative integer exponents of variables.

- **Key Concepts:** Degree, roots (zeros), factors.
- **Theorems:**
 1. **Remainder Theorem:** If a polynomial $P(x)$ is divided by $(x - a)$, the remainder is $P(a)$.
 2. **Factor Theorem:** $(x - a)$ is a factor of $P(x)$ if and only if $P(a) = 0$.
- **Technique:** Factorization, synthetic division, applying remainder/factor theorems.
- **Pitfall:** Errors in algebraic manipulation or sign errors.

Powers

Powers (or exponents) indicate the number of times a base number is multiplied by itself.

- **Rules of Exponents:**
 1. $x^a \cdot x^b = x^{a+b}$
 2. $x^a / x^b = x^{a-b}$
 3. $(x^a)^b = x^{ab}$
 4. $(xy)^a = x^a y^a$
 5. $(x/y)^a = x^a / y^a$
 6. $x^0 = 1$ (for $x \neq 0$)
 7. $x^{-a} = 1/x^a$
 8. $x^{1/n} = \sqrt[n]{x}$
- **Pitfall:** Incorrectly applying rules, especially with negative exponents or fractions.
- **Technique:** Simplify expressions using exponent rules, convert to common base if possible.

Prime Numbers

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

- **Properties:**
 1. 2 is the only even prime number.
 2. Every integer greater than 1 is either prime or can be uniquely expressed as a product of primes (Fundamental Theorem of Arithmetic).
- **Technique:** To check if N is prime, test divisibility by primes up to $\sqrt{N}$.
- **Pitfall:** Confusing 1 as a prime number (it's not).

Probability

Probability is the measure of the likelihood that an event will occur.

- **Formula:** $P(E) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$

- **Properties:**

1. $0 \leq P(E) \leq 1$
2. $P(\text{not } E) = 1 - P(E)$
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
4. For independent events: $P(A \cap B) = P(A)P(B)$

- **Technique:** Clearly define the sample space and the event. Use counting techniques (P&C) to find favorable and total outcomes.

- **Pitfall:** Incorrectly identifying sample space or favorable outcomes, or assuming independence when events are dependent.

Probability Density Function

For continuous random variables, the Probability Density Function (PDF) describes the relative likelihood for the random variable to take on a given value. The probability of the variable falling within a range is given by the integral of the PDF over that range.

- **Core Idea:** For continuous variables, $P(a \leq X \leq b) = \int_a^b f(x)dx$.

- **Property:** $\int_{-\infty}^{\infty} f(x)dx = 1$

- **Technique:** Understand that for continuous variables, $P(X = x) = 0$. Focus on areas under the curve.

- **Pitfall:** Treating PDF values directly as probabilities (they are not).

Profit Loss

Refer to "Cost Market Price" section for details.

Quadratic Equations

A quadratic equation is a polynomial equation of the second degree, typically written as $ax^2 + bx + c = 0$.

- **Formulas:**

1. **Quadratic Formula (roots):** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

2. **Discriminant:** $D = b^2 - 4ac$

- If $D > 0$, two distinct real roots.
- If $D = 0$, two equal real roots.
- If $D < 0$, no real roots (complex roots).

3. **Sum of roots:** $\alpha + \beta = -b/a$

4. **Product of roots:** $\alpha\beta = c/a$

- **Technique:** Factorization, completing the square, or using the quadratic formula.

- **Pitfall:** Sign errors in the quadratic formula or discriminant.

Radar Chart

A radar chart (or spider chart) displays multivariate data in the form of a two-dimensional chart of three or more quantitative variables represented on axes starting from the same point.

- **Core Idea:** Comparing multiple attributes of one or more items.

- **Technique:** Compare the area covered by different polygons or the values along each axis.

- **Pitfall:** Overcrowding with too many variables or items, making comparisons difficult.

Ratio Proportion

Ratio is a comparison of two quantities by division. Proportion states that two ratios are equal.

- **Formulas:**

1. **Ratio:** $a : b = a/b$

2. **Proportion:** $a : b :: c : d \implies a/b = c/d \implies ad = bc$ (product of extremes = product of means)

3. **Direct Proportion:** $y = kx$

4. **Inverse Proportion:** $y = k/x$

- **Technique:** Simplify ratios, use cross-multiplication for proportions, apply unitary method for direct/inverse proportion problems.
- **Pitfall:** Incorrectly setting up ratios or proportions, especially in word problems.

Scatter Plot

A scatter plot uses Cartesian coordinates to display values for two variables for a set of data. The data is displayed as a collection of points, each having the value of one variable determining the position on the horizontal axis and the value of the other variable determining the position on the vertical axis.

- **Core Idea:** Visualizing the relationship (correlation) between two variables.
- **Technique:** Observe the trend of points (positive, negative, or no correlation), strength of correlation (tightness of points).
- **Pitfall:** Inferring causation from correlation.

Seating Arrangement

Seating arrangement problems involve arranging people or objects in a specific order (linear, circular, etc.) based on given conditions.

- **Core Idea:** Logical deduction and application of permutation principles.
- **Technique:** Draw diagrams, start with fixed positions or most restrictive conditions, use elimination. For circular arrangements, fix one person's position first.
- **Pitfall:** Missing a condition, or misinterpreting "left/right" in circular arrangements.

Sequence Series

A sequence is an ordered list of numbers. A series is the sum of the terms of a sequence. (Refer to Arithmetic Series for specific formulas).

- **Core Idea:** Identifying patterns and calculating sums.
- **Types:** Arithmetic, Geometric, Harmonic, special series (squares, cubes).
- **Geometric Series:**
 1. n -th term: $a_n = ar^{n-1}$
 2. Sum of n terms: $S_n = \frac{a(r^n - 1)}{r - 1}$ (for $r \neq 1$)
 3. Sum to infinity: $S_\infty = \frac{a}{1 - r}$ (for $|r| < 1$)
- **Technique:** Determine the type of sequence, find the common difference/ratio, and apply the correct formula.
- **Pitfall:** Confusing arithmetic and geometric series formulas.

Set Theory

Set theory deals with collections of objects (sets) and their properties and operations.

- **Key Concepts:** Union ($\cup$), Intersection ($\cap$), Complement (A' or A^c), Difference ($A - B$), Subset ($\subseteq$), Cardinality ($|A|$).
- **Formulas:**
 1. $|A \cup B| = |A| + |B| - |A \cap B|$
 2. $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$
- **Technique:** Use Venn diagrams to visualize and solve problems involving multiple sets.
- **Pitfall:** Double-counting or missing elements in set operations.

Shortest Path

While a CS algorithm topic, in QA, it might involve finding the shortest distance between two points on a simple grid or a small, explicitly given graph.

- **Core Idea:** Finding the minimum distance or cost to travel between two nodes.
- **Technique:** For simple graphs, inspect paths and sum edge weights. For grids, use Manhattan distance or Euclidean distance depending on movement rules.
- **Pitfall:** Missing a shorter path or incorrectly calculating path length.

Speed Time Distance

These problems relate speed, time, and distance, often involving relative motion, trains, or boats.

- **Formulas:**

1. Distance = Speed $\times$ Time
2. Average Speed = $\frac{\text{Total Distance}}{\text{Total Time}}$
3. **Relative Speed (same direction):** $|S_1 - S_2|$
4. **Relative Speed (opposite direction):** $S_1 + S_2$
5. Conversion: $1 \text{ km/hr} = 5/18 \text{ m/s}$

- **Technique:** Ensure consistent units. For relative speed, determine if objects are moving towards or away from each other.
- **Pitfall:** Unit conversion errors, or misapplying relative speed concepts.

Squares

A square is a regular quadrilateral, which means it has four equal sides and four equal angles (90° each).

- **Formulas:**

1. **Area:** s^2
2. **Perimeter:** $4s$
3. **Diagonal:** $s\sqrt{2}$

- **Properties:** All sides equal, all angles 90° , diagonals bisect each other at 90° .
- **Technique:** Apply formulas directly. Recognize perfect squares.
- **Pitfall:** Confusing square properties with other quadrilaterals.

Statistics

Statistics involves the collection, analysis, interpretation, presentation, and organization of data.

- **Key Measures:**

1. **Mean:** Average of all values.
2. **Median:** Middle value when data is ordered.
3. **Mode:** Most frequent value.
4. **Range:** Max value - Min value.
5. **Standard Deviation:** Measure of data dispersion (basic understanding).

- **Technique:** Calculate measures of central tendency and dispersion. Interpret basic statistical graphs.
- **Pitfall:** Confusing mean, median, and mode.

System of Equations

A system of equations is a set of two or more equations containing common variables, which are to be solved simultaneously.

- **Core Idea:** Finding values for variables that satisfy all equations simultaneously.

- **Techniques:**

1. **Substitution:** Solve one equation for a variable, substitute into others.
2. **Elimination:** Add/subtract equations to eliminate a variable.
3. **Matrix methods:** For larger systems (less common in basic QA).

- **Pitfall:** Algebraic errors during substitution or elimination.

Tables / Tabular Data

Tables organize data into rows and columns, providing a structured way to present information.

- **Core Idea:** Extracting specific data points and performing calculations based on row/column totals or individual entries.
- **Technique:** Carefully read row and column headers, identify the required data, and perform calculations (sum, average, percentage, ratio).
- **Pitfall:** Misreading data from the wrong row/column, or incorrect calculations.

Triangles

A triangle is a polygon with three edges and three vertices. It is one of the basic shapes in geometry.

- **Properties:**

1. Sum of angles = 180° .
2. Triangle Inequality: Sum of any two sides is greater than the third side.
3. Types: Equilateral, Isosceles, Scalene, Right-angled, Acute, Obtuse.

- **Formulas:**

1. **Area:** $\frac{1}{2} \times \text{base} \times \text{height}$ (or Heron's formula)
2. **Pythagorean Theorem (for right triangle):** $a^2 + b^2 = c^2$

- **Technique:** Draw diagrams, identify triangle type, apply relevant theorems (e.g., Pythagorean, similar triangles).
- **Pitfall:** Assuming a triangle is right-angled or isosceles without sufficient information.

Trigonometry

Trigonometry deals with the relationships between the sides and angles of triangles, particularly right-angled triangles.

- **Ratios (SOH CAH TOA):**

1. $\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$
2. $\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$
3. $\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{\sin \theta}{\cos \theta}$

- **Identities:**

1. $\sin^2 \theta + \cos^2 \theta = 1$
2. $\sec^2 \theta - \tan^2 \theta = 1$
3. $\csc^2 \theta - \cot^2 \theta = 1$

- **Standard Angles:** Memorize values for $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$.
- **Technique:** Draw right triangles, label sides, apply appropriate ratios.
- **Pitfall:** Confusing ratios, or incorrect use of identities.

Unit Digit

Finding the unit digit of large powers or complex expressions.

- **Core Idea:** The unit digit of powers of a number follows a cycle.
- **Technique:** Identify the cyclicity of the unit digit of the base number (e.g., 2: 2,4,8,6; 3: 3,9,7,1; 4: 4,6; 5: 5; 6: 6; 7: 7,9,3,1; 8: 8,4,2,6; 9: 9,1). Divide the exponent by the cycle length and use the remainder.
- **Pitfall:** Errors in determining the cycle or using the remainder.

Venn Diagram

Venn diagrams are graphical representations used to show relationships between sets. (Refer to Set Theory for formulas).

- **Core Idea:** Visualizing set operations and relationships.
- **Technique:** Draw overlapping circles (or other shapes) to represent sets. Fill in the numbers in each region based on the given information, starting from the innermost intersection.
- **Pitfall:** Incorrectly placing numbers in regions or misinterpreting the meaning of overlapping areas.

Volume

Volume is the amount of three-dimensional space occupied by an object or substance.

- **Formulas:**

1. **Cube:** a^3
2. **Cuboid:** $l \times w \times h$
3. **Cylinder:** $\pi r^2 h$
4. **Cone:** $\frac{1}{3} \pi r^2 h$
5. **Sphere:** $\frac{4}{3} \pi r^3$

6. **Hemisphere:** $\frac{2}{3}\pi r^3$

- **Technique:** Identify the 3D shape, recall the correct formula, ensure consistent units.
- **Pitfall:** Confusing volume with surface area, or using incorrect dimensions.

Work Time

Work-time problems involve calculating the time taken to complete a task by individuals or groups, often working at different rates.

- **Core Idea:** Work = Rate $\times$ Time. Rate is typically "work per unit time".
- **Formulas:**
 1. If A takes x days, A's 1-day work = $1/x$.
 2. If A and B work together, their combined 1-day work = $1/x + 1/y$.
 3. Efficiency $\propto \frac{1}{\text{Time}}$
 4. Men $\times$ Days $\times$ Hours/Work = Constant (for multiple workers)
- **Technique:** Convert all work rates to a common unit (e.g., work per day). Use LCM method for efficiency.
- **Pitfall:** Incorrectly adding/subtracting rates, or misinterpreting "together" vs. "alone" work.

Quick Formula Reference

- **Absolute Value:** $|x| = a \implies x = \pm a$; $|x| < a \implies -a < x < a$ or $|x| > a \implies x > a$ or $x < -a$
- **Algebraic Identities:** $(a+b)^2 = a^2 + 2ab + b^2$, $a^2 - b^2 = (a-b)(a+b)$, $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- **Alligation:** $\frac{Q_C}{Q_D} = \frac{P_D - P_M}{P_M - P_C}$
- **Area:** Square s^2 , Rectangle lw , Circle πr^2 , Triangle $\frac{1}{2}bh$
- **Arithmetic Series:** $a_n = a_1 + (n-1)d$, $S_n = \frac{n}{2}(a_1 + a_n)$
- **Average:** $\frac{\sum x}{n}$
- **Bayes Theorem:** $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- **Cartesian Coordinates:** Distance $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, Midpoint $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$
- **Circle:** Circumference $2\pi r$, Area πr^2
- **Clock Angle:** $|30H - \frac{11}{2}M|$
- **Combinations:** $C(n, k) = \frac{n!}{k!(n-k)!}$
- **Permutations:** $P(n, k) = \frac{n!}{(n-k)!}$
- **Compound Interest:** $A = P(1 + R/100)^T$
- **Conditional Probability:** $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- **Cone:** Volume $\frac{1}{3}\pi r^2 h$, CSA $\pi r l$, TSA $\pi r(l + r)$
- **Cost/Profit/Loss:** Profit/Loss % = $\frac{\text{Profit/Loss}}{\text{CP}} \times 100$, Discount % = $\frac{\text{Discount}}{\text{MP}} \times 100$
- **Cube:** Volume a^3 , Surface Area $6a^2$
- **Factors:** For $N = p_1^{a_1} \dots p_k^{a_k}$, No. of factors $(a_1 + 1) \dots (a_k + 1)$
- **Geometric Series:** $a_n = ar^{n-1}$, $S_n = \frac{a(r^n - 1)}{r - 1}$
- **LCM HCF:** $\text{LCM}(a, b) \times \text{HCF}(a, b) = a \times b$
- **Lines:** Slope $m = \frac{y_2 - y_1}{x_2 - x_1}$, $y = mx + c$, Parallel $m_1 = m_2$, Perpendicular $m_1 m_2 = -1$
- **Logarithms:** $\log(xy) = \log x + \log y$, $\log(x/y) = \log x - \log y$, $\log x^n = n \log x$
- **Percentage:** Value = Base $\times \frac{\text{Percentage}}{100}$
- **Probability:** $P(E) = \frac{\text{Favorable}}{\text{Total}}$
- **Quadratic Equation:** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, Sum of roots $-b/a$, Product of roots c/a
- **Ratio Proportion:** $a : b :: c : d \implies ad = bc$
- **Set Theory:** $|A \cup B| = |A| + |B| - |A \cap B|$
- **Speed Time Distance:** $D = S \times T$, Relative Speed (same) $|S_1 - S_2|$, (opposite) $S_1 + S_2$
- **Trigonometry:** $\sin \theta = O/H$, $\cos \theta = A/H$, $\tan \theta = O/A$, $\sin^2 \theta + \cos^2 \theta = 1$
- **Volume:** Cube a^3 , Cuboid lwh , Cylinder $\pi r^2 h$, Cone $\frac{1}{3}\pi r^2 h$, Sphere $\frac{4}{3}\pi r^3$
- **Work Time:** Work = Rate $\times$ Time

Important Tips for GATE

1. **Master Fundamentals:** Ensure a strong grasp of basic arithmetic, algebra, geometry, and data interpretation. Many complex problems are built upon these foundational concepts.
2. **Memorize Formulas:** Quantitative Aptitude is heavily formula-driven. Create flashcards or a dedicated notebook for all key formulas and identities. Regular revision is crucial.
3. **Practice Data Interpretation:** DI questions are common and can be time-consuming. Practice reading various charts (bar, pie, line, scatter, radar) and tables accurately and quickly. Pay attention to scales and units.
4. **Time Management:** Aptitude questions often have quick solutions if the right approach is identified. Don't get stuck on one question; if it's taking too long, mark it for review and move on.
5. **Unit Consistency:** Always check and convert units to be consistent within a problem (e.g., km/hr to m/s, cm to meters) to avoid common calculation errors.
6. **Approximation Skills:** For questions with close options or complex calculations, develop the ability to estimate and approximate to quickly narrow down choices.
7. **Identify Question Type:** Quickly categorize the problem (e.g., P&C, Profit/Loss, Speed-Time) to recall the relevant formulas and problem-solving strategies.
8. **Review Common Pitfalls:** Be aware of typical mistakes like sign errors in algebra, confusing permutations with combinations, or misinterpreting relative speed. Consciously check for these during practice.

9.1

No Classified Topic (1)

9.1.1 GATE CSE 2026 | Set 2 | GA | Question: 4



The values of Stock A and Stock B on a particular day are Rs. 50 and Rs. 80, respectively. An investor invests Rs. 100 in Stock A and Rs. 80 in Stock B. He sells all the stocks the next day when the value of Stock A is Rs. 55 and Stock B is Rs. 70. The profit made by the investor is Rs. _____

- A. 0 B. 5 C. 10 D. 20

gatecse-2026-set2 quantitative-aptitude one-mark

Answer key

9.2

Absolute Value (5)

Practice Test: Test 1 (10Q)

9.2.1 Absolute Value: GATE CSE 2014 | Set 2 | GA | Question: 8



If x is real and $|x^2 - 2x + 3| = 11$, then possible values of $|-x^3 + x^2 - x|$ include

- A. 2,4 B. 2,14 C. 4,52 D. 14,52

gatecse-2014-set2 quantitative-aptitude normal absolute-value

Answer key

9.2.2 Absolute Value: GATE CSE 2017 | Set 1 | GA | Question: 8



The expression $\frac{(x+y)-|x-y|}{2}$ is equal to :

- A. The maximum of x and y B. The minimum of x and y
C. 1 D. None of the above

gatecse-2017-set1 general-aptitude quantitative-aptitude maxima-minima absolute-value easy

Answer key

9.2.3 Absolute Value: GATE2011 AG: GA-7



Given that $f(y) = \frac{|y|}{y}$, and q is non-zero real number, the value of $|f(q) - f(-q)|$ is

- A. 0 B. -1
C. 1 D. 2

general-aptitude quantitative-aptitude gate2011-ag absolute-value

Answer key

9.2.4 Absolute Value: GATE2013 AE: GA-8



If $|-2X + 9| = 3$ then the possible value of $|-X| - X^2$ would be:

- A. 30 B. -30 C. -42 D. 42

gate2013-ae quantitative-aptitude absolute-value

Answer key

9.2.5 Absolute Value: GATE2013 CE: GA-7



If $|4X - 7| = 5$ then the values of $2|X| - |-X|$ is:

- A. $2, \left(\frac{1}{3}\right)$ B. $\left(\frac{1}{2}\right), 3$ C. $\left(\frac{3}{2}\right), 9$ D. $\left(\frac{2}{3}\right), 9$

gate2013-ce quantitative-aptitude absolute-value

Answer key

9.3 Algebra (1)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(12Q\)](#)

9.3.1 Algebra: GATE2011 MN: GA-61



If $\frac{(2y+1)}{(y+2)} < 1$, then which of the following alternatives gives the CORRECT range of y ?

- A. $-2 < y < 2$ B. $-2 < y < 1$
C. $-3 < y < 1$ D. $-4 < y < 1$

quantitative-aptitude gate2011-mn algebra

Answer key

9.4 Alligation Mixture (2)

Practice Test: [Test 1 \(8Q\)](#)

9.4.1 Alligation Mixture: GATE CSE 2011 | Question: 65



A container originally contains 10 litres of pure spirit. From this container, 1 litre of spirit replaced with 1 litre of water. Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?

- A. 7.58 litres B. 7.84 litres
C. 7 litres D. 7.29 litres

gatecse-2011 quantitative-aptitude normal alligation-mixture

Answer key

9.4.2 Alligation Mixture: GATE2010 TF: GA-9



A tank has 100 liters of water. At the end of every hour, the following two operations are performed in sequence: *i*) water equal to $m\%$ of the current contents of the tank is added to the tank, *ii*) water equal to $n\%$ of the current contents of the tank is removed from the tank. At the end of 5 hours, the tank contains exactly 100 liters of water. The relation between m and n is :

- A. $m = n$ B. $m > n$
C. $m < n$ D. None of the previous

general-aptitude quantitative-aptitude gate2010-tf alligation-mixture

Answer key

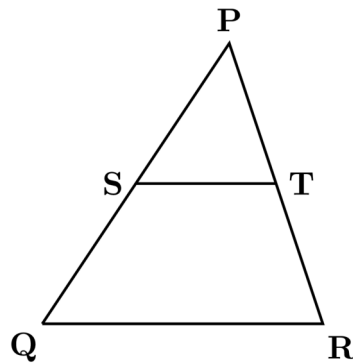
9.5 Area (1)

9.5.1 Area: GATE CSE 2025 | Set 1 | GA | Question: 7



In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR. What is the ratio of the area of the triangle PST to the area of the trapezium SQRT?

Note: The figure shown is representative.



- A. $\frac{1}{3}$ B. $\frac{1}{4}$ C. $\frac{2}{5}$ D. $\frac{1}{2}$

gatecse2025-set1 quantitative-aptitude geometry area two-marks

Answer key

9.6

Arithmetic Series (3)

Practice Test: Test 1 (10Q)

9.6.1 Arithmetic Series: GATE CSE 2013 | Question: 58



What will be the maximum sum of 44, 42, 40, ... ?

- A. 502 B. 504 C. 506 D. 500

gatecse-2013 quantitative-aptitude easy arithmetic-series

Answer key

9.6.2 Arithmetic Series: GATE CSE 2015 | Set 2 | GA | Question: 6



If the list of letters P, R, S, T, U is an arithmetic sequence, which of the following are also in arithmetic sequence?

- I. $2P, 2R, 2S, 2T, 2U$
 II. $P - 3, R - 3, S - 3, T - 3, U - 3$
 III. P^2, R^2, S^2, T^2, U^2

- A. I only B. I and II C. II and III D. I and III

gatecse-2015-set2 quantitative-aptitude normal arithmetic-series

Answer key

9.6.3 Arithmetic Series: GATE2011 AG: GA-6



The sum of n terms of the series $4 + 44 + 444 + \dots$ is

- A. $\frac{4}{81} [10^{n+1} - 9n - 1]$
 B. $\frac{4}{81} [10^{n-1} - 9n - 1]$

C. $\frac{4}{81}[10^{n+1} - 9n - 10]$

D. $\frac{4}{81}[10^n - 9n - 10]$

general-aptitude quantitative-aptitude gate2011-ag arithmetic-series

Answer key

9.7 Average (1)

Practice Test: Test 1 (13Q)

9.7.1 Average: GATE CSE 2025 | Set 1 | GA | Question: 3



The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?

- A. 25 B. 28 C. 30 D. 32

gatecse2025-set1 quantitative-aptitude average one-mark

Answer key

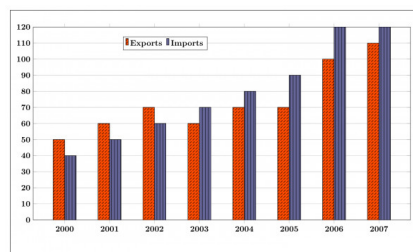
9.8 Bar Graph (4)

Practice Tests: Test 1 (15Q) Test 2 (4Q)

9.8.1 Bar Graph: GATE CSE 2015 | Set 3 | GA | Question: 10



The exports and imports (in crores of Rs.) of a country from the year 2000 to 2007 are given in the following bar chart. In which year is the combined percentage increase in imports and exports the highest?



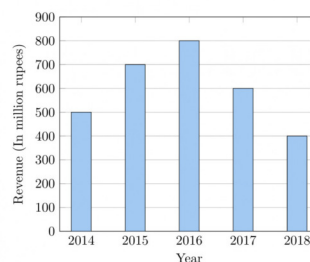
gatecse-2015-set3 quantitative-aptitude data-interpretation normal numerical-answers bar-graph

Answer key

9.8.2 Bar Graph: GATE CSE 2020 | GA | Question: 10



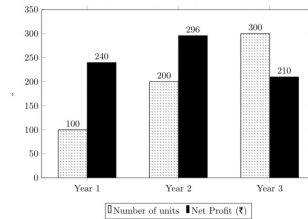
The total revenue of a company during 2014 – 2018 is shown in the bar graph. If the total expenditure of the company in each year is 500 million rupees, then the aggregate profit or loss (in percentage) on the total expenditure of the company during 2014 – 2018 is _____.



- A. 16.67% profit B. 16.67% loss
C. 20% profit D. 20% loss

Answer key

9.8.3 Bar Graph: GATE CSE 2021 | Set 2 | GA | Question: 9



The number of units of a product sold in three different years and the respective net profits are presented in the figure above. The cost/unit in Year 3 was ₹ 1, which was half the cost/unit in Year 2. The cost/unit in Year 3 was one-third of the cost/unit in Year 1. Taxes were paid on the selling price at 10%, 13%, and 15% respectively for the three years. Net profit is calculated as the difference between the selling price and the sum of cost and taxes paid in that year.

The ratio of the selling price in Year 2 to the selling price in Year 3 is _____.

- A. 4 : 3 B. 1 : 1 C. 3 : 4 D. 1 : 2

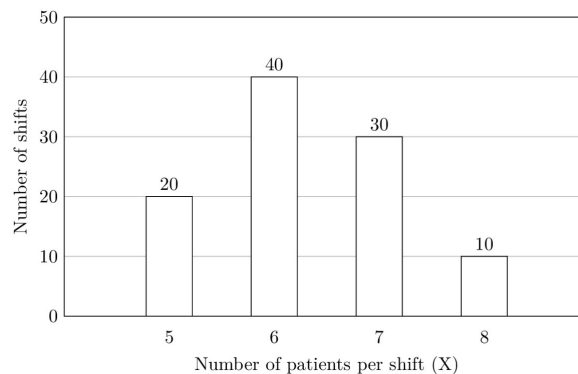
Answer key

9.8.4 Bar Graph: GATE DA 2025 | GA | Question: 10



The number of patients per shift (X) consulting Dr. Gita in her past 100 shifts is shown in the figure. If the amount she earns is ₹ $1000(X - 0.2)$, what is the average amount (in ₹) she has earned per shift in the past 100 shifts?

Note: The figure shown is representative.



- A. 6,100 B. 6,300 C. 6,000 D. 6,500

Answer key

9.9

Bayes Theorem (1)

9.9.1 Bayes Theorem: GATE CSE 2012 | Question: 63



An automobile plant contracted to buy shock absorbers from two suppliers X and Y . X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X 's shock absorbers, 96% are reliable. Of Y 's shock absorbers, 72% are reliable.

The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

A. 0.288

B. 0.334

C. 0.667

D. 0.720

gatecse-2012 quantitative-aptitude bayes-theorem probability normal conditional-probability

Answer key

9.10

Cartesian Coordinates (4)

Practice Test: Test 1 (10Q)

9.10.1 Cartesian Coordinates: GATE CSE 2020 | GA | Question: 9



Two straight lines are drawn perpendicular to each other in $X - Y$ plane. If α and β are the acute angles the straight lines make with the X -axis, then $\alpha + \beta$ is _____.

A. 60° B. 90° C. 120° D. 180°

gatecse-2020 quantitative-aptitude geometry cartesian-coordinates two-marks

Answer key

9.10.2 Cartesian Coordinates: GATE CSE 2022 | GA | Question: 2



A function $y(x)$ is defined in the interval $[0, 1]$ on the x -axis as

$$y(x) = \begin{cases} 2 & \text{if } 0 \leq x < \frac{1}{3} \\ 3 & \text{if } \frac{1}{3} \leq x < \frac{3}{4} \\ 1 & \text{if } \frac{3}{4} \leq x \leq 1 \end{cases}$$

Which one of the following is the area under the curve for the interval $[0, 1]$ on the x -axis?

A. $\frac{5}{6}$ B. $\frac{6}{5}$ C. $\frac{13}{6}$ D. $\frac{6}{13}$

gatecse-2022 quantitative-aptitude cartesian-coordinates area one-mark

Answer key

9.10.3 Cartesian Coordinates: GATE2012 AE: GA-9



Two points $(4, p)$ and $(0, q)$ lie on a straight line having a slope of $3/4$. The value of $(p - q)$ is

A. -3 B. 0 C. 3 D. 4

gate2012-ae quantitative-aptitude cartesian-coordinates geometry

Answer key

9.10.4 Cartesian Coordinates: GATE2014 AE: GA-4



If $y = 5x^2 + 3$, then the tangent at $x = 0, y = 3$

A. passes through $x = 0, y = 0$ B. has a slope of $+1$ C. is parallel to the x -axisD. has a slope of -1

gate2014-ae quantitative-aptitude geometry cartesian-coordinates

Answer key

9.11

Circle (3)

Practice Test: Test 1 (10Q)

9.11.1 Circle: GATE CSE 2018 | GA | Question: 3



The area of a square is d . What is the area of the circle which has the diagonal of the square as its diameter?

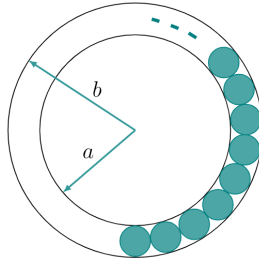
A. πd B. πd^2 C. $\frac{1}{4}\pi d^2$ D. $\frac{1}{2}\pi d$

Answer key

9.11.2 Circle: GATE CSE 2020 | GA | Question: 8



The figure below shows an annular ring with outer and inner as b and a , respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted area available in annular space is _____.



A. $\pi[(b^2 - a^2) - \frac{n}{4}(b - a)^2]$
 C. $\pi[(b^2 - a^2) + \frac{n}{4}(b - a)^2]$

B. $\pi[(b^2 - a^2) - n(b - a)^2]$
 D. $\pi[(b^2 - a^2) + n(b - a)^2]$

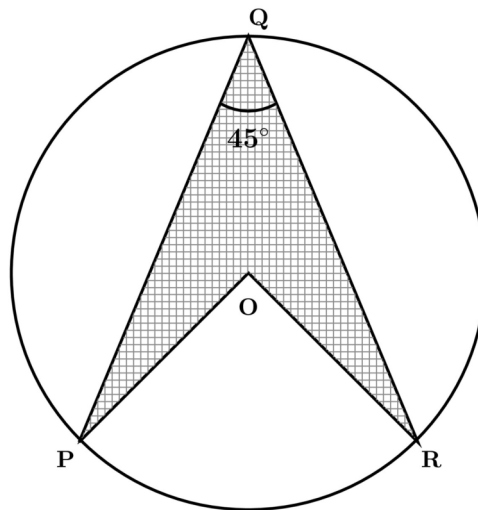
gatecse-2020 quantitative-aptitude geometry circle area two-marks

Answer key

9.11.3 Circle: GATE DA 2026 | GA | Question: 10



In the given figure, P, Q , and R are three points on a circle of radius 10 cm with O as its center, $\overline{PQ} = \overline{RQ}$, and $\angle PQR = 45^\circ$. The figure is representative. The area of the shaded region $PQRO$ is _____ cm^2 .



A. 50

B. $25\sqrt{2}$

C. $50\sqrt{2}$

D. 100

gateda-2026 quantitative-aptitude geometry circle area two-marks

Answer key

9.12

Clock Time (1)

Practice Test: Test 1 (11Q)

9.12.1 Clock Time: GATE CSE 2014 | Set 2 | GA | Question: 10



At what time between 6 a. m. and 7 a. m. will the minute hand and hour hand of a clock make an angle

closest to 60° ?

- A. 6 : 22 a.m. B. 6 : 27 a.m. C. 6 : 38 a.m. D. 6 : 45 a.m.

gatecse-2014-set2 quantitative-aptitude normal clock-time

Answer key 

9.13 Combinatory (5)

 **Practice Test:** Test 1 (13Q)

9.13.1 Combinatory: GATE CSE 2010 | Question: 65



Given digits 2, 2, 3, 3, 3, 4, 4, 4, 4 how many distinct 4 digit numbers greater than 3000 can be formed?

- A. 50 B. 51 C. 52 D. 54

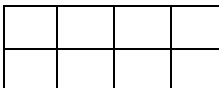
gatecse-2010 quantitative-aptitude combinatory normal

Answer key 

9.13.2 Combinatory: GATE CSE 2016 | Set 2 | GA | Question: 09



In a 2×4 rectangle grid shown below, each cell is rectangle. How many rectangles can be observed in the grid?



- A. 21 B. 27 C. 30 D. 36

gatecse-2016-set2 quantitative-aptitude normal combinatory

Answer key 

9.13.3 Combinatory: GATE CSE 2017 | Set 1 | GA | Question: 9



Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?

- A. 21 B. 18 C. 16 D. 14

gatecse-2017-set1 combinatory quantitative-aptitude

Answer key 

9.13.4 Combinatory: GATE2011 MN: GA-59



In how many ways 3 scholarships can be awarded to 4 applicants, when each applicant can receive any number of scholarships?

- A. 4 B. 12 C. 64 D. 81

quantitative-aptitude gate2011-mn combinatory

Answer key 

9.13.5 Combinatory: GATE2012 AR: GA-5



Ten teams participate in a tournament. Every team plays each of the other teams twice. The total number of matches to be played is

- A. 20 B. 45 C. 60 D. 90

gate2012-ar quantitative-aptitude combinatory

Answer key 

 **Practice Test:** Test 1 (5Q)

9.14.1 Compound Interest: GATE DS&AI 2024 | GA | Question: 8



Person 1 and Person 2 invest in three mutual funds A, B, and C. The amounts they invest in each of these mutual funds are given in the table.

	Mutual fund A	Mutual fund B	Mutual fund C
Person1	10,000	20,000	20,000
Person2	20,000	15,000	15,000

At the end of one year, the total amount that Person 1 gets is ₹500 more than Person 2. The annual rate of return for the mutual funds B and C is 15% each. What is the annual rate of return for the mutual fund A?

- A. 7.5% B. 10% C. 15% D. 20%

gate-ds-ai-2024 quantitative-aptitude compound-interest two-marks

Answer key 

9.14.2 Compound Interest: GATE2010 MN: GA-5



A person invest Rs.1000 at 10% annual compound interest for 2 years. At the end of two years, the whole amount is invested at an annual simple interest of 12% for 5 years. The total value of the investment finally is :

- A. 1776 B. 1760 C. 1920 D. 1936

general-aptitude quantitative-aptitude gate2010-mn compound-interest

Answer key 

9.14.3 Compound Interest: GATE2014 AG: GA-5



The population of a new city is 5 million and is growing at 20% annually. How many years would it take to double at this growth rate?

- A. 3 – 4 years B. 4 – 5 years C. 5 – 6 years D. 6 – 7 years

gate2014-ag quantitative-aptitude compound-interest normal

Answer key 

 **Practice Test:** Test 1 (6Q)

9.15.1 Conditional Probability: GATE2013 AE: GA-10



In a factory, two machines M_1 and M_2 manufacture 60% and 40% of the autocomponents respectively. Out of the total production, 2% of M_1 and 3% of M_2 are found to be defective. If a randomly drawn autocomponent from the combined lot is found defective, what is the probability that it was manufactured by M_2 ?

- A. 0.35 B. 0.45 C. 0.5 D. 0.4

gate2013-ae quantitative-aptitude conditional-probability

Answer key 

9.15.2 Conditional Probability: GATE2014 AG: GA-10



10% of the population in a town is HIV^+ . A new diagnostic kit for HIV detection is available; this kit correctly identifies HIV^+ individuals 95% of the time, and HIV^- individuals 89% of the time. A particular patient is tested using this kit and is found to be positive. The probability that the individual is actually positive is

Answer key

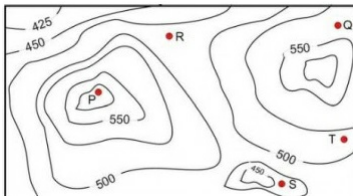
9.16

Contour Plots (2)

9.16.1 Contour Plots: GATE CSE 2017 | Set 1 | GA | Question: 10



A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot. If in a flood, the water level rises to 525 m, which of the villages P, Q, R, S, T get submerged?



- A. P, Q B. P, Q, T C. R, S, T D. Q, R, S

gatecse-2017-set1 general-aptitude quantitative-aptitude data-interpretation normal contour-plots

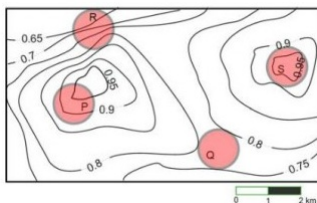
Answer key

9.16.2 Contour Plots: GATE CSE 2017 | Set 2 | GA | Question: 10



An air pressure contour line joins locations in a region having the same atmospheric pressure. The following is an air pressure contour plot of a geographical region. Contour lines are shown at 0.05 bar intervals in this plot.

If the possibility of a thunderstorm is given by how fast air pressure rises or drops over a region, which of the following regions is most likely to have a thunderstorm?



- A. P B. Q C. R D. S

gatecse-2017-set2 quantitative-aptitude data-interpretation normal contour-plots

Answer key

9.17

Cost Market Price (4)

Practice Test: Test 1 (7Q)

9.17.1 Cost Market Price: GATE CSE 2011 | Question: 63



The variable cost (V) of manufacturing a product varies according to the equation $V = 4q$, where q is the quantity produced. The fixed cost (F) of production of same product reduces with q according to the equation $F = \frac{100}{q}$. How many units should be produced to minimize the total cost ($V + F$)?

- A. 5 B. 4 C. 7 D. 6

gatecse-2011 quantitative-aptitude cost-market-price normal

Answer key

9.17.2 Cost Market Price: GATE CSE 2012 | Question: 56

The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of ₹50 per unit. The number of units to be produced by the firm such that the profit is maximized is

- A. 5 B. 10 C. 15 D. 25

gatecse-2012 quantitative-aptitude cost-market-price normal

Answer key

9.17.3 Cost Market Price: GATE CSE 2019 | GA | Question: 4

Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs. 150 more. The cost of the gift was Rs. ____

- A. 666 B. 3000 C. 6000 D. 12000

gatecse-2019 general-aptitude quantitative-aptitude cost-market-price one-mark

Answer key

9.17.4 Cost Market Price: GATE2014 AE: GA-5

A foundry has a fixed daily cost of Rs 50,000 whenever it operates and a variable cost of RS $800Q$, where Q is the daily production in tonnes. What is the cost of production in Rs per tonne for a daily production of 100 tonnes.

gate2014-ae quantitative-aptitude cost-market-price numerical-answers

Answer key

9.18 Cubes (2)**9.18.1 Cubes: GATE CSE 2016 | Set 1 | GA | Question: 05**

A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.

- a. 56 b. 64 c. 72 d. 96

gatecse-2016-set1 quantitative-aptitude geometry normal cubes

Answer key

9.18.2 Cubes: GATE CSE 2021 | Set 2 | GA | Question: 3

If θ is the angle, in degrees, between the longest diagonal of the cube and any one of the edges of the cube, then, $\cos \theta =$

- A. $\frac{1}{2}$ B. $\frac{1}{\sqrt{3}}$ C. $\frac{1}{\sqrt{2}}$ D. $\frac{\sqrt{3}}{2}$

gatecse-2021-set2 quantitative-aptitude mensuration cubes one-mark

Answer key

9.19 Currency Notes (1)**9.19.1 Currency Notes: GATE2012 CY: GA-10**

Raju has 14 currency notes in his pocket consisting of only Rs. 20 notes and Rs. 10 notes. The total money value of the notes is Rs. 230. The number of Rs. 10 notes that Raju has is

- A. 5 B. 6 C. 9 D. 10

gate2012-cy quantitative-aptitude numerical-computation currency-notes

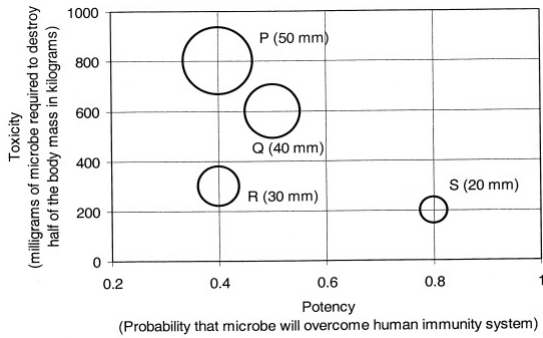
Answer key

9.20 Data Interpretation (3)

9.20.1 Data Interpretation: GATE CSE 2011 | Question: 62



P, Q, R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below:



A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

- A. P B. Q C. R D. S

gatecse-2011 quantitative-aptitude data-interpretation normal

Answer key

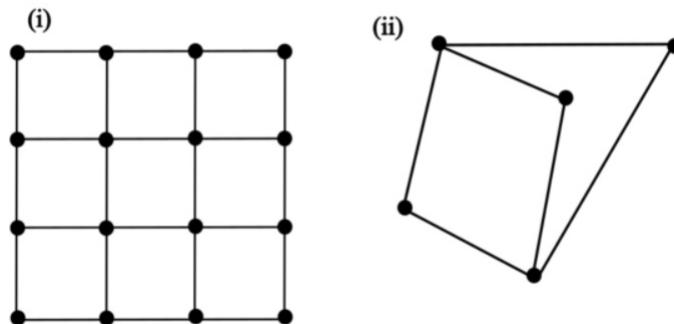
9.20.2 Data Interpretation: GATE CSE 2026 | Set 2 | GA | Question: 8

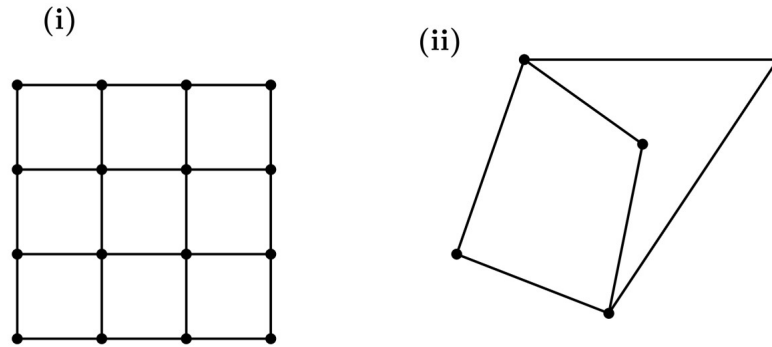


Figures (i) and (ii) represent intercity highway systems. The black dots represent cities and the line segments between them represent intercity highways.

A salesperson needs to make a trip. She needs to start from a city, visit each of the remaining cities exactly once, and finally return to the same city from which she started.

Which one of the following options is then true?





- A. Such a trip is possible for (i), but not for (ii).
 B. Such a trip is possible for (ii), but not for (i).
 C. Such a trip is possible for both (i) and (ii).
 D. Such a trip is possible neither for (i) nor for (ii).

gatecse-2026-set2 quantitative-aptitude data-interpretation two-marks

Answer key

9.20.3 Data Interpretation: GATE DA 2025 | GA | Question: 7



Weight of a person can be expressed as a function of their age. The function usually varies from person to person. Suppose this function is identical for two brothers, and it monotonically increases till the age of 50 years and then it monotonically decreases. Let a_1 and a_2 (in years) denote the ages of the brothers and $a_1 < a_2$. Which one of the following statements is correct about their age on the day when they attain the same weight?

- A. $a_1 < a_2 < 50$
 B. $a_1 < 50 < a_2$
 C. $50 < a_1 < a_2$
 D. Either $a_1 = 50$ or $a_2 = 50$

gateda-2025 quantitative-aptitude data-interpretation two-marks

Answer key

9.21 Digital Image Processing (1)

9.21.1 Digital Image Processing: GATE DA 2025 | GA | Question: 3



A 4×4 digital image has pixel intensities (U) as shown in the figure. The number of pixels with $U \leq 4$ is:

0	1	0	2
4	7	3	3
5	5	4	4
6	7	3	2

- A. 3
 B. 8
 C. 11
 D. 9

gateda-2025 quantitative-aptitude digital-image-processing easy one-mark

Answer key

9.22 Factors (3)

Practice Test: Test 1 (5Q)

9.22.1 Factors: GATE CSE 2013 | Question: 62



Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

A. $\left(\frac{13}{90}\right)$

B. $\left(\frac{12}{90}\right)$

C. $\left(\frac{78}{90}\right)$

D. $\left(\frac{77}{90}\right)$

gatecse-2013 quantitative-aptitude easy probability factors

Answer key

9.22.2 Factors: GATE CSE 2018 | GA | Question: 4



What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

A. 3047

B. 6047

C. 7987

D. 63847

gatecse-2018 quantitative-aptitude factors one-mark

Answer key

9.22.3 Factors: GATE2010 MN: GA-8



Consider the set of integers $\{1, 2, 3, \dots, 5000\}$. The number of integers that is divisible by neither 3 nor 4 is :

A. 1668

B. 2084

C. 2500

D. 2916

general-aptitude quantitative-aptitude gate2010-mn factors

Answer key

9.23

Fractions (1)

9.23.1 Fractions: GATE CSE 2018 | GA | Question: 6



In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift Rs 750 to each male senior citizen in the town and Rs 1000 to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only $\frac{8}{9}$ of the eligible men and $\frac{2}{3}$ of the eligible women claimed the gift. How much money (in Rupees) did the philanthropist give away in total?

A. 1,50,000

B. 2,00,000

C. 1,75,000

D. 1,51,000

gatecse-2018 quantitative-aptitude fractions normal two-marks

Answer key

9.24

Functions (7)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(7Q\)](#)

9.24.1 Functions: GATE CSE 2015 | Set 2 | GA | Question: 9



If p, q, r, s are distinct integers such that:

$$f(p, q, r, s) = \max(p, q, r, s)$$

$$g(p, q, r, s) = \min(p, q, r, s)$$

$$h(p, q, r, s) = \text{remainder of } \frac{(p \times q)}{(r \times s)} \text{ if } (p \times q) > (r \times s)$$

$$\text{or remainder of } \frac{(r \times s)}{(p \times q)} \text{ if } (r \times s) > (p \times q)$$

$$\text{Also a function } fgh(p, q, r, s) = f(p, q, r, s) \times g(p, q, r, s) \times h(p, q, r, s)$$

Also the same operations are valid with two variable functions of the form $f(p, q)$

What is the value of $fg(h(2, 5, 7, 3), 4, 6, 8)$?

gatecse-2015-set2 functions normal numerical-answers

Answer key

9.24.2 Functions: GATE CSE 2015 | Set 3 | GA | Question: 5



A function $f(x)$ is linear and has a value of 29 at $x = -2$ and 39 at $x = 3$. Find its value at $x = 5$.

- A. 59 B. 45 C. 43 D. 35

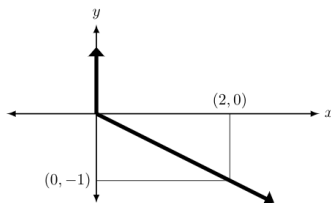
gatecse-2015-set3 quantitative-aptitude normal functions

Answer key

9.24.3 Functions: GATE CSE 2015 | Set 3 | GA | Question: 8



Choose the most appropriate equation for the function drawn as thick line, in the plot below.



- A. $x = y - |y|$ B. $x = -(y - |y|)$ C. $x = y + |y|$ D. $x = -(y + |y|)$

gatecse-2015-set3 quantitative-aptitude normal functions

Answer key

9.24.4 Functions: GATE CSE 2023 | GA | Question: 7



Consider two functions of time (t),

$$f(t) = 0.01t^2$$

$$g(t) = 4t$$

where $0 < t < \infty$.

Now consider the following two statements:

- For some $t > 0$, $g(t) > f(t)$.
- There exists a T , such that $f(t) > g(t)$ for all $t > T$.

Which one of the following options is TRUE?

- A. only (i) is correct B. only (ii) is correct
C. both (i) and (ii) are correct D. neither (i) nor (ii) is correct

gatecse-2023 quantitative-aptitude functions two-marks

Answer key

9.24.5 Functions: GATE CSE 2023 | GA | Question: 9



$f(x)$ and $g(y)$ are functions of x and y , respectively, and $f(x) = g(y)$ for all values of x and y . Which one of the following options is necessarily TRUE for a x and y ?

- A. $f(x) = 0$ and $g(y) = 0$ B. $f(x) = g(y) = \text{constant}$
C. $f(x) \neq \text{constant}$ and $g(y) \neq \text{constant}$ D. $f(x) + g(y) = f(x) - g(y)$

gatecse-2023 quantitative-aptitude functions two-marks

Answer key

9.24.6 Functions: GATE2010 MN: GA-10



Given the following four functions $f_1(n) = n^{100}$, $f_2(n) = (1.2)^n$, $f_3(n) = 2^{n/2}$, $f_4(n) = 3^{n/3}$ which function will have the largest value for sufficiently large values of n (*i. e.* $n \rightarrow \infty$)?

A. f_4 B. f_3 C. f_2 D. f_1

general-aptitude quantitative-aptitude gate2010-mn functions

Answer key

9.24.7 Functions: GATE2012 AR: GA-7



Let $f(x) = x - [x]$, where $x \geq 0$ and $[x]$ is the greatest integer not larger than x . Then $f(x)$ is a

- A. monotonically increasing function
- B. monotonically decreasing function
- C. linearly increasing function between two integers
- D. linearly decreasing function between two integers

gate2012-ar quantitative-aptitude functions normal

Answer key

9.25

Geometry (5)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(4Q\)](#)

9.25.1 Geometry: GATE CSE 2014 | Set 1 | GA | Question: 10



When a point inside of a tetrahedron (a solid with four triangular surfaces) is connected by straight lines to its corners, how many (new) internal planes are created with these lines?

gatecse-2014-set1 quantitative-aptitude geometry combinatory normal numerical-answers

Answer key

9.25.2 Geometry: GATE CSE 2024 | Set 1 | GA | Question: 7



A rectangular paper sheet of dimensions $54 \text{ cm} \times 4 \text{ cm}$ is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken.

Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

- A. $1/\pi$
- B. $2/\pi$
- C. $3/\pi$
- D. $4/\pi$

gatecse-2024-set1 quantitative-aptitude geometry two-marks

Answer key

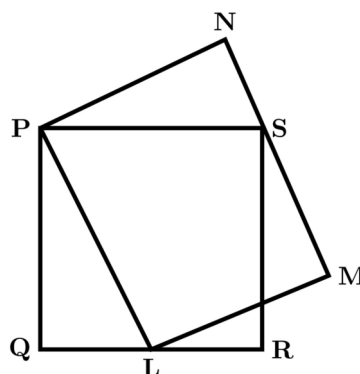
9.25.3 Geometry: GATE CSE 2025 | Set 2 | GA | Question: 9



In the Given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side QR. Side MN of the rectangle passes through the corner S of the square.

What is the area (in cm^2) of the rectangle PLMN?

Note: The figure shown is representative.



A. $2\sqrt{2}$

B. 2

C. 8

D. 4

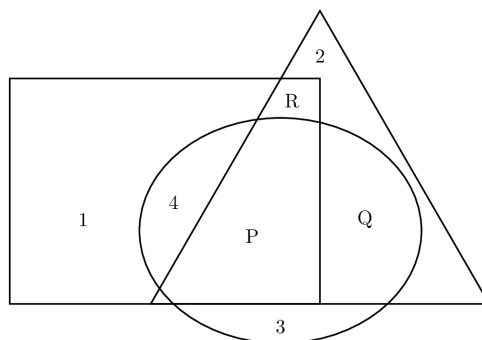
gatecse2025-set2 quantitative-aptitude geometry two-marks

Answer key

9.25.4 Geometry: GATE DA 2025 | GA | Question: 4



In the given figure, the numbers associated with the rectangle, triangle, and ellipse are 1, 2, and 3, respectively. Which one among the given options is the most appropriate combination of P, Q, and R?

A. $P = 6; Q = 5; R = 3$ C. $P = 3; Q = 6; R = 6$ B. $P = 5; Q = 6; R = 3$ D. $P = 5; Q = 3; R = 6$

gateda-2025 quantitative-aptitude geometry one-mark

Answer key

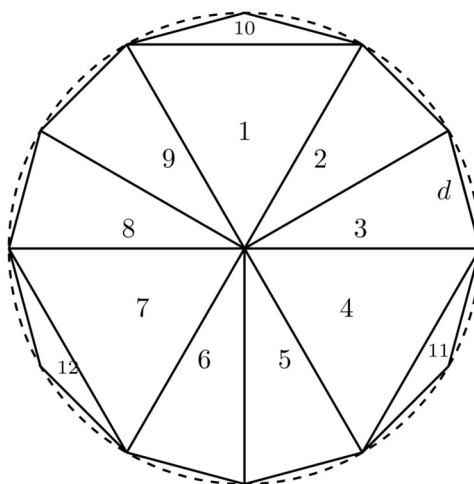
9.25.5 Geometry: GATE DA 2025 | GA | Question: 8



A regular dodecagon (12-sided regular polygon) is inscribed in a circle of radius r cm as shown in the figure. The side of the dodecagon is d cm. All the triangles (numbered 1 to 12) in the figure are used to form squares of side r cm and each numbered triangle is used only once to form a square.

The number of squares that can be formed and the number of triangles required to form each square, respectively, are:

Note: The figure shown is representative.



A. 3;4

B. 4;3

C. 3;3

D. 3;2

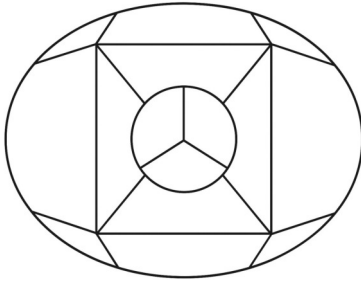
gateda-2025 quantitative-aptitude geometry two-marks

Answer key

9.26.1 Graph Coloring: GATE DS&AI 2024 | GA | Question: 2



The 15 parts of the given figure are to be painted such that no two adjacent parts with shared boundaries (excluding corners) have the same color. The minimum number of colors required is



- A. 4 B. 3 C. 5 D. 6

gate-ds-ai-2024 quantitative-aptitude graph-coloring one-mark easy

Answer key

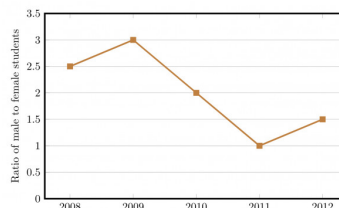
9.27 Line Graph (5)

Practice Test: Test 1 (12Q)

9.27.1 Line Graph: GATE CSE 2014 | Set 2 | GA | Question: 9



The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students doubled in 2009, by what percent did the number of male students increase in 2009?



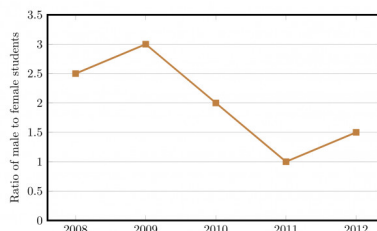
gatecse-2014-set2 quantitative-aptitude data-interpretation numerical-answers normal line-graph

Answer key

9.27.2 Line Graph: GATE CSE 2014 | Set 3 | GA | Question: 9



The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students in 2011 and 2012 is equal, what is the ratio of male students in 2012 to male students in 2011?



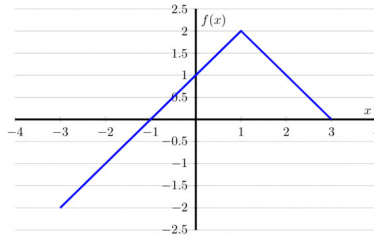
- A. 1 : 1 B. 2 : 1 C. 1.5 : 1 D. 2.5 : 1

gatecse-2014-set3 quantitative-aptitude data-interpretation normal line-graph

Answer key

9.27.3 Line Graph: GATE CSE 2016 | Set 2 | GA | Question: 10





A. $f(x) = 1 - |x - 1|$
 C. $f(x) = 2 - |x - 1|$

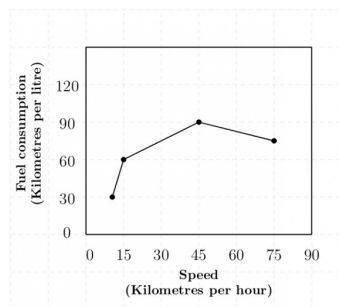
B. $f(x) = 1 + |x - 1|$
 D. $f(x) = 2 + |x - 1|$

gatecse-2016-set2 quantitative-aptitude data-interpretation normal line-graph

Answer key

9.27.4 Line Graph: GATE2011 AG: GA-9

The fuel consumed by a motorcycle during a journey while traveling at various speeds is indicated in the graph below.



The distances covered during four laps of the journey are listed in the table below

Lap	Distance (kilometres)	Average Speed (kilometres per hour)
P	15	15
Q	75	45
R	40	75
S	10	10

From the given data, we can conclude that the fuel consumed per kilometre was least during the lap

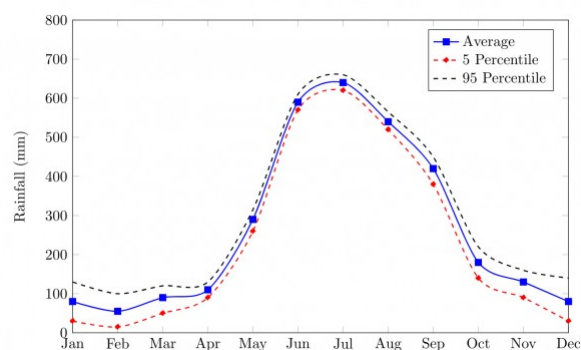
- A. *P* B. *Q* C. *R* D. *S*

general-aptitude quantitative-aptitude gate2011-ag data-interpretation line-graph

Answer key

9.27.5 Line Graph: GATE2014 AE: GA-10

The monthly rainfall chart based on 50 years of rainfall in Agra is shown in the following figure.



Which of the following are true? (k percentile is the value such that k percent of the data fall below that value)

- i. On average, it rains more in July than in December
- ii. Every year, the amount of rainfall in August is more than that in January
- iii. July rainfall can be estimated with better confidence than February rainfall
- iv. In August, there is at least 500 mm of rainfall

- A. (i) and (ii) B. (i) and (iii) C. (ii) and (iii) D. (iii) and (iv)

gate2014-ae quantitative-aptitude data-interpretation line-graph

Answer key

9.28

Logarithms (6)

 Practice Tests: Test 1 (15Q) Test 2 (4Q)

9.28.1 Logarithms: GATE CSE 2011 | Question: 57



If $\log(P) = (1/2)\log(Q) = (1/3)\log(R)$, then which of the following options is **TRUE**?

- A. $P^2 = Q^3R^2$ B. $Q^2 = PR$ C. $Q^2 = R^3P$ D. $R = P^2Q^2$

gatecse-2011 quantitative-aptitude normal logarithms

Answer key

9.28.2 Logarithms: GATE CSE 2018 | GA | Question: 7



If $pqr \neq 0$ and $p^{-x} = \frac{1}{q}, q^{-y} = \frac{1}{r}, r^{-z} = \frac{1}{p}$, what is the value of the product xyz ?

- A. -1 B. $\frac{1}{pqr}$ C. 1 D. pqr

gatecse-2018 quantitative-aptitude ratio-proportion two-marks logarithms

Answer key

9.28.3 Logarithms: GATE CSE 2024 | Set 1 | GA | Question: 5



For positive non-zero real variables p and q , if

$$\log(p^2 + q^2) = \log p + \log q + 2\log 3,$$

then, the value of $\frac{p^4 + q^4}{p^2 q^2}$ is

- A. 79 B. 81 C. 9 D. 83

gatecse-2024-set1 quantitative-aptitude logarithms one-mark

Answer key

9.28.4 Logarithms: GATE CSE 2024 | Set 2 | GA | Question: 4



For positive non-zero real variables x and y , if

$$\ln\left(\frac{x+y}{2}\right) = \frac{1}{2}[\ln(x) + \ln(y)]$$

then, the value of $\frac{x}{y} + \frac{y}{x}$ is

- A. 1 B. $1/2$ C. 2 D. 4

gatecse-2024-set2 quantitative-aptitude logarithms one-mark

Answer key

9.28.5 Logarithms: GATE DA 2026 | GA | Question: 4



Consider two distinct positive real numbers m, n , with $m > n$. Let $x = n^{\log_{10}(m)}$ and $y = m^{\log_{10}(n)}$. The relation between x and y is ____.

- A. $x > y$ B. $x < y$ C. $x = y$ D. $x = \log_{10}(y)$

gateda-2026 quantitative-aptitude logarithms one-mark

Answer key

9.28.6 Logarithms: GATE2012 AR: GA-6



A value of x that satisfies the equation $\log x + \log(x-7) = \log(x+11) + \log 2$ is

- A. 1 B. 2 C. 7 D. 11

gate2012-ar quantitative-aptitude logarithms

Answer key

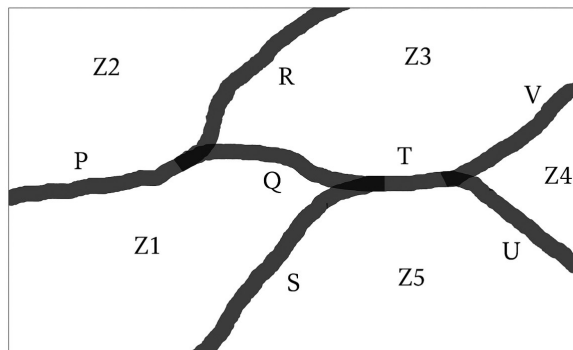
9.29 Maps (1)

9.29.1 Maps: GATE CSE 2025 | Set 2 | GA | Question: 10



The diagram below shows a river system consisting of 7 segments, marked P, Q, R, S, T, U , and V . It splits the land into 5 zones, marked $Z1, Z2, Z3, Z4$, and $Z5$. We need to connect these zones using the least number of bridges. Out of the following options, which one is correct?

Note: The figure shown is representative.



- A. Bridges on P, Q and T B. Bridges on P, Q, S and T
C. Bridges on Q, R, T and V D. Bridges on P, Q, S, U and V

gatecse2025-set2 quantitative-aptitude maps two-marks

Answer key

9.30 Maxima Minima (4)

Practice Test: Test 1 (5Q)

9.30.1 Maxima Minima: GATE CSE 2012 | Question: 62



A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

- A. 8 meters B. 10 meters C. 12 meters D. 14 meters

gatecse-2012 quantitative-aptitude normal maxima-minima

Answer key

9.30.2 Maxima Minima: GATE CSE 2017 | Set 2 | GA | Question: 9



The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

- A. 0 B. 1 C. 2 D. 3

gatecse-2017-set2 quantitative-aptitude normal maxima-minima calculus

Answer key

9.30.3 Maxima Minima: GATE2010 TF: GA-5



Consider the function $f(x) = \max(7 - x, x + 3)$. In which range does f take its minimum value?

- A. $-6 \leq x < -2$ B. $-2 \leq x < 2$
C. $2 \leq x < 6$ D. $6 \leq x < 10$

general-aptitude quantitative-aptitude gate2010-tf maxima-minima functions

Answer key

9.30.4 Maxima Minima: GATE2013 CE: GA-6



X and Y are two positive real numbers such that $2X + Y \leq 6$ and $X + 2Y \leq 8$. For which of the following values of

(X, Y) the function $f(X, Y) = 3X + 6Y$ will give maximum value ?

- A. $\left(\frac{4}{3}, \frac{10}{3}\right)$ B. $\left(\frac{8}{3}, \frac{20}{3}\right)$ C. $\left(\frac{8}{3}, \frac{10}{3}\right)$ D. $\left(\frac{4}{3}, \frac{20}{3}\right)$

gate2013-ce quantitative-aptitude maxima-minima

Answer key

9.31

Mensuration (1)

9.31.1 Mensuration: GATE DA 2025 | GA | Question: 5



A rectangle has a length L and a width W , where $L > W$. If the width, W , is increased by 10%, which one of the following statements is correct for all values of L and W ?

- A. Perimeter increases by 10%. B. Length of the diagonals increases by 10%.
C. Area increases by 10%. D. The rectangle becomes a square.

gateda-2025 quantitative-aptitude mensuration one-mark

Answer key

9.32

Modular Arithmetic (1)

9.32.1 Modular Arithmetic: GATE2012 CY: GA-1



If $(1.001)^{1259} = 3.52$ and $(1.001)^{2062} = 7.85$, then $(1.001)^{3321} =$

- A. 2.23 B. 4.33 C. 11.37 D. 27.64

gate2012-cy quantitative-aptitude modular-arithmetic

Answer key

9.33

Number Representation (1)

9.33.1 Number Representation: GATE DA 2026 | GA | Question: 2



The product of the digits of a three-digit number is 70. The sum of the digits of this three-digit number is

- A. 12 B. 14 C. 16 D. 18

gateda-2026 quantitative-aptitude one-mark number-representation

Answer key

9.34

Number Series (5)

 **Practice Test:** Test 1 (10Q)

9.34.1 Number Series: GATE CSE 2013 | Question: 61



Find the sum of the expression

$$\frac{1}{\sqrt{1+\sqrt{2}}} + \frac{1}{\sqrt{2+\sqrt{3}}} + \frac{1}{\sqrt{3+\sqrt{4}}} + \dots + \frac{1}{\sqrt{80+\sqrt{81}}}$$

- A. 7 B. 8 C. 9 D. 10

gatecse-2013 quantitative-aptitude normal number-series

Answer key

9.34.2 Number Series: GATE CSE 2014 | Set 2 | GA | Question: 5



The value of $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}}$ is

- A. 3.464 B. 3.932 C. 4.000 D. 4.444

gatecse-2014-set2 quantitative-aptitude easy number-series

Answer key

9.34.3 Number Series: GATE CSE 2014 | Set 3 | GA | Question: 4



Which number does not belong in the series below?

2, 5, 10, 17, 26, 37, 50, 64

- A. 17 B. 37 C. 64 D. 26

gatecse-2014-set3 quantitative-aptitude number-series easy

Answer key

9.34.4 Number Series: GATE2010 TF: GA-7



Consider the series $\frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{8} + \frac{1}{9} - \frac{1}{16} + \frac{1}{32} + \frac{1}{27} - \frac{1}{64} + \dots$. The sum of the infinite series above is:

- A. ∞ B. $\frac{5}{6}$ C. $\frac{1}{2}$ D. 0

general-aptitude quantitative-aptitude gate2010-tf number-series

Answer key

9.34.5 Number Series: GATE2013 EE: GA-10



Find the sum to ' n ' terms of the series $10 + 84 + 734 + \dots$

- A. $\frac{9(9^n+1)}{10} + 1$ B. $\frac{9(9^n-1)}{8} + 1$ C. $\frac{9(9^n-1)}{8} + n$ D. $\frac{9(9^n-1)}{8} + n^2$

gate2013-ee quantitative-aptitude number-series

Answer key

9.35

Number System (4)

 **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (2Q)

9.35.1 Number System: GATE CSE 2014 | Set 3 | GA | Question: 10



Consider the equation: $(7526)_8 - (Y)_8 = (4364)_8$, where $(X)_N$ stands for X to the base N . Find Y .

A. 1634

B. 1737

C. 3142

D. 3162

gatecse-2014-set3 quantitative-aptitude number-system normal digital-logic

Answer key

9.35.2 Number System: GATE CSE 2017 | Set 2 | GA | Question: 8



X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have

A. 90 digits

B. 91 digits

C. 92 digits

D. 93 digits

gatecse-2017-set2 quantitative-aptitude number-system

Answer key

9.35.3 Number System: GATE DA 2026 | GA | Question: 9



P, Q, R, S, X , and Y are distinct single-digit whole numbers taking values from 0 to 9.

PQ is a two-digit number with Q being in the units place and P in the tens place. Similarly, RS is a two-digit number.

It is known that PQ and RS are consecutive numbers and $(PQ)^2 + (RS)^2 = XYP$, with XYP being a three-digit number.

The value of Y is _____

A. 4

B. 5

C. 6

D. 7

gateda-2026 quantitative-aptitude two-marks number-system

Answer key

9.35.4 Number System: GATE2010 MN: GA-9



A positive integer m in base 10 when represented in base 2 has the representation p and in base 3 has the representation q . We get $p - q = 990$ where the subtraction is done in base 10. Which of the following is necessarily true?

A. $m \geq 14$ B. $9 \leq m \leq 13$ C. $6 \leq m \leq 8$ D. $m < 6$

general-aptitude quantitative-aptitude gate2010-mn number-system

Answer key

9.36

Number Theory (2)

Practice Test: Test 1 (7Q)

9.36.1 Number Theory: GATE CSE 2025 | Set 2 | GA | Question: 3



If $P e^x = Q e^{-x}$ for all real values of x , which one of the following statements is true?

A. $P = Q = 0$ B. $P = Q = 1$ C. $P = 1; Q = -1$ D. $\frac{P}{Q} = 0$

gatecse2025-set2 quantitative-aptitude mathematical-logic one-mark number-theory

Answer key

9.36.2 Number Theory: GATE2012 AE: GA-8



If a prime number on division by 4 gives a remainder of 1, then that number can be expressed as

A. sum of squares of two natural numbers

B. sum of cubes of two natural numbers

C. sum of square roots of two natural numbers

D. sum of cube roots of two natural numbers

[Answer key](#)

9.37

Numerical Computation (9)

[Practice Tests:](#) [Test 1 \(15Q\)](#) [Test 2 \(3Q\)](#)

9.37.1 Numerical Computation: GATE CSE 2014 | Set 1 | GA | Question: 4



If $\left(z + \frac{1}{z}\right)^2 = 98$, compute $\left(z^2 + \frac{1}{z^2}\right)$.

gatecse-2014-set1 quantitative-aptitude easy numerical-answers numerical-computation

[Answer key](#)

9.37.2 Numerical Computation: GATE CSE 2014 | Set 2 | GA | Question: 4



What is the average of all multiples of 10 from 2 to 198?

- A. 90 B. 100 C. 110 D. 120

gatecse-2014-set2 quantitative-aptitude easy numerical-computation factors

[Answer key](#)

9.37.3 Numerical Computation: GATE CSE 2017 | Set 1 | GA | Question: 4



Find the smallest number y such that $y \times 162$ is a perfect cube.

- A. 24 B. 27 C. 32 D. 36

gatecse-2017-set1 general-aptitude quantitative-aptitude easy numerical-computation

[Answer key](#)

9.37.4 Numerical Computation: GATE CSE 2017 | Set 2 | GA | Question: 4



A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have?

- A. 12 B. 15 C. 18 D. 19

gatecse-2017-set2 quantitative-aptitude numerical-computation

[Answer key](#)

9.37.5 Numerical Computation: GATE CSE 2026 | Set 1 | GA | Question: 3



Consider a knock-out women's badminton singles tournament where there are no ties. The loser in each game is eliminated from the tournament. Every player plays until she is defeated or remains the last undefeated player. The last undefeated player is declared the winner of the tournament. If there are 64 players in the beginning of the tournament, how many games should be played in total to declare the winner of the tournament?

- A. 127 B. 64 C. 63 D. 32

gatecse-2026-set1 quantitative-aptitude numerical-computation easy one-mark

[Answer key](#)

9.37.6 Numerical Computation: GATE CSE 2026 | Set 1 | GA | Question: 4



A student needs to enroll for a minimum of 60 credits. A student cannot enroll for more than 70 credits. The credits are divided amongst project and three distinct sets of courses namely, core courses, specialization courses, and elective courses. It is compulsory for a student to enroll for exactly 15 credits of core courses and exactly 20 credits of project. In addition, a student has to enroll for a minimum of 10 credits of specialization

courses. The maximum credits of elective courses that a student can enroll for is ____

- A. 10 B. 15 C. 20 D. 25

gatecse-2026-set1 quantitative-aptitude numerical-computation one-mark

Answer key 

9.37.7 Numerical Computation: GATE CSE 2026 | Set 1 | GA | Question: 8



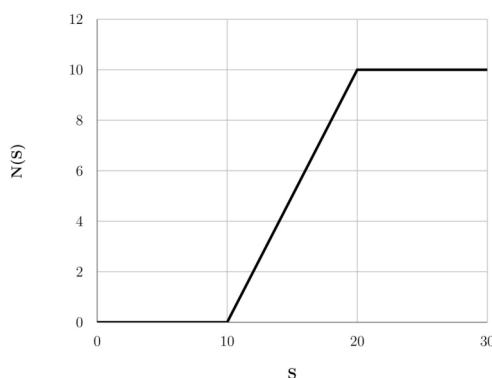
For positive real numbers S and K , the function $H_K(S)$ is defined as:

$H_K(S) = \max(S - K, 0)$. The max function is defined as:

$$\max(a, b) = \begin{cases} a, & \text{when } a > b \\ b, & \text{when } a \leq b \end{cases}$$

The graph below shows the plot of a function $N(S)$ versus S .

$N(S)$ can be expressed as ____.



- A. $H_{10}(S) - H_{20}(S)$ B. $H_{10}(S) - 2H_{20}(S)$
C. $-H_{10}(S) + H_{20}(S)$ D. $H_{15}(S) - H_{20}(S)$

gatecse-2026-set1 quantitative-aptitude numerical-computation two-marks

Answer key 

9.37.8 Numerical Computation: GATE2010 TF: GA-10



A student is answering a multiple choice examination with 65 questions with a marking scheme as follows:

i) 1 marks for each correct answer, ii) $-\frac{1}{4}$ for a wrong answer, iii) $-\frac{1}{8}$ for a question that has not been attempted. If the student gets 37 marks in the test then the least possible number of questions the student has NOT answered is:

- A. 6 B. 5 C. 7 D. 4

general-aptitude quantitative-aptitude gate2010-tf numerical-computation

Answer key 

9.37.9 Numerical Computation: GATE2013 CE: GA-1



A number is as much greater than 75 as it is smaller than 117. The number is:

- A. 91 B. 93 C. 89 D. 96

gate2013-ce quantitative-aptitude numerical-computation

Answer key 

9.38.1 Percentage: GATE CSE 2014 | Set 1 | GA | Question: 8



Round-trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is $Rs\ 100$, a group of 5 tourists purchasing round-trip tickets will be charged Rs _____

gatecse-2014-set1 quantitative-aptitude easy numerical-answers percentage

Answer key

9.38.2 Percentage: GATE CSE 2014 | Set 3 | GA | Question: 8



The Gross Domestic Product (GDP) in Rupees grew at 7% during 2012 – 2013. For international comparison, the GDP is compared in US Dollars (USD) after conversion based on the market exchange rate. During the period 2012 – 2013 the exchange rate for the USD increased from $Rs. 50/USD$ to $Rs. 60/USD$. India's GDP in USD during the period 2012 – 2013

- A. increased by 5% B. decreased by 13%
C. decreased by 20% D. decreased by 11%

gatecse-2014-set3 quantitative-aptitude normal percentage

Answer key

9.38.3 Percentage: GATE CSE 2024 | Set 1 | GA | Question: 4



The number of coins of ₹1, ₹5, and ₹10 denominations that a person has are in the ratio 5 : 3 : 13. Of the total amount, the percentage of money in ₹5 coins is

- A. 21% B. $14\frac{2}{7}\%$ C. 10% D. 30%

gatecse-2024-set1 quantitative-aptitude percentage one-mark

Answer key

9.38.4 Percentage: GATE2011 AG: GA-4



There are two candidates P and Q in an election. During the campaign, 40% of the voters promised to vote for P , and rest for Q . However, on the day of election 15% of the voters went back on their promise to vote for P and instead voted for Q . 25% of the voters went back on their promise to vote for Q and instead voted for P . Suppose, P lost by 2 votes, then what was the total number of voters?

- A. 100 B. 110 C. 90 D. 95

general-aptitude quantitative-aptitude gate2011-ag percentage

Answer key

9.38.5 Percentage: GATE2012 AE: GA-7



The total runs scored by four cricketers P, Q, R and S in years 2009 and 2010 are given in the following table;

Player	2009	2010
P	802	1008
Q	765	912
R	429	619
S	501	701

The player with the lowest percentage increase in total runs is

- A. P B. Q C. R D. S

gate2012-ae quantitative-aptitude percentage

Answer key

9.38.6 Percentage: GATE2012 CY: GA-8



The data given in the following table summarizes the monthly budget of an average household.

Category	Amount (Rs.)
Food	4000
Rent	2000
Savings	1500
Other expenses	1800
Clothing	1200

The approximate percentage of the monthly budget **NOT** spent on savings is

- A. 10% B. 14% C. 81% D. 86%

gate2012-cy quantitative-aptitude percentage

Answer key

9.38.7 Percentage: GATE2013 EE: GA-2



In the summer of 2012, in New Delhi, the mean temperature of Monday to Wednesday was 41°C and of Tuesday to Thursday was 43°C . If the temperature on Thursday was 15% higher than that of Monday, then the temperature in $^{\circ}\text{C}$ on Thursday was

- A. 40 B. 43 C. 46 D. 49

gate2013-ee quantitative-aptitude percentage

Answer key

9.38.8 Percentage: GATE2014 AE: GA-9



One percent of the people of country X are taller than 6 ft. Two percent of the people of country Y are taller than 6 ft. There are thrice as many people in country X as in country Y . Taking both countries together, what is the percentage of people taller than 6 ft?

- A. 3.0 B. 2.5 C. 1.5 D. 1.25

gate2014-ae percentage quantitative-aptitude

Answer key

9.39 Permutation and Combination (4)

Practice Test: Test 1 (14Q)

9.39.1 Permutation and Combination: GATE CSE 2024 | Set 2 | GA | Question: 2



Two wizards try to create a spell using all the four elements, *water*, *air*, *fire*, and *earth*. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work.

How many attempts does each wizard make before coming to this conclusion, independently?

- A. 24 B. 48 C. 16 D. 12

gatecse-2024-set2 quantitative-aptitude permutation-and-combination one-mark

Answer key

9.39.2 Permutation and Combination: GATE CSE 2025 | Set 1 | GA | Question: 10



A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. The order in which the scoops are purchased is inconsequential. If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?

- A. 4 B. 20 C. 24 D. 48

gatecse2025-set1 quantitative-aptitude permutation-and-combination two-marks

Answer key

9.39.3 Permutation and Combination: GATE DA 2026 | GA | Question: 7



Five integers are picked from 0 to 20, with possible repetitions, such that their mean is 12, median is 18, and they have a single mode of 20.

Ignoring permutations, the number of ways to pick these five integers is _____

- A. 0 B. 1 C. 2 D. 3

gateda-2026 quantitative-aptitude permutation-and-combination two-marks

Answer key

9.39.4 Permutation and Combination: GATE DS&AI 2024 | GA | Question: 3



How many 4-digit positive integers divisible by 3 can be formed using only the digits {1, 3, 4, 6, 7}, such that no digit appears more than once in a number?

- A. 24 B. 48 C. 72 D. 12

gate-ds-ai-2024 quantitative-aptitude permutation-and-combination one-mark

Answer key

9.40

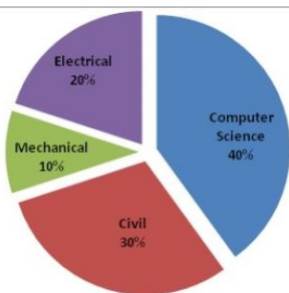
Pie Chart (5)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(7Q\)](#)

9.40.1 Pie Chart: GATE CSE 2015 | Set 1 | GA | Question: 9



The pie chart below has the breakup of the number of students from different departments in an engineering college for the year 2012. The proportion of male to female students in each department is 5 : 4. There are 40 males in Electrical Engineering. What is the difference between the numbers of female students in the civil department and the female students in the Mechanical department?



gatecse-2015-set1 quantitative-aptitude data-interpretation numerical-answers pie-chart

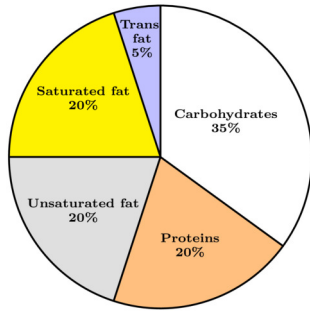
Answer key

9.40.2 Pie Chart: GATE CSE 2024 | Set 1 | GA | Question: 8



The pie chart presents the percentage contribution of different macronutrients to a typical 2,000 kcal diet of a person.

Macronutrient energy contribution



The typical energy density (kcal/g) of these macronutrients is given in the table.

Macronutrient	Energy density (kcal/g)
Carbohydrates	4
Proteins	4
Unsaturated fat	9
Saturated fat	9
Trans fat	9

The total fat (all three types), in grams, this person consumes is

- A. 44.4 B. 77.8 C. 100 D. 3,600

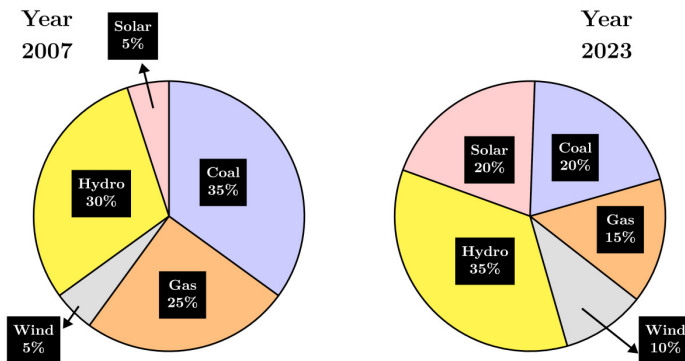
gatecse-2024-set1 quantitative-aptitude pie-chart two-marks

Answer key

9.40.3 Pie Chart: GATE CSE 2024 | Set 2 | GA | Question: 8



The pie charts depict the shares of various power generation technologies in the total electricity generation of a country for the years 2007 and 2023.



The renewable sources of electricity generation consist of Hydro, Solar and Wind. Assuming that the total electricity generated remains the same from 2007 to 2023, what is the percentage increase in the share of the renewable sources of electricity generation over this period?

- A. 25% B. 50% C. 77.5% D. 62.5%

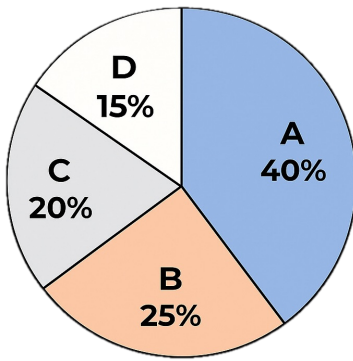
gatecse-2024-set2 quantitative-aptitude pie-chart two-marks

Answer key

9.40.4 Pie Chart: GATE DS&AI 2024 | GA | Question: 5



In an election, the share of valid votes received by the four candidates A, B, C, and D is represented by the pie chart shown. The total number of votes cast in the election were 1,15,000, out of which 5,000 were invalid.



Based on the data provided, the total number of valid votes received by the candidates B and C is

- A. 45,000 B. 49,500 C. 51,750 D. 54,000

gate-ds-ai-2024 quantitative-aptitude pie-chart one-mark

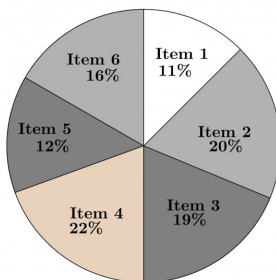
Answer key

9.40.5 Pie Chart: GATE2014 AG: GA-8

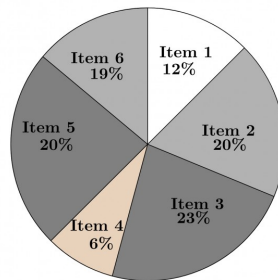


The total exports and revenues from the exports of a country are given in the two pie charts below. The pie chart for exports shows the quantity of each item as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 5 lakh tonnes and the total revenues are 250 crore rupees. What is the ratio of the revenue generated through export of Item 1 per kilogram to the revenue generated through export of Item 4 per kilogram?

Exports



Revenues



- A. 1 : 2 B. 2 : 1 C. 1 : 4 D. 4 : 1

gate2014-ag quantitative-aptitude data-interpretation pie-chart ratio-proportion normal

Answer key

9.41

Polynomials (1)

Practice Test: Test 1 (5Q)

9.41.1 Polynomials: GATE CSE 2016 | Set 1 | GA | Question: 09



If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?

- A. $(x^3 + 8)$ B. $(x - 1)$ C. $(2x - 5)$ D. $(x + 1)$

gatecse-2016-set1 quantitative-aptitude polynomials normal

Answer key

9.42

Powers (1)

9.42.1 Powers: GATE DA 2025 | GA | Question: 9



If a real variable x satisfies $3^{x^2} = 27 \times 9^x$, then the value of $\frac{2^{x^2}}{(2^x)^2}$ is:

- A. 2^{-1} B. 2^0 C. 2^3 D. 2^{15}

gateda-2025 quantitative-aptitude powers two-marks

Answer key

9.43

Prime Numbers (1)

9.43.1 Prime Numbers: GATE CSE 2025 | Set 2 | GA | Question: 5



Let p_1 and p_2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of p_1 and p_2 ?

- A. $p_1 + p_2$ not a prime number B. $p_1 p_2$ is not a prime number
C. $p_1 + p_2 + 1$ is a prime number D. $p_1 p_2 + 1$ is a prime number

gatecse2025-set2 quantitative-aptitude easy prime-numbers one-mark

Answer key

9.44

Probability (17)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(10Q\)](#)

9.44.1 Probability: GATE CSE 2015 | Set 1 | GA | Question: 10



The probabilities that a student passes in mathematics, physics and chemistry are m, p and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m, p, c :

- I. $p + m + c = 27/20$
II. $p + m + c = 13/20$
III. $(p) \times (m) \times (c) = 1/10$

- A. Only relation I is true. B. Only relation II is true.
C. Relations II and III are true. D. Relations I and III are true.

gatecse-2015-set1 quantitative-aptitude probability

Answer key

9.44.2 Probability: GATE CSE 2015 | Set 1 | GA | Question: 3



Given Set $A = 2, 3, 4, 5$ and Set $B = 11, 12, 13, 14, 15$, two numbers are randomly selected, one from each set. What is the probability that the sum of the two numbers equals 16?

- A. 0.20 B. 0.25 C. 0.30 D. 0.33

gatecse-2015-set1 quantitative-aptitude probability normal

Answer key

9.44.3 Probability: GATE CSE 2017 | Set 1 | GA | Question: 5



The probability that a k -digit number does NOT contain the digits 0, 5, or 9 is

- A. 0.3^k B. 0.6^k C. 0.7^k D. 0.9^k

gatecse-2017-set1 general-aptitude quantitative-aptitude probability easy

Answer key

9.44.4 Probability: GATE CSE 2017 | Set 2 | GA | Question: 5

There are 3 red socks, 4 green socks and 3 blue socks. You choose 2 socks. The probability that they are of the same colour is

- A. $\frac{1}{5}$ B. $\frac{7}{30}$ C. $\frac{1}{4}$ D. $\frac{4}{15}$

gatecse-2017-set2 quantitative-aptitude probability

Answer key

9.44.5 Probability: GATE CSE 2018 | GA | Question: 10

A six sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?

- A. Three green faces and four red faces. B. Four green faces and three red faces.
C. Five green faces and two red faces. D. Six green faces and one red face

gatecse-2018 quantitative-aptitude probability normal two-marks

Answer key

9.44.6 Probability: GATE CSE 2021 | Set 1 | GA | Question: 8

There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag.

The probability that at least two chocolates are identical is _____

- A. 0.3024 B. 0.4235 C. 0.6976 D. 0.8125

gatecse-2021-set1 quantitative-aptitude probability two-marks

Answer key

9.44.7 Probability: GATE CSE 2022 | GA | Question: 8

A box contains five balls of same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball.

First ball is drawn. What is the probability of getting an orange ball in the next draw?

- A. $\frac{1}{2}$ B. $\frac{8}{25}$ C. $\frac{19}{50}$ D. $\frac{23}{50}$

gatecse-2022 quantitative-aptitude probability two-marks

Answer key

9.44.8 Probability: GATE CSE 2025 | Set 1 | GA | Question: 8

A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once?

- A. $\frac{75}{216}$ B. $\frac{1}{6}$ C. $\frac{1}{18}$ D. $\frac{25}{216}$

gatecse2025-set1 quantitative-aptitude probability easy two-marks

Answer key

9.44.9 Probability: GATE CSE 2026 | Set 1 | GA | Question: 10

An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the number appearing in the second roll is an integer multiple of the number appearing in the first roll is _____

- A. $\frac{1}{6}$ B. $\frac{5}{18}$ C. $\frac{7}{18}$ D. $\frac{5}{6}$

Answer key

9.44.10 Probability: GATE CSE 2026 | Set 2 | GA | Question: 10



An unbiased six-faced dice whose faces are marked with numbers 1, 2, 3, 4, 5, and 6 is rolled twice in succession and the number on the top face is recorded each time. The probability that the sum of the two recorded numbers is a prime number is _____

- A. $\frac{3}{36}$ B. $\frac{13}{36}$ C. $\frac{15}{36}$ D. $\frac{19}{36}$

Answer key

9.44.11 Probability: GATE CSE 2026 | Set 2 | GA | Question: 3



A day can only be cloudy or sunny. The probability of a day being cloudy is 0.5, independent of the condition on other days. What is the probability that in any given four days, there will be three cloudy days and one sunny day?

- A. $\frac{1}{4}$ B. $\frac{3}{4}$ C. $\frac{2}{3}$ D. $\frac{3}{8}$

Answer key

9.44.12 Probability: GATE DS&AI 2024 | GA | Question: 7



The probability of a boy or a girl being born is $\frac{1}{2}$. For a family having only three children, what is the probability of having two girls and one boy?

- A. $\frac{3}{8}$ B. $\frac{1}{8}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

Answer key

9.44.13 Probability: GATE2012 AE: GA-6



Two policemen, A and B , fire once each at the same time at an escaping convict. The probability that A hits the convict is three times the probability that B hits the convict. If the probability of the convict not getting injured is 0.5, the probability that B hits the convict is

- A. 0.14 B. 0.22 C. 0.33 D. 0.40

Answer key

9.44.14 Probability: GATE2012 AR: GA-9



A smuggler has 10 capsules in which five are filled with narcotic drugs and the rest contain the original medicine. All the 10 capsules are mixed in a single box, from which the customs officials picked two capsules at random and tested for the presence of narcotic drugs. The probability that the smuggler will be caught is

- A. 0.50 B. 0.67 C. 0.78 D. 0.82

Answer key

9.44.15 Probability: GATE2012 CY: GA-7



A and B are friends. They decide to meet between 1:00 pm and 2:00 pm on a given day. There is a condition that whoever arrives first will not wait for the other for more than 15 minutes. The probability that they will meet on that day is

A. $\frac{1}{4}$ B. $\frac{1}{16}$ C. $\frac{7}{16}$ D. $\frac{9}{16}$

gate2012-cy quantitative-aptitude probability

Answer key

9.44.16 Probability: GATE2013 EE: GA-6



What is the chance that a leap year, selected at random, will contain 53 Saturdays?

A. $\frac{2}{7}$ B. $\frac{3}{7}$ C. $\frac{1}{7}$ D. $\frac{5}{7}$

gate2013-ee quantitative-aptitude probability

Answer key

9.44.17 Probability: GATE2014 AG: GA-4



In any given year, the probability of an earthquake greater than Magnitude 6 occurring in the Garhwal Himalayas is 0.04. The average time between successive occurrences of such earthquakes is ____ years.

gate2014-ag quantitative-aptitude probability numerical-answers normal

Answer key

9.45

Profit Loss (3)

Practice Test: Test 1 (14Q)

9.45.1 Profit Loss: GATE CSE 2021 | Set 1 | GA | Question: 7



Items	Cost (₹)	Profit \%	Marked Price (₹)
P	5,400	— — —	5,860
Q	— — —	25	10,000

Details of prices of two items P and Q are presented in the above table. The ratio of cost of item P to cost of item Q is 3 : 4. Discount is calculated as the difference between the marked price and the selling price. The profit percentage is calculated as the ratio of the difference between selling price and cost, to the cost

$$(\text{Profit}\% = \frac{\text{Selling price} - \text{Cost}}{\text{Cost}} \times 100)$$

The discount on item Q , as a percentage of its marked price, is _____

A. 25

B. 12.5

C. 10

D. 5

gatecse-2021-set1 quantitative-aptitude profit-loss two-marks

Answer key

9.45.2 Profit Loss: GATE CSE 2024 | Set 2 | GA | Question: 7



A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of

A. 1% profit

B. 2% profit

C. 1% loss

D. 2% loss

gatecse-2024-set2 quantitative-aptitude profit-loss two-marks

Answer key

9.45.3 Profit Loss: GATE2013 CE: GA-9



A firm is selling its product at Rs. 60 per unit. The total cost of production is Rs. 100 and firm is earning total profit of Rs. 500. Later, the total cost increased by 30%. By what percentage the price should be increased to maintained the same profit level.

A. 5

B. 10

C. 15

D. 30

quantitative-aptitude gate2013-ce profit-loss

Answer key

9.46

Quadratic Equations (6)

Practice Test: Test 1 (13Q)

9.46.1 Quadratic Equations: GATE CSE 2014 | Set 1 | GA | Question: 5



The roots of $ax^2 + bx + c = 0$ are real and positive. a, b and c are real. Then $ax^2 + b|x| + c = 0$ has

A. no roots

B. 2 real roots

C. 3 real roots

D. 4 real roots

gatecse-2014-set1 quantitative-aptitude quadratic-equations normal

Answer key

9.46.2 Quadratic Equations: GATE CSE 2016 | Set 2 | GA | Question: 05



In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$

A. n^4 B. 4^n C. 2^{2n-1} D. 4^{n-1}

gatecse-2016-set2 quantitative-aptitude quadratic-equations normal

Answer key

9.46.3 Quadratic Equations: GATE CSE 2021 | Set 2 | GA | Question: 4



If $\left(x - \frac{1}{2}\right)^2 - \left(x - \frac{3}{2}\right)^2 = x + 2$, then the value of x is:

A. 2

B. 4

C. 6

D. 8

gatecse-2021-set2 quantitative-aptitude quadratic-equations one-mark

Answer key

9.46.4 Quadratic Equations: GATE CSE 2022 | GA | Question: 3



Let r be a root of the equation $x^2 + 2x + 6 = 0$.

Then the value of the expression $(r+2)(r+3)(r+4)(r+5)$ is

A. 51

B. -51

C. 126

D. -126

gatecse-2022 quantitative-aptitude quadratic-equations one-mark

Answer key

9.46.5 Quadratic Equations: GATE2011 MN: GA-62



A student attempted to solve a quadratic equation in x twice. However, in the first attempt, he incorrectly wrote the constant term and ended up with the roots as $(4, 3)$. In the second attempt, he incorrectly wrote down the coefficient of x and got the roots as $(3, 2)$. Based on the above information, the roots of the correct quadratic equation are

A. $(-3, 4)$ B. $(3, -4)$ C. $(6, 1)$ D. $(4, 2)$

gate2011-mn quadratic-equations quantitative-aptitude

Answer key

9.46.6 Quadratic Equations: GATE2013 EE: GA-8



The set of values of p for which the roots of the equation $3x^2 + 2x + p(p-1) = 0$ are of opposite sign is

- A. $(-\infty, 0)$ B. $(0, 1)$ C. $(1, \infty)$ D. $(0, \infty)$

gate2013-ee quantitative-aptitude quadratic-equations

Answer key

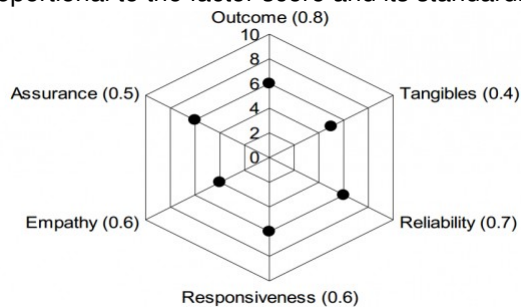
9.47

Radar Chart (1)

9.47.1 Radar Chart: GATE2011 GG: GA-9



The quality of services delivered by a company consists of six factors as shown below in the radar diagram. The dots in the figure indicate the score for each factor on a scale of 0 to 10. The standardized coefficient for each factor is given in the parentheses. The contribution of each factor to the overall service quality is directly proportional to the factor score and its standardized coefficient.



The lowest contribution among all the above factors to the overall quality of services delivered by the company is

- A. 10% B. 20% C. 24% D. 40%

gate2011-gg difficult quantitative-aptitude data-interpretation radar-chart

Answer key

9.48

Ratio Proportion (9)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(3Q\)](#)

9.48.1 Ratio Proportion: GATE CSE 2016 | Set 1 | GA | Question: 10



In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is _____.

- A. 40.00 B. 46.02 C. 60.01 D. 92.02

gatecse-2016-set1 quantitative-aptitude ratio-proportion normal

Answer key

9.48.2 Ratio Proportion: GATE CSE 2018 | GA | Question: 8



In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees in the party?

- A. 2:3 B. 1:1 C. 3:2 D. 2:1

gatecse-2018 quantitative-aptitude ratio-proportion two-marks

Answer key

9.48.3 Ratio Proportion: GATE CSE 2021 | Set 1 | GA | Question: 1

The ratio of boys to girls in a class is 7 to 3.

Among the options below, an acceptable value for the total number of students in the class is:

- A. 21 B. 37 C. 50 D. 73

gatecse-2021-set1 quantitative-aptitude ratio-proportion one-mark

Answer key

9.48.4 Ratio Proportion: GATE CSE 2021 | Set 2 | GA | Question: 8

The number of students in three classes is in the ratio 3 : 13 : 6. If 18 students are added to each class, the ratio changes to 15 : 35 : 21.

The total number of students in all the three classes in the beginning was:

- A. 22 B. 66 C. 88 D. 110

gatecse-2021-set2 ratio-proportion quantitative-aptitude two-marks

Answer key

9.48.5 Ratio Proportion: GATE CSE 2024 | Set 1 | GA | Question: 2

If two distinct non-zero real variables x and y are such that $(x + y)$ is proportional to $(x - y)$ then the value of $\frac{x}{y}$

- A. depends on xy B. depends only on x and not on y
C. depends only on y and not on x D. is a constant

gatecse-2024-set1 quantitative-aptitude ratio-proportion one-mark

Answer key

9.48.6 Ratio Proportion: GATE2011 AG: GA-8

Three friends, R , S and T shared toffee from a bowl. R took $\frac{1}{3}$ rd of the toffees, but returned four to the bowl. S took $\frac{1}{4}$ th of what was left but returned three toffees to the bowl. T took half of the remainder but returned two back into the bowl. If the bowl had 17 toffees left, how many toffees were originally there in the bowl?

- A. 38 B. 31 C. 48 D. 41

general-aptitude quantitative-aptitude gate2011-ag ratio-proportion

Answer key

9.48.7 Ratio Proportion: GATE2011 GG: GA-4

If m students require a total of m pages of stationery in m days, then 100 students will require 100 pages of stationery in

- A. 100 days B. $m/100$ days C. $100/m$ days D. m days

gate2011-gg quantitative-aptitude ratio-proportion

Answer key

9.48.8 Ratio Proportion: GATE2013 AE: GA-1

If $3 \leq X \leq 5$ and $8 \leq Y \leq 11$ then which of the following options is TRUE?

- A. $\left(\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{5}\right)$ B. $\left(\frac{3}{11} \leq \frac{X}{Y} \leq \frac{5}{8}\right)$
C. $\left(\frac{3}{11} \leq \frac{X}{Y} \leq \frac{8}{5}\right)$ D. $\left(\frac{3}{5} \leq \frac{X}{Y} \leq \frac{8}{11}\right)$

gate2013-ae quantitative-aptitude ratio-proportion normal

Answer key

9.48.9 Ratio Proportion: GATE2014 AE: GA-8



The smallest angle of a triangle is equal to two thirds of the smallest angle of a quadrilateral. The ratio between the angles of the quadrilateral is $3 : 4 : 5 : 6$. The largest angle of the triangle is twice its smallest angle. What is the sum, in degrees, of the second largest angle of the triangle and the largest angle of the quadrilateral?

gate2014-ae quantitative-aptitude ratio-proportion numerical-answers

Answer key

9.49

Sequence Series (4)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(1Q\)](#)

9.49.1 Sequence Series: GATE CSE 2012 | Question: 65



Given the sequence of terms, AD CG FK JP, the next term is

- A. OV B. OW C. PV D. PW

gatecse-2012 quantitative-aptitude sequence-series easy

Answer key

9.49.2 Sequence Series: GATE CSE 2018 | GA | Question: 5



What is the missing number in the following sequence?

2, 12, 60, 240, 720, 1440, _____, 0

- A. 2880 B. 1440 C. 720 D. 0

gatecse-2018 quantitative-aptitude sequence-series easy one-mark

Answer key

9.49.3 Sequence Series: GATE CSE 2023 | GA | Question: 3



A series of natural numbers $F_1, F_2, F_3, F_4, F_5, F_6, F_7, \dots$ obeys $F_{n+1} = F_n + F_{n-1}$ for all integers $n \geq 2$.

If $F_6 = 37$, and $F_7 = 60$, then what is F_1 ?

- A. 4 B. 5 C. 8 D. 9

gatecse-2023 quantitative-aptitude sequence-series one-mark

Answer key

9.49.4 Sequence Series: GATE DS&AI 2024 | GA | Question: 4



The sum of the following infinite series is

$$2 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{8} + \frac{1}{9} + \frac{1}{16} + \frac{1}{27} + \dots$$

- A. $11/3$ B. $7/2$ C. $13/4$ D. $9/2$

gate-ds-ai-2024 quantitative-aptitude sequence-series one-mark

Answer key

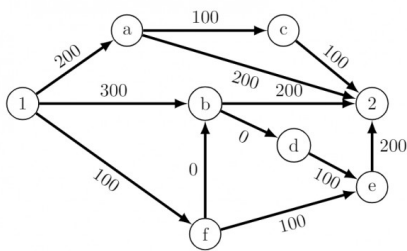
9.50

Shortest Path (1)

9.50.1 Shortest Path: GATE CSE 2020 | GA | Question: 5



There are multiple routes to reach from node 1 to node 2, as shown in the network.



The cost of travel on an edge between two nodes is given in rupees. Nodes 'a', 'b', 'c', 'd', 'e', and 'f' are toll booths. The toll price at toll booths marked 'a' and 'e' is Rs. 200, and is Rs. 100 for the other toll booths. Which is the cheapest route from node 1 to node 2?

- A. $1 - a - c - 2$ B. $1 - f - b - 2$ C. $1 - b - 2$ D. $1 - f - e - 2$

gatecse-2020 quantitative-aptitude graph-theory shortest-path one-mark

Answer key

9.51 Speed Time Distance (4)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(11Q\)](#)

9.51.1 Speed Time Distance: GATE CSE 2013 | Question: 64



A tourist covers half of his journey by train at 60 km/h , half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h . The average speed of the tourist in km/h during his entire journey is

- A. 36 B. 30 C. 24 D. 18

gatecse-2013 quantitative-aptitude easy speed-time-distance

Answer key

9.51.2 Speed Time Distance: GATE CSE 2019 | GA | Question: 3



Two cars at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h . The number of hours it takes for the distance between the two cars to be 20 km is _____.

- A. 1 B. 2 C. 3 D. 6

gatecse-2019 general-aptitude quantitative-aptitude speed-time-distance one-mark

Answer key

9.51.3 Speed Time Distance: GATE2013 AE: GA-6



Velocity of an object fired directly in upward direction is given by $V = 80 - 32t$, where t (time) is in seconds. When will the velocity be between 32 m/sec and 64 m/sec ?

- A. $\left(1, \frac{3}{2}\right)$ B. $\left(\frac{1}{2}, 1\right)$
C. $\left(\frac{1}{2}, \frac{3}{2}\right)$ D. $(1, 3)$

gate2013-ae quantitative-aptitude speed-time-distance

Answer key

9.51.4 Speed Time Distance: GATE2013 EE: GA-9



A car travels 8 km in the first quarter of an hour, 6 km in the second quarter and 16 km in the third quarter. The average speed of the car in km per hour over the entire journey is

- A. 30 B. 36 C. 40 D. 24

gate2013-ee speed-time-distance quantitative-aptitude

9.52.1 Statistics: GATE CSE 2012 | Question: 64



Which of the following assertions are **CORRECT**?

- P: Adding 7 to each entry in a list adds 7 to the mean of the list
- Q: Adding 7 to each entry in a list adds 7 to the standard deviation of the list
- R: Doubling each entry in a list doubles the mean of the list
- S: Doubling each entry in a list leaves the standard deviation of the list unchanged

A. P, Q B. Q, R C. P, R D. R, S

gatecse-2012 quantitative-aptitude statistics normal

9.52.2 Statistics: GATE CSE 2024 | Set 1 | GA | Question: 3



Consider the following sample of numbers:

9, 18, 11, 14, 15, 17, 10, 69, 11, 13

The median of the sample is

A. 13.5

B. 14

C. 11

D. 18.7

gatecse-2024-set1 quantitative-aptitude statistics one-mark

9.52.3 Statistics: GATE2012 AE: GA-5



The arithmetic mean of five different natural numbers is 12. The largest possible value among the numbers is

A. 12

B. 40

C. 50

D. 60

gate2012-ae statistics quantitative-aptitude

9.53.1 System of Equations: GATE2011 GG: GA-6



The number of solutions for the following system of inequalities is

- $X_1 \geq 0$
- $X_2 \geq 0$
- $X_1 + X_2 \leq 10$
- $2X_1 + 2X_2 \geq 22$

A. 0

B. infinite

C. 1

D. 2

gate2011-gg quantitative-aptitude system-of-equations

9.54.1 Tables: GATE CSE 2025 | Set 2 | GA | Question: 8



Which one of the following options is correct for the given data in the table?

Iteration (i)	0	1	2	3
Input (I)	20	-4	10	15
Output (X)	20	16	26	41
Output (Y)	20	-80	-800	-12000

- A. $X(i) = X(i-1) + I(i); Y(i) = Y(i-1)I(i); i > 0$
 B. $X(i) = X(i-1)I(i); Y(i) = Y(i-1) + I(i); i > 0$
 C. $X(i) = X(i-1)I(i); Y(i) = Y(i-1)I(i); i > 0$
 D. $X(i) = X(i-1) + I(i); Y(i) = Y(i-1)I(i-1); i > 0$

gatecse2025-set2 quantitative-aptitude tables two-marks

Answer key

9.55

Tabular Data (7)

 **Practice Test:** Test 1 (12Q)

9.55.1 Tabular Data: GATE CSE 2014 | Set 1 | GA | Question: 9



In a survey, 300 respondents were asked whether they own a vehicle or not. If yes, they were further asked to mention whether they own a car or scooter or both. Their responses are tabulated below. What percent of respondents do not own a scooter?

		Men	Women
Own vehicle	Car	40	34
Own vehicle	Scooter	30	20
Own vehicle	Both	60	46
Do not own vehicle		20	50

gatecse-2014-set1 quantitative-aptitude normal numerical-answers data-interpretation tabular-data

Answer key

9.55.2 Tabular Data: GATE CSE 2014 | Set 3 | GA | Question: 5



The table below has question-wise data on the performance of students in an examination. The marks for each question are also listed. There is no negative or partial marking in the examination.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

What is the average of the marks obtained by the class in the examination?

- A. 1.34 B. 1.74 C. 3.02 D. 3.91

gatecse-2014-set3 quantitative-aptitude normal data-interpretation tabular-data

Answer key

9.55.3 Tabular Data: GATE CSE 2015 | Set 1 | GA | Question: 6



The number of students in a class who have answered correctly, wrongly, or not attempted each question in an exam, are listed in the table below. The marks for each question are also listed. There is no negative or partial marking.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

What is the average of the marks obtained by the class in the examination?

- A. 2.290 B. 2.970 C. 6.795 D. 8.795

gatecse-2015-set1 quantitative-aptitude easy data-interpretation tabular-data

Answer key

9.55.4 Tabular Data: GATE CSE 2016 | Set 1 | GA | Question: 06



A shaving set company sells 4 different types of razors- Elegance, Smooth, Soft and Executive.

Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78 and Executive at Rs. 173 per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter/Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

- A. Elegance B. Executive C. Smooth D. Soft

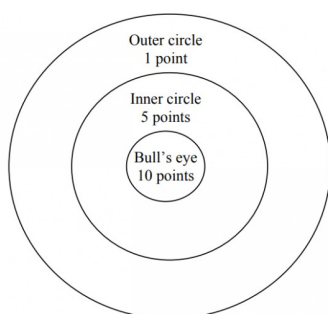
gatecse-2016-set1 quantitative-aptitude data-interpretation easy tabular-data

Answer key

9.55.5 Tabular Data: GATE2011 MN: GA-64



Four archers P, Q, R, and S try to hit a bull's eye during a tournament consisting of seven rounds. As illustrated in the figure below, a player receives 10 points for hitting the bull's eye, 5 points for hitting within the inner circle and 1 point for hitting within the outer circle.



The final scores received by the players during the tournament are listed in the table below.

Round	P	Q	R	S
1	1	5	1	10
2	5	10	10	1
3	1	1	1	5
4	10	10	1	1
5	1	5	5	10
6	10	5	1	1
7	5	10	1	1

The most accurate and the most consistent players during the tournament are respectively

- A. P and S B. Q and R C. Q and Q D. R and Q

gate2011-mn data-interpretation quantitative-aptitude tabular-data

Answer key

9.55.6 Tabular Data: GATE2013 AE: GA-7



Following table gives data on tourist from different countries visiting India in the year 2011

Country	Number of tourists
USA	2000
England	3500
Germany	1200
Italy	1100
Japan	2400
Australia	2300
France	1000

Which two countries contributed to the one third of the total number of tourists who visited India in 2011?

- A. USA and Japan B. USA and Australia C. England and France D. Japan and Australia

gate2013-ae quantitative-aptitude data-interpretation normal tabular-data

Answer key

9.55.7 Tabular Data: GATE2013 CE: GA-8



Following table provides figures(in rupees) on annual expenditure of a firm for two years -2010 and 2011.

Category	2010	2011
Raw material	5200	6240
Power & fuel	7000	9450
Salary & wages	9000	12600
Plant & machinery	20000	25000
Advertising	15000	19500
Research & Development	22000	26400

In 2011, which of the two categories have registered increase by same percentage?

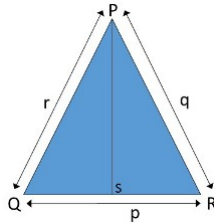
- A. Raw material and Salary & wages. B. Salary & wages and Advertising.
C. Power & fuel and Advertising. D. Raw material and research & Development.

quantitative-aptitude gate2013-ce data-interpretation normal tabular-data

9.56.1 Triangles: GATE CSE 2015 | Set 2 | GA | Question: 8



In a triangle PQR , PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60^\circ$. What is the length of PS ?



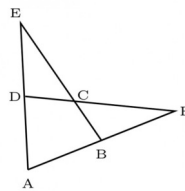
- A. $\left(\frac{(q+r)}{qr}\right)$
 B. $\left(\frac{qr}{q+r}\right)$
 C. $\sqrt{(q^2 + r^2)}$
 D. $\left(\frac{(q+r)^2}{qr}\right)$

gatecse-2015-set2 quantitative-aptitude geometry difficult triangles

9.56.2 Triangles: GATE CSE 2018 | GA | Question: 9



In the figure below, $\angle DEC + \angle BFC$ is equal to _____



- A. $\angle BCD - \angle BAD$
 B. $\angle BAD + \angle BCF$
 C. $\angle BAD + \angle BCD$
 D. $\angle CBA + \angle ADC$

gatecse-2018 quantitative-aptitude geometry normal triangles two-marks

9.56.3 Triangles: GATE CSE 2022 | GA | Question: 9



The corners and mid-points of the sides of a triangle are named using the distinct letters P, Q, R, S, T and U, but not necessarily in the same order. Consider the following statements:

- The line joining P and R is parallel to the line joining Q and S.
- P is placed on the side opposite to the corner T.
- S and U cannot be placed on the same side.

Which one of the following statements is correct based on the above information?

- A. P cannot be placed at a corner
 B. S cannot be placed at a corner
 C. U cannot be placed at a mid-point
 D. R cannot be placed at a corner

gatecse-2022 quantitative-aptitude geometry triangles two-marks

9.57.1 Venn Diagram: GATE CSE 2010 | Question: 59

25 persons are in a room. 15 of them play hockey, 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is:

- A. 2 B. 17 C. 13 D. 3

gatecse-2010 quantitative-aptitude easy set-theory&algebra venn-diagram

Answer key

9.57.2 Venn Diagram: GATE CSE 2016 | Set 2 | GA | Question: 06

Among 150 faculty members in an institute, 55 are connected with each other through Facebook and 85 are connected through Whatsapp. 30 faculty members do not have Facebook or Whatsapp accounts. The numbers of faculty members connected only through Facebook accounts is _____.

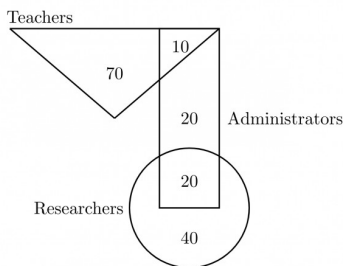
- A. 35 B. 45 C. 65 D. 90

gatecse-2016-set2 quantitative-aptitude venn-diagram easy

Answer key

9.57.3 Venn Diagram: GATE CSE 2019 | GA | Question: 7

In the given diagram, teachers are represented in the triangle, researchers in the circle and administrators in the rectangle. Out of the total number of the people, the percentage of administrators shall be in the range of _____.



- A. 0 to 15 B. 16 to 30 C. 31 to 45 D. 46 to 60

gatecse-2019 general-aptitude quantitative-aptitude venn-diagram two-marks

Answer key

9.57.4 Venn Diagram: GATE CSE 2019 | GA | Question: 9

In a college, there are three student clubs, 60 students are only in the Drama club, 80 students are only in the Dance club, 30 students are only in Maths club, 40 students are in both Drama and Dance clubs, 12 students are in both Dance and Maths clubs, 7 students are in both Drama and Maths clubs, and 2 students are in all clubs. If 75% of the students in the college are not in any of these clubs, then the total number of students in the college is _____.

- A. 1000 B. 975 C. 900 D. 225

gatecse-2019 general-aptitude quantitative-aptitude venn-diagram two-marks

Answer key

9.57.5 Venn Diagram: GATE CSE 2024 | Set 2 | GA | Question: 3

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is $\frac{1}{4}^{\text{th}}$ of the number of students who like other branches. The number of students who like both their core and other branches is 500.

The number of students who like their core branches is

- A. 1,800 B. 3,500 C. 1,600 D. 1,500

gatecse-2024-set2 quantitative-aptitude venn-diagram one-mark

Answer key**9.57.6 Venn Diagram: GATE2010 TF: GA-8**

A gathering of 50 linguists discovered that 4 knew Kannada, Telugu and Tamil, 7 knew only Telugu and Tamil, 5 knew only Kannada and Tamil, 6 knew only Telugu and Kannada. If the number of linguists who knew Tamil is 24 and those who knew Kannada is also 24, how many linguists knew only Telugu?

- A. 9 B. 10 C. 11 D. 8

general-aptitude quantitative-aptitude gate2010-tf venn-diagram

Answer key**9.57.7 Venn Diagram: GATE2011 GG: GA-7**

In a class of 300 students in an M.Tech programme, each student is required to take at least one subject from the following three:

- M600: Advanced Engineering Mathematics
- C600: Computational Methods for Engineers
- E600: Experimental Techniques for Engineers

The registration data for the M.Tech class shows that 100 students have taken M600, 200 students have taken C600, and 60 students have taken E600. What is the maximum possible number of students in the class who have taken all the above three subjects?

- A. 20 B. 30 C. 40 D. 50

gate2011-gg quantitative-aptitude set-theory&algebra venn-diagram

Answer key**9.58 Volume (1)****9.58.1 Volume: GATE CSE 2021 | Set 1 | GA | Question: 6**

We have 2 rectangular sheets of paper, M and N, of dimensions $6\text{ cm} \times 1\text{ cm}$ each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is _____.

- A. $\frac{\pi}{2}$ B. $\frac{3}{\pi}$ C. $\frac{9}{\pi}$ D. 3π

gatecse-2021-set1 quantitative-aptitude mensuration volume two-marks

Answer key**9.59 Work Time (3)**

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(8Q\)](#)

9.59.1 Work Time: GATE CSE 2010 | Question: 64

5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long it will take to build the wall?

- A. 20 days B. 18 days C. 16 days D. 15 days

gatecse-2010 quantitative-aptitude normal work-time

Answer key**9.59.2 Work Time: GATE CSE 2011 | Question: 64**

A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders.

Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10^{th} day. What is the minimum number of trucks required so that there will be no pending order at the end of 5^{th} day?

- A. 4 B. 5 C. 6 D. 7

gatecse-2011 quantitative-aptitude normal work-time

Answer key

9.59.3 Work Time: GATE CSE 2013 | Question: 65



The current erection cost of a structure is Rs. 13,200. If the labour wages per day increase by $1/5$ of the current wages and the working hours decrease by $1/24$ of the current period, then the new cost of erection in Rs. is

- A. 16,500 B. 15,180 C. 11,000 D. 10,120

gatecse-2013 quantitative-aptitude normal work-time

Answer key

Answer Keys

9.1.1	A	9.2.1	D	9.2.2	B	9.2.3	D	9.2.4	B
9.2.5	B	9.3.1	B	9.4.1	D	9.4.2	B	9.5.1	A
9.6.1	C	9.6.2	B	9.6.3	C	9.7.1	C	9.8.1	2006
9.8.2	C	9.8.3	A	9.8.4	A	9.9.1	B	9.10.1	B
9.10.2	C	9.10.3	C	9.10.4	C	9.11.1	D	9.11.2	A
9.11.3	C	9.12.1	A	9.13.1	B	9.13.2	C	9.13.3	D
9.13.4	C	9.13.5	D	9.14.1	B	9.14.2	D	9.14.3	A
9.15.1	C	9.15.2	0.4895 : 0.4897	9.16.1	C	9.16.2	C	9.17.1	A
9.17.2	A	9.17.3	C	9.17.4	1300	9.18.1	D	9.18.2	B
9.19.1	A	9.20.1	D	9.20.2	A	9.20.3	B	9.21.1	C
9.22.1	D	9.22.2	C	9.22.3	C	9.23.1	B	9.24.1	8
9.24.2	C	9.24.3	B	9.24.4	C	9.24.5	B	9.24.6	A
9.24.7	C	9.25.1	6	9.25.2	A	9.25.3	D	9.25.4	A
9.25.5	A	9.26.1	A	9.27.1	140	9.27.2	C	9.27.3	C
9.27.4	B	9.27.5	B	9.28.1	B	9.28.2	C	9.28.3	A
9.28.4	C	9.28.5	C	9.28.6	D	9.29.1	C	9.30.1	B
9.30.2	C	9.30.3	C	9.30.4	A	9.31.1	C	9.32.1	D
9.33.1	B	9.34.1	B	9.34.2	C	9.34.3	C	9.34.4	B
9.34.5	D	9.35.1	C	9.35.2	A	9.35.3	C	9.35.4	B
9.36.1	A	9.36.2	X	9.37.1	96	9.37.2	B	9.37.3	D
9.37.4	B	9.37.5	C	9.37.6	D	9.37.7	A	9.37.8	A
9.37.9	D	9.38.1	850	9.38.2	D	9.38.3	C	9.38.4	A
9.38.5	B	9.38.6	D	9.38.7	C	9.38.8	D	9.39.1	A
9.39.2	B	9.39.3	B	9.39.4	B	9.40.1	32	9.40.2	C
9.40.3	D	9.40.4	B	9.40.5	D	9.41.1	B	9.42.1	C
9.43.1	B	9.44.1	D	9.44.2	A	9.44.3	C	9.44.4	D

9.44.5	C
9.44.10	C
9.44.15	C
9.45.3	A
9.46.5	C
9.48.3	C
9.48.8	B
9.49.4	B
9.51.4	C
9.54.1	A
9.55.5	B
9.56.3	B
9.57.5	A
9.59.2	C

9.44.6	C
9.44.11	A
9.44.16	A
9.46.1	D
9.46.6	B
9.48.4	C
9.48.9	180
9.50.1	B
9.52.1	C
9.55.1	48
9.55.6	C
9.57.1	D
9.57.6	C
9.59.3	B

9.44.7	D
9.44.12	A
9.44.17	25
9.46.2	B
9.47.1	A
9.48.5	D
9.49.1	A
9.51.1	C
9.52.2	A
9.55.2	C
9.55.7	D
9.57.2	A
9.57.7	B

9.44.8	A
9.44.13	A
9.45.1	C
9.46.3	B
9.48.1	B
9.48.6	C
9.49.2	B
9.51.2	B
9.52.3	C
9.55.3	C
9.56.1	B
9.57.3	C
9.58.1	C

9.44.9	C
9.44.14	C
9.45.2	C
9.46.4	D
9.48.2	B
9.48.7	D
9.49.3	A
9.51.3	C
9.53.1	A
9.55.4	B
9.56.2	A
9.57.4	C
9.59.1	D



Transformation of shapes: translation, rotation, scaling, mirroring, assembling, and grouping, Paper folding, cutting, and patterns in 2 and 3 dimensions

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	1	1	0	1	0	0	1	1	2	1	0
2 Marks Count	1	0	1	0	2	1	1	1	0	1	0
Total Marks	3	1	2	1	4	2	3	3	2	3	1

Welcome to the "General Aptitude: Spatial Aptitude" chapter of your GATE Computer Science exam preparation. This section is designed to equip you with the essential concepts, strategies, and quick references needed to excel in spatial reasoning questions. Spatial aptitude tests your ability to visualize, interpret, and manipulate objects and patterns in two and three dimensions. For GATE CS, mastering this area is crucial not just for the General Aptitude section, but also for developing fundamental visualization skills that are indirectly beneficial in areas like computer graphics, data structures, and algorithms. Typically, you can expect 1-3 questions from Spatial Aptitude in the General Aptitude section, contributing 2-5 marks. Questions often involve identifying patterns, mentally rotating shapes, understanding mirror images, or assembling figures, usually presented as Multiple Choice Questions (MCQs).

Topic-wise Key Concepts

1. 3D Structure

Definition and Core Idea: This topic involves understanding and visualizing three-dimensional objects from their two-dimensional representations (like orthographic or isometric views) or vice-versa. It requires the ability to mentally construct, deconstruct, and manipulate 3D shapes, identifying their hidden faces, edges, and vertices.

Important Formulas, Theorems, and Results:

- **Euler's Formula for Polyhedra:** For any simple (convex) polyhedron, the number of vertices (V), edges (E), and faces (F) are related by the formula:

$$V - E + F = 2$$

This formula is fundamental for checking the consistency of polyhedral structures.

- **Common Polyhedra Properties:**

- **Cube:** $V = 8, E = 12, F = 6$
- **Cuboid:** $V = 8, E = 12, F = 6$
- **Triangular Prism:** $V = 6, E = 9, F = 5$
- **Square Pyramid:** $V = 5, E = 8, F = 5$
- **Tetrahedron:** $V = 4, E = 6, F = 4$

Key Properties and Identities:

- **Orthographic Projections:** These are 2D views (Front, Top, Side) that show the true shape and size of one face of an object, without perspective distortion.
- **Isometric Views:** A 3D representation where all three axes are equally foreshortened, giving a sense of depth and volume. Parallel lines remain parallel.
- **Nets of 3D Shapes:** A 2D pattern that can be folded to form a 3D shape. Understanding nets is crucial for visualizing how faces connect.
- **Hidden Lines:** In 2D representations of 3D objects, dashed lines often represent edges that are not directly visible from the chosen viewpoint.

Common Pitfalls or Tricky Points:

- Misinterpreting hidden lines or confusing different orthographic views.
- Incorrectly counting faces, edges, or vertices, especially when parts are hidden or joined.
- Difficulty in visualizing the folding/unfolding of complex nets.
- Confusing isometric projection with perspective projection.

Standard Problem-Solving Techniques or Shortcuts:

- **Mental Rotation and Manipulation:** Practice mentally rotating and viewing objects from different angles.
- **Feature Tracking:** Focus on a unique corner, edge, or patterned face and track its position across different views or during folding.
- **Elimination Strategy:** Quickly eliminate options that clearly violate geometric principles or don't match specific features.
- **Auxiliary Views:** If struggling, mentally sketch an additional view to clarify the structure.
- **Nets to 3D:** For nets, identify adjacent faces and opposite faces. Opposite faces never share an edge.

2. Assembling

Definition and Core Idea: This topic involves combining given 2D or 3D components to form a complete shape or object, or identifying which set of components can form a specified shape. It tests your spatial reasoning to fit parts together precisely without overlap or gaps.

Important Formulas, Theorems, and Results: No specific mathematical formulas are typically used. The core idea relies on geometric principles of congruence, tessellation, and conservation of area/volume.

Key Properties and Identities:

- **Congruence:** Pieces must match exactly in shape and size to fit together.
- **Tessellation:** The ability of shapes to fit together without gaps or overlaps to cover a surface (for 2D assembling).
- **Conservation of Area/Volume:** The total area (for 2D) or volume (for 3D) of the assembled figure must equal the sum of the areas/volumes of its constituent pieces.
- **Edge/Face Matching:** The edges or faces that meet must be identical in length and shape.
- **Symmetry:** Recognizing symmetry in the target shape or individual pieces can aid assembly.

Common Pitfalls or Tricky Points:

- Overlooking small details or subtle differences in piece shapes.
- Assuming pieces can be resized, distorted, or flipped (unless explicitly allowed).
- Not considering all possible orientations (rotations and reflections) for each piece.
- Difficulty in visualizing how irregular shapes fit into complex outlines.

Standard Problem-Solving Techniques or Shortcuts:

- **Start with Distinctive Pieces:** Begin by placing pieces with unique shapes, sharp corners, or specific patterns that have limited fitting options.
- **Work from the Edges/Corners:** For outline-based problems, try to fill the outer boundaries first.
- **Identify Complementary Shapes:** Look for pieces whose contours fit together like puzzle pieces.
- **Mental Rotation and Translation:** Actively rotate and slide pieces mentally to test fits.
- **Elimination:** If a piece clearly doesn't fit in any orientation, eliminate the option containing it.

3. Assembling Pieces

Definition and Core Idea: This topic is a specific type of assembling, often focusing on arranging a set of smaller, usually irregular, 2D pieces to form a larger, specified 2D figure. It's akin to solving a jigsaw puzzle or tangram, requiring visualization of how shapes fit together, including rotation and reflection.

Important Formulas, Theorems, and Results: No specific formulas. The principles are geometric matching and conservation of area.

Key Properties and Identities:

- **Area Conservation:** The sum of the areas of the individual pieces must equal the area of the final assembled figure.
- **Edge and Vertex Matching:** Edges of adjacent pieces must align perfectly, and vertices must meet precisely.
- **No Overlap or Gaps:** The assembled figure must be solid, with no parts of pieces overlapping and no empty spaces within the boundary.
- **Orientation Flexibility:** Pieces can usually be rotated and/or reflected unless specified otherwise.

Common Pitfalls or Tricky Points:

- Incorrectly assuming a piece can be flipped (reflected) when only rotation is allowed, or vice-versa.
- Misjudging the size or shape of internal gaps that need to be filled.
- Difficulty in visualizing how multiple irregular pieces combine to form a regular outline.
- Overlooking that pieces might need to be rotated by angles other than 90 degrees.

Standard Problem-Solving Techniques or Shortcuts:

- **Focus on Unique Features:** Identify pieces with distinct curves, sharp angles, or long straight edges.
- **Largest Piece First:** Often, starting with the largest or most complex piece helps define the initial structure.
- **Work from the Outside In:** Try to form the outer boundary of the target figure first.
- **Consider Negative Space:** Sometimes, it's easier to visualize the shape of the 'hole' that needs to be filled by the remaining pieces.
- **Trial and Error (Mental):** Mentally try fitting pieces, eliminating options as you go.

4. Counting Figure

Definition and Core Idea: This involves systematically counting the number of specific geometric shapes (e.g., triangles, squares, rectangles, cubes) within a complex, often overlapping, figure. The core idea is to develop a structured approach to avoid double-counting or missing any figures.

Important Formulas, Theorems, and Results:

- **Number of Squares in an $n \times n$ Grid:**

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

For example, in a 3×3 grid, it's $3^2 + 2^2 + 1^2 = 9 + 4 + 1 = 14$ squares.

- **Number of Rectangles in an $n \times m$ Grid:**

$$\left(\frac{n(n+1)}{2} \right) \times \left(\frac{m(m+1)}{2} \right)$$

This counts all possible rectangles, including squares.

- **Number of Triangles in a Simple Base with n Segments:** If a base line is divided into n segments and a common vertex is connected to all endpoints, the number of triangles formed is:

$$\frac{n(n+1)}{2}$$

This applies to triangles sharing a common vertex and lying on the same base line. More complex configurations require specific counting methods.

- **Number of Cubes in a Stack:** For a stack of unit cubes, count cubes layer by layer, or by identifying visible and hidden cubes based on the structure.

Key Properties and Identities:

- **Hierarchical Counting:** Often, figures can be counted by their size (e.g., 1-unit squares, 2x2 squares, etc.) or by the number of smaller units they encompass.
- **Systematic Listing:** Labeling vertices or sections can help in systematically listing and counting figures without repetition.
- **Symmetry:** If a figure is symmetrical, counting one part and multiplying can be a shortcut, but be careful with figures on the axis of symmetry.

Common Pitfalls or Tricky Points:

- Double-counting figures or missing smaller, embedded figures.
- Not having a systematic approach, leading to confusion and errors.
- Difficulty in visualizing overlapping shapes, especially triangles or rectangles.
- For 3D figures, forgetting to count hidden cubes or structures.

Standard Problem-Solving Techniques or Shortcuts:

- **Size-Based Counting:** Count all figures of the smallest size, then combinations of two, three, and so on, up to the largest possible figure.
- **Vertex/Line-Based Counting:** For triangles, identify a common vertex and count the number of base segments. For rectangles, count horizontal and vertical lines.
- **Decomposition:** Break down complex figures into simpler, countable components.

- **Labeling:** Assign numbers or letters to distinct regions or vertices to help track counted figures.
- **Pattern Recognition:** For grids, apply the formulas directly.

5. Grouping

Definition and Core Idea: This topic involves identifying a common characteristic or underlying rule among a set of figures and grouping them accordingly, or identifying the 'odd one out' that does not share the common property. It primarily tests pattern recognition and classification based on visual attributes.

Important Formulas, Theorems, and Results: No specific formulas. Relies purely on observational skills and logical deduction.

Key Properties and Identities:

- **Number of Elements/Sides:** Counting internal components, sides of a polygon, or lines.
- **Symmetry:** Presence or absence of rotational, reflectional, or point symmetry.
- **Orientation:** The direction or alignment of figures or their internal components.
- **Shading/Filling Patterns:** Whether parts are filled, hollow, or have specific textures.
- **Relative Positions:** The spatial relationship between different elements within a figure (e.g., inside/outside, left/right).
- **Transformations:** Whether figures are related by rotation, reflection, or translation.
- **Connectivity:** Whether elements are connected or separate.

Common Pitfalls or Tricky Points:

- Focusing on superficial differences rather than fundamental commonalities.
- Missing subtle patterns or rules that apply to most figures.
- Not considering all possible attributes (e.g., only looking at shape, not shading or orientation).
- Assuming a simple rule when a more complex one is at play.

Standard Problem-Solving Techniques or Shortcuts:

- **Attribute Listing:** For each figure, list all observable attributes (e.g., number of sides, type of lines, shading, orientation of internal arrow).
- **Compare and Contrast:** Systematically compare each figure with the others to find commonalities or unique features.
- **Hypothesis Testing:** Formulate a hypothesis about a potential grouping rule and test it against all figures.
- **Look for Deviations:** When finding the odd one out, identify the property that one figure lacks or possesses uniquely.
- **Consider Transformations:** Check if figures can be grouped based on being rotations or reflections of each other.

6. Image Rotation

Definition and Core Idea: This topic involves mentally rotating a 2D or 3D object to match a given orientation or to predict its appearance after a specified rotation. It tests your ability to visualize how shapes and their internal components change position relative to each other during rotation.

Important Formulas, Theorems, and Results: No specific formulas, but understanding angular rotation (e.g., 90°, 180°, 270° clockwise/anticlockwise) is key.

Key Properties and Identities:

- **Relative Positions:** The spatial relationship between different parts of the object remains constant during rotation. Only the object's overall orientation changes.
- **Clockwise vs. Anticlockwise:** Distinguish between the direction of rotation.
- **Degrees of Rotation:** Standard rotations are often multiples of 90 degrees. A 180-degree rotation results in an inverted image (upside down). A 360-degree rotation returns the object to its original orientation.
- **Fixed Point/Axis:** Rotation occurs around a central point (for 2D) or an axis (for 3D).

Common Pitfalls or Tricky Points:

- Incorrectly rotating parts of the image independently instead of the whole figure.
- Confusing clockwise with anticlockwise rotation.
- Misjudging the exact angle of rotation, especially for non-90-degree rotations.
- Difficulty in tracking multiple features simultaneously during rotation.

Standard Problem-Solving Techniques or Shortcuts:

- **Focus on a Distinctive Feature:** Pick a unique part of the figure (e.g., an arrow, a specific corner, a shaded region) and track its movement during rotation.
- **Mental Overlay:** Imagine overlaying the rotated figure onto the original to check alignment.
- **Use Reference Points:** Mentally fix a point on the figure and observe its path during rotation.
- **Elimination:** Quickly rule out options that are clearly not rotations (e.g., reflections, altered shapes).
- **Practice with Physical Objects:** If struggling, use a physical object (like a matchbox or a drawn shape on paper) and rotate it to build intuition.

7. Mirror Image

Definition and Core Idea: This topic involves determining the appearance of an object or figure when reflected across a mirror line (which can be vertical, horizontal, or diagonal). The core idea is to understand the concept of lateral inversion.

Important Formulas, Theorems, and Results: No specific formulas. The principle is geometric reflection.

Key Properties and Identities:

- **Lateral Inversion (Vertical Mirror):** For a vertical mirror, left becomes right, and right becomes left. The top and bottom parts remain in their respective positions.
- **Vertical Inversion (Horizontal Mirror):** For a horizontal mirror, top becomes bottom, and bottom becomes top. The left and right parts remain in their respective positions.
- **Distance Preservation:** Every point in the original figure is the same perpendicular distance from the mirror line as its corresponding point in the mirror image.
- **Symmetry:** If a figure has reflectional symmetry along the mirror line, its mirror image will be identical to the original.
- **Text/Numbers:** Letters and numbers appear reversed in a mirror image (e.g., 'b' becomes 'd', 'p' becomes 'q', '6' becomes '9'). Some letters (A, H, I, M, O, T, U, V, W, X, Y) are symmetrical and appear unchanged in a vertical mirror.

Common Pitfalls or Tricky Points:

- Confusing mirror image with rotation (especially 180-degree rotation, which looks similar to a horizontal mirror image followed by a vertical mirror image).
- Incorrectly reflecting across diagonal mirror lines, which requires more complex visualization.
- Misinterpreting internal patterns or text within the figure.
- Forgetting that the 'depth' or 'front-back' orientation is also inverted in a 3D mirror image.

Standard Problem-Solving Techniques or Shortcuts:

- **Imagine Folding:** Mentally fold the paper along the mirror line. The image should appear on the other side.
- **Point-by-Point Reflection:** For complex figures, reflect key points or vertices one by one.
- **Feature Tracking:** Pick a distinctive feature (e.g., an arrow, a shaded corner) and visualize its reflected position.
- **Elimination:** Rule out options that show rotation or distortion instead of reflection.
- **Practice with Letters/Numbers:** Familiarize yourself with the mirror images of common letters and digits.

8. Paper Folding

Definition and Core Idea: This topic involves visualizing a sequence of folds made on a piece of paper and then predicting the pattern of holes or cuts when the paper is unfolded, or vice-versa. The core idea is to mentally reverse the folding process, understanding how cuts/holes multiply and reflect across each fold line.

Important Formulas, Theorems, and Results: No specific formulas. Relies on principles of symmetry and reflection.

Key Properties and Identities:

- **Symmetry:** Each fold line acts as a mirror line. When the paper is unfolded, any cut or hole will be reflected across that fold line.
- **Number of Layers:** A cut made through multiple layers of folded paper will result in multiple holes upon unfolding, corresponding to the number of layers cut.
- **Order of Unfolding:** The paper must be unfolded in the reverse order of folding. The last fold made is the first one to be unfolded.
- **Position of Cuts:** The position of a cut relative to the edges and previous fold lines is crucial.

Common Pitfalls or Tricky Points:

- Incorrectly reversing the order of folds during unfolding.
- Missing reflections across fold lines, especially when multiple folds are involved.
- Misjudging the number of layers a cut passes through.
- Difficulty in visualizing the exact shape and position of holes after multiple reflections.
- Confusing the orientation of the paper after each fold.

Standard Problem-Solving Techniques or Shortcuts:

- **Work Backward:** Start from the final folded and cut state and mentally unfold the paper step-by-step, applying reflection at each fold line.
- **Draw Each Stage:** If the problem is complex, quickly sketch the paper at each stage of unfolding, marking the reflections of the cuts.
- **Identify Fold Lines as Mirror Lines:** Each fold line becomes an axis of symmetry for the cuts/holes when unfolded.
- **Focus on a Single Cut/Hole:** Track how one specific cut or hole multiplies and reflects across the folds.
- **Eliminate Options:** Use the first few unfolding steps to eliminate options that don't match the expected pattern.

9. Patterns In Three Dimensions

Definition and Core Idea: This topic involves identifying sequences or transformations of 3D objects, predicting the next object in a series, or identifying relationships between 3D patterns. It extends 2D pattern recognition skills to three dimensions, often involving rotation, translation, and changes in internal features of 3D shapes.

Important Formulas, Theorems, and Results: No specific formulas. Relies on understanding 3D geometry and transformations.

Key Properties and Identities:

- **3D Transformations:**
 - **Rotation:** Around X, Y, or Z axes (e.g., a cube rotating on its edge or face).
 - **Translation:** Movement along an axis.
 - **Scaling:** Change in size (less common in GATE spatial aptitude).
- **Internal Features:** Patterns, markings, or smaller objects within the 3D structure can also follow a sequence of changes.
- **Structural Changes:** Addition or removal of smaller cubes or components from a larger 3D structure.
- **Perspective Changes:** How the 2D representation of a 3D object changes with different viewpoints.

Common Pitfalls or Tricky Points:

- Difficulty in visualizing complex 3D changes from static 2D representations.
- Missing subtle transformations or changes in internal elements.
- Confusing rotations around different axes.
- Overlooking hidden changes in the back or underside of the 3D object.

Standard Problem-Solving Techniques or Shortcuts:

- **Focus on One Aspect at a Time:** Analyze the pattern of change for one specific feature (e.g., the top face, a specific corner, an internal dot) before combining observations.
- **Mentally Manipulate:** Actively imagine the 3D objects moving or transforming.
- **Look for Consistent Rules:** Identify a clear, repeatable rule or sequence of operations that applies to all given figures in the series.
- **Use a Reference Point/Axis:** Mentally fix a point or an axis on the object to track its movement or rotation.
- **Elimination:** Rule out options that clearly do not follow the established pattern.

10. Patterns In Two Dimensions

Definition and Core Idea: This topic involves identifying the underlying rule or sequence in a series of 2D figures and predicting the next figure, or finding the odd one out. It requires recognizing transformations (rotation, reflection, translation), changes in number, size, shading, or arrangement of elements within the figures.

Important Formulas, Theorems, and Results: No specific formulas. Relies on logical deduction and visual pattern recognition.

Key Properties and Identities:

- **Transformations:**
 - **Rotation:** Clockwise or anticlockwise, by specific degrees (e.g., 45°, 90°, 180°).

- **Reflection:** Across a horizontal, vertical, or diagonal axis.
- **Translation:** Movement of the entire figure or its components in a specific direction.
- **Changes in Attributes:**
 - **Number:** Increase or decrease in the count of elements.
 - **Size/Shape:** Enlargement, reduction, or alteration of shapes.
 - **Shading/Filling:** Alternating or progressive changes in shading.
 - **Arrangement:** Changes in the relative positions of components.
 - **Connectivity:** Elements joining or separating.
- **Alternating Patterns:** A pattern that alternates between two or more states.
- **Progressive Patterns:** A pattern that changes incrementally with each step.

Common Pitfalls or Tricky Points:

- Overlooking subtle changes in size, orientation, or shading.
- Assuming a simple pattern when a more complex, multi-layered rule is present.
- Not considering all possible transformations (e.g., only looking for rotation, ignoring reflection).
- Getting confused by multiple elements changing simultaneously.

Standard Problem-Solving Techniques or Shortcuts:

- **Systematic Analysis:** Break down the figure into its constituent elements. Analyze how each element (e.g., outer shape, inner shape, arrow, dot, shading) changes from one figure to the next.
- **Track One Element at a Time:** If multiple elements are changing, focus on one element's transformation throughout the series before moving to the next.
- **Look for Cycles:** Identify if a pattern repeats after a certain number of figures.
- **Identify the Rule:** Formulate a clear rule (e.g., "the inner square rotates 90 degrees clockwise, and the outer circle alternates shading").
- **Test the Rule:** Apply your hypothesized rule to the subsequent figures in the series to confirm its validity.
- **Elimination:** Use the identified pattern to quickly eliminate options that do not conform.

Quick Formula Reference

This section consolidates the key formulas and mathematical results important for Spatial Aptitude problems.

- **Euler's Formula for Polyhedra:** For any simple polyhedron, the relationship between vertices (V), edges (E), and faces (F) is given by:

$$V - E + F = 2$$

- **Number of Squares in an $n \times n$ Grid:** The total number of squares (of all sizes) in an $n \times n$ grid is:

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

- **Number of Rectangles in an $n \times m$ Grid:** The total number of rectangles (including squares) in an $n \times m$ grid is:

$$\left(\frac{n(n+1)}{2} \right) \times \left(\frac{m(m+1)}{2} \right)$$

- **Number of Triangles in a Simple Base with n Segments:** If a base line is divided into n segments and a common vertex is connected to all endpoints, the number of triangles formed is:

$$\frac{n(n+1)}{2}$$

Important Tips for GATE

Mastering Spatial Aptitude for GATE requires not just understanding concepts but also developing effective exam strategies. Here are some practical tips:

1. **Visualize Actively, Don't Just Observe:** Spatial aptitude is about mental manipulation. Don't just look at the figures; actively imagine them moving, rotating, folding, or reflecting. Close your eyes briefly if it helps to form a clearer mental image.

2. **Adopt a Systematic Approach:** Avoid haphazard guessing. For each problem type (e.g., counting, paper folding, pattern series), develop and follow a structured method. This prevents errors like double-counting or missing steps.
3. **Focus on Key Features and Reference Points:** When dealing with rotations, reflections, or complex patterns, identify a unique or distinctive element (e.g., a specific corner, an arrow, a shaded region). Track its movement or transformation throughout the sequence or across options.
4. **Leverage the Elimination Strategy:** In multiple-choice questions, often several options can be quickly ruled out because they clearly violate a fundamental rule (e.g., a reflection instead of a rotation, an extra piece in an assembly). This narrows down your choices and increases your chances.
5. **Practice Under Time Constraints:** Spatial aptitude questions can be time-consuming due to the mental effort required. Practice solving problems within strict time limits to improve your speed and efficiency. The more you practice, the faster your visualization skills will become.
6. **Clearly Differentiate Mirror Image from Rotation:** These are common sources of confusion. Remember that a mirror image involves lateral inversion (left-right flip for a vertical mirror), while rotation changes orientation without flipping. A 180-degree rotation can sometimes look similar to a mirror image, so pay close attention.
7. **Don't Hesitate to Sketch (Mentally or Physically):** For complex problems like paper folding or 3D structures, a quick mental or physical sketch of intermediate steps can significantly clarify the problem and prevent errors. This is especially useful for tracking reflections in paper folding.
8. **Start with Simpler Problems to Build Confidence:** If you find a particular type of spatial problem challenging, begin with simpler variations to build your foundational understanding and confidence before tackling more complex ones. Gradually increase the difficulty.

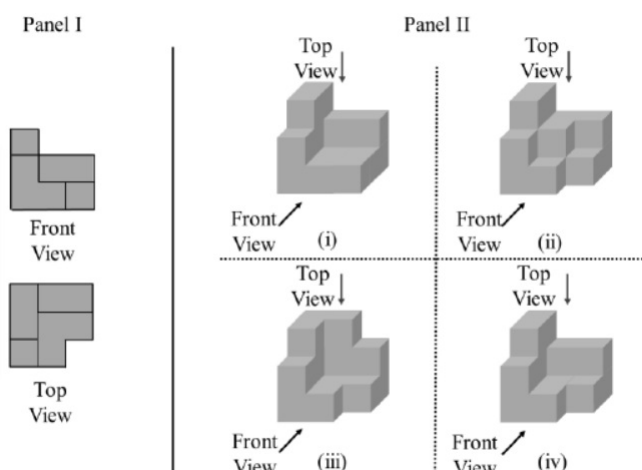
10.1

3d Structure (1)

10.1.1 3d Structure: GATE CSE 2026 | Set 1 | GA | Question: 7



In Panel I of the figure below, the front view and top view of a structure are shown. Which one of the 3D structures shown in Panel II possesses the views shown in Panel I?



- A. (i) B. (ii) C. (iii) D. (iv)

gatescse-2026-set1 spatial-aptitude 3d-structure two-marks

Answer key

10.2

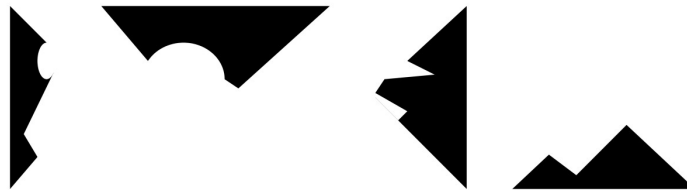
Assembling (1)

10.2.1 Assembling: GATE DA 2026 | GA | Question: 3



The four pieces of a puzzle are shown in the figure below.





Which one of the figures labelled as P, Q, R, and S can be constructed by using each of the four pieces only once without overlaps?



P



Q



R



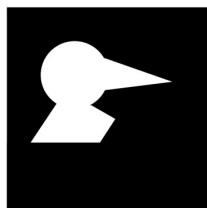
S



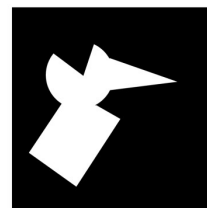
P



Q



R



S

A. P

B. Q

C. R

D. S

gateda-2026 spatial-aptitude assembling one-mark

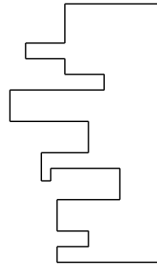
Answer key 

10.3

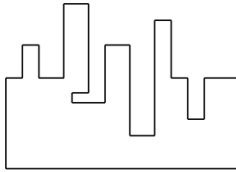
Assembling Pieces (1)

10.3.1 Assembling Pieces: GATE CSE 2021 | Set 2 | GA | Question: 7

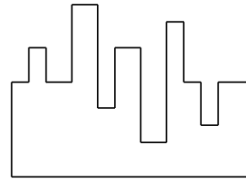




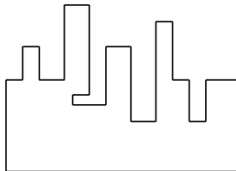
A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.



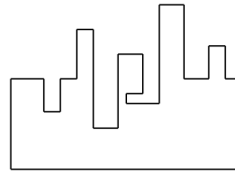
A.



B.



C.



D.

gatecse-2021-set2 spatial-aptitude assembling-pieces two-marks

Answer key

10.4

Image Rotation (3)

Practice Test: Test 1 (7Q)

10.4.1 Image Rotation: GATE CSE 2022 | GA | Question: 5



A palindrome is a word that reads the same forwards and backwards. In a game of words, a player has the following two plates painted with letters.



From the additional plates given in the options, which one of the combinations of additional plates would allow the player to construct a five-letter palindrome. The player should use all the five plates exactly once. The plates can be rotated in their plane.



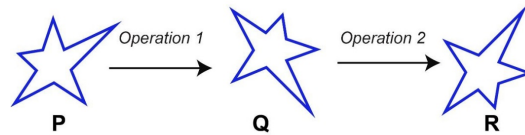
gatecse-2022 spatial-aptitude image-rotation one-mark

Answer key

10.4.2 Image Rotation: GATE CSE 2023 | GA | Question: 10



Which one of the options best describes the transformation of the 2-dimensional figure **P** to **Q**, and then to **R**, as shown?



- A. Operation 1: A clockwise rotation by 90° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a horizontal line
- B. Operation 1: A counter clockwise rotation by 90° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a horizontal line
- C. Operation 1: A clockwise rotation by 90° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a vertical line
- D. Operation 1: A counter clockwise rotation by 180° about an axis perpendicular to the plane of the figure
Operation 2: A reflection along a vertical line

gatecse-2023 spatial-aptitude image-rotation two-marks

Answer key

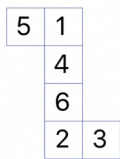
10.4.3 Image Rotation: GATE DS&AI 2024 | GA | Question: 9



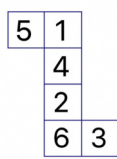
Three different views of a dice are shown in the figure below.



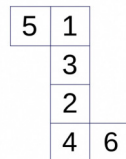
The piece of paper that can be folded to make this dice is



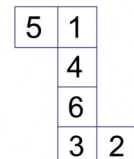
A.



B.



C.



D.

gate-ds-ai-2024 spatial-aptitude image-rotation two-marks

Answer key

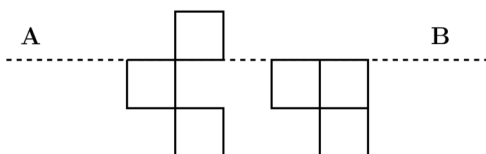
10.5 Mirror Image (1)

Practice Test: Test 1 (10Q)

10.5.1 Mirror Image: GATE CSE 2024 | Set 1 | GA | Question: 10



The least number of squares to be added in the figure to make A B a line of symmetry is



A. 6

B. 4

C. 5

D. 7

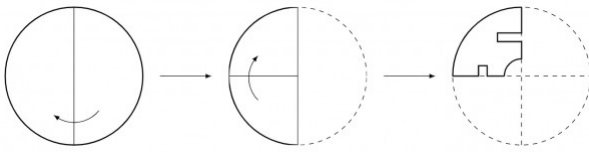
gatecse-2024-set1 spatial-aptitude mirror-image two-marks

Answer key

10.6 Paper Folding (5)

Practice Tests: Test 1 (15Q) Test 2 (5Q)

10.6.1 Paper Folding: GATE CSE 2021 | Set 1 | GA | Question: 4



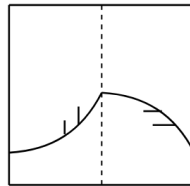
A circular sheet of paper is folded along the lines in the directions shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like _____.

- A.
- B.
- C.
- D.

gatecse-2021-set1 spatial-aptitude paper-folding one-mark

Answer key

10.6.2 Paper Folding: GATE CSE 2021 | Set 2 | GA | Question: 2



A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.

- A.
- B.
- C.
- D.

gatecse-2021-set2 spatial-aptitude paper-folding one-mark

Answer key

10.6.3 Paper Folding: GATE CSE 2024 | Set 1 | GA | Question: 9



A rectangular paper of $20 \text{ cm} \times 8 \text{ cm}$ is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is

- A. 18 B. 24 C. 20 D. 21

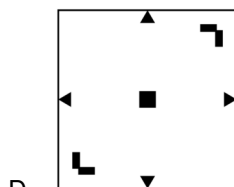
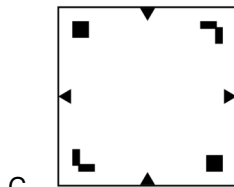
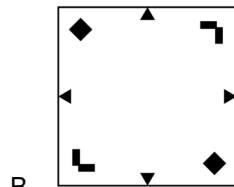
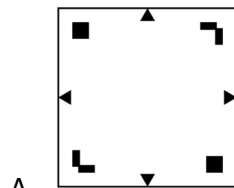
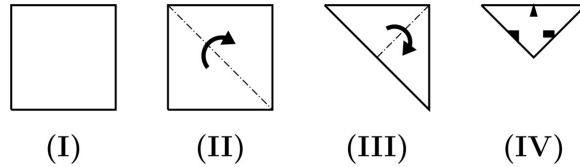
gatecse-2024-set1 spatial-aptitude paper-folding two-marks

10.6.4 Paper Folding: GATE CSE 2025 | Set 1 | GA | Question: 9



A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded?

Note: The figures shown are representative.

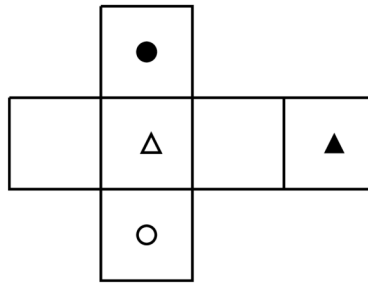


10.6.5 Paper Folding: GATE CSE 2025 | Set 2 | GA | Question: 4



The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube?

Note: The figures shown are representative.



- A.
- B.
- C.
- D.

gatecse2025-set2 spatial-aptitude paper-folding easy one-mark

Answer key

10.7

Patterns In Three Dimensions (3)

Practice Test: Test 1 (10Q)

10.7.1 Patterns In Three Dimensions: GATE CSE 2023 | GA | Question: 5



Looking at the surface of a smooth 3-dimensional object from the outside, which one of the following options is **TRUE**?

- A. The surface of the object must be concave everywhere.
- B. The surface of the object must be convex everywhere.
- C. The surface of the object may be concave in some places and convex in other places.
- D. The object can have edges, but no corners.

gatecse-2023 spatial-aptitude patterns-in-three-dimensions one-mark

Answer key

10.7.2 Patterns In Three Dimensions: GATE CSE 2024 | Set 2 | GA | Question: 9



A cube is to be cut into 8 pieces of equal size and shape. Here, each cut should be straight and it should not stop till it reaches the other end of the cube.

The minimum number of such cuts required is

- A. 3
- B. 4
- C. 7
- D. 8

Answer key

10.7.3 Patterns In Three Dimensions: GATE DS&AI 2024 | GA | Question: 10



Visualize two identical right circular cones such that one is inverted over the other and they share a common circular base. If a cutting plane passes through the vertices of the assembled cones, what shape does the outer boundary of the resulting cross-section make?

- A. A rhombus B. A triangle C. An ellipse D. A hexagon

Answer key

10.8 Patterns In Two Dimensions (4)

Practice Test: Test 1 (10Q)

10.8.1 Patterns In Two Dimensions: GATE CSE 2021 | Set 1 | GA | Question: 2



A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon.

Which one of the following is NOT a convex polygon?

- A.  B.  C.  D. 

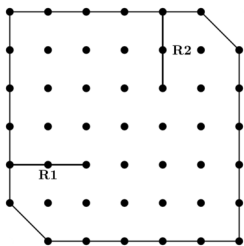
Answer key

10.8.2 Patterns In Two Dimensions: GATE CSE 2022 | GA | Question: 10



A plot of land must be divided between four families. They want their individual plots to be similar in shape, not necessarily equal in area. The land has equally spaced poles, marked as dots in the below figure. Two ropes, R_1 and R_2 , are already present and cannot be moved.

What is the least number of **additional** straight ropes needed to create the desired plots? A single rope can pass through three poles that are aligned in a straight line.



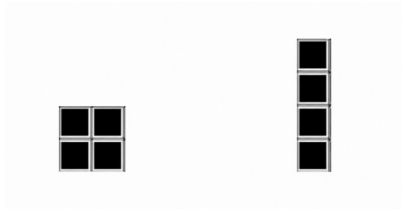
- A. 2 B. 4 C. 5 D. 3

Answer key

10.8.3 Patterns In Two Dimensions: GATE CSE 2026 | Set 1 | GA | Question: 2



The figure shows two 4-tile patterns.



Either one or both of the patterns can be used any number of times and in any orientation to construct a new pattern. Which one of the options below cannot be constructed by using only these two 4-tile patterns assuming there are no overlaps among them?

- A.
- B.
- C.
- D.
- E.
- F.
- G.
- H.

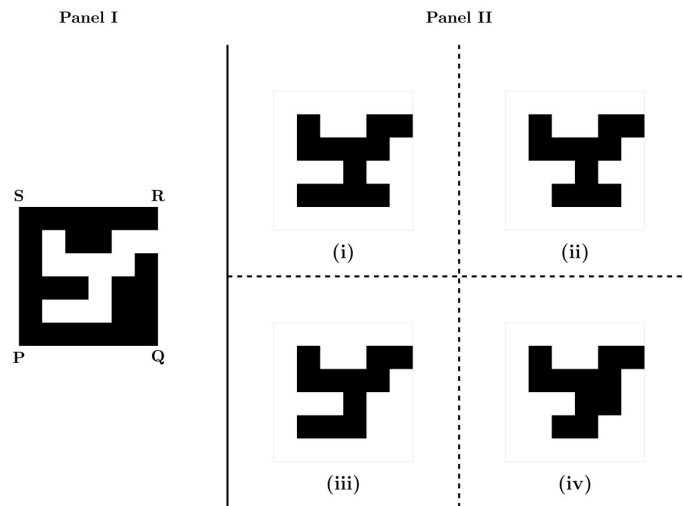
gatecse-2026-set1 spatial-aptitude patterns-in-two-dimensions one-mark

Answer key

10.8.4 Patterns In Two Dimensions: GATE CSE 2026 | Set 2 | GA | Question: 2



A black square PQRS has been cut into two parts. One part of it is shown in Panel I. Which one of the shapes in Panel II is the other part?



A. (i)

B. (ii)

C. (iii)

D. (iv)

gatecse-2026-set2 spatial-aptitude one-mark patterns-in-two-dimensions

Answer key 

Answer Keys

10.1.1	D
10.4.3	A
10.6.4	A
10.8.1	A

10.2.1	A
10.5.1	A
10.6.5	A
10.8.2	D

10.3.1	A
10.6.1	A
10.7.1	C
10.8.3	C

10.4.1	B
10.6.2	B
10.7.2	A
10.8.4	C

10.4.2	A
10.6.3	A
10.7.3	A



Syllabus: English grammar, Sentence completion. Verbal analogies, Word groups. Instructions, Critical reasoning and Verbal deduction

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	1	1	2	2	1	1	2	1	2	2	1
2 Marks Count	0	0	1	0	1	1	1	1	1	1	0
Total Marks	1	1	4	2	3	3	4	3	4	4	1

Welcome to the "General Aptitude: Verbal Aptitude" chapter of your GATE Computer Science exam preparation. This section is designed to equip you with the essential linguistic and reasoning skills necessary to excel in the verbal aptitude segment of the GATE exam. Verbal Aptitude assesses your comprehension of English language, grammar, vocabulary, and logical reasoning abilities, which are crucial not just for the exam but also for effective communication in your professional career. Typically, verbal aptitude questions constitute a significant portion (around 5-8 marks out of 15 for General Aptitude) and can range from direct vocabulary and grammar questions to more complex reading comprehension and logical deduction problems. Mastering these topics can significantly boost your overall GATE score.

Topic-wise Key Concepts

Antonyms

Definition and core idea: Antonyms are words that have opposite meanings. Understanding antonyms enhances vocabulary and comprehension by highlighting contrasts in meaning.

- **Key properties and identities:**

- Antonyms can be absolute (e.g., hot/cold) or relative (e.g., happy/sad).
- Often formed by adding prefixes (e.g., *un-*, *dis-*, *in-*, *im-*, *non-*) to a word.
- Context is crucial; a word can have different antonyms depending on its usage.

- **Common pitfalls:**

- Choosing a word that is merely different, not opposite.
- Ignoring prefixes/suffixes that might indicate an antonym.
- Misinterpreting the nuance of the given word.

- **Standard problem-solving techniques:**

1. Understand the exact meaning of the given word.
2. Look for prefixes that reverse meaning (e.g., *capable* -> *incapable*).
3. Consider the context if the word has multiple meanings.
4. Eliminate options that are synonyms or unrelated.
5. If unsure, try to use the options in a sentence to see which one creates an opposite meaning.

Articles

Definition and core idea: Articles are words (a, an, the) that define whether a noun is specific or unspecific. They are determiners that precede nouns.

- **Important formulas, theorems, and results:**

1. **Indefinite Article 'a':** Used before singular, countable nouns that begin with a consonant sound. E.g., *a book*, *a university* (starts with 'y' sound).
2. **Indefinite Article 'an':** Used before singular, countable nouns that begin with a vowel sound. E.g., *an apple*, *an hour* (starts with 'o' sound).
3. **Definite Article 'the':** Used before specific or unique nouns, or nouns that have been previously mentioned. E.g., *the sun*, *the book I read yesterday*.
4. **Zero Article:** No article is used before plural countable nouns, uncountable nouns, proper nouns, or abstract nouns when speaking generally. E.g., *water is essential*, *students learn*.

- **Key properties and identities:**

- 'A' and 'an' are used for general or non-specific items.
- 'The' is used for specific or previously identified items.
- Sound, not spelling, determines 'a' vs. 'an'.

- **Common pitfalls:**

- Confusing 'a' and 'an' based on spelling rather than sound.

- Overusing 'the' with general nouns or proper nouns.
- Omitting articles where they are required for singular countable nouns.
- **Standard problem-solving techniques:**
 1. Identify if the noun is singular/plural, countable/uncountable.
 2. Determine if the noun is specific or general in the context.
 3. Pay attention to the initial sound of the word following the article.
 4. Check for common phrases or idioms that require specific article usage.

Comparative Forms

Definition and core idea: Comparative forms of adjectives and adverbs are used to compare two things, while superlative forms are used to compare three or more things.

- **Important formulas, theorems, and results:**
 1. **One-syllable adjectives/adverbs:** Add *-er* for comparative, *-est* for superlative. E.g., *tall* → *taller* → *tallest*.
 2. **Two-syllable adjectives ending in -y:** Change *-y* to *-i* and add *-er/-est*. E.g., *happy* → *happier* → *happiest*.
 3. **Two or more syllable adjectives/adverbs:** Use *more* for comparative, *most* for superlative. E.g., *beautiful* → *more beautiful* → *most beautiful*.
 4. **Irregular forms:** Some words have irregular comparative/superlative forms. E.g., *good* → *better* → *best*; *bad* → *worse* → *worst*; *much/many* → *more* → *most*.
 5. **Comparison structure:** Comparative forms are often followed by *than*. E.g., *X is taller than Y*.
- **Key properties and identities:**
 - Do not use double comparatives/superlatives (e.g., "more happier" is incorrect).
 - Ensure logical comparison (comparing like with like).
- **Common pitfalls:**
 - Using "more" or "most" with *-er* or *-est* endings.
 - Incorrectly forming irregular comparatives/superlatives.
 - Making illogical comparisons (e.g., "My car is faster than John").
- **Standard problem-solving techniques:**
 1. Count the syllables of the adjective/adverb.
 2. Check for irregular forms.
 3. Ensure the comparison is grammatically and logically sound.

English Grammar

Definition and core idea: English grammar refers to the set of structural rules governing the composition of clauses, phrases, and words in the English language. It ensures clarity and correctness in communication.

- **Important formulas, theorems, and results:**
 1. **Basic Sentence Structure:** *Subject + Verb + Object* (SVO) is the most common structure. E.g., *She reads a book*.
 2. **Parts of Speech:** Nouns, Pronouns, Verbs, Adjectives, Adverbs, Prepositions, Conjunctions, Interjections. Each has specific roles.
 3. **Subject-Verb Agreement:** The verb must agree in number with its subject. E.g., *He runs* (singular); *They run* (plural).
 4. **Tense Consistency:** Maintain a consistent tense throughout a sentence or paragraph unless there's a specific reason to change.
 5. **Parallelism:** Similar items in a series or comparison should be in the same grammatical form. E.g., *Helikes swimming, hiking, and cycling*.
- **Key properties and identities:**
 - Grammar provides the framework for meaningful sentences.
 - Correct grammar prevents ambiguity and misinterpretation.
- **Common pitfalls:**
 - Errors in subject-verb agreement, pronoun agreement, and tense usage.
 - Lack of parallelism.
 - Misplaced modifiers.
 - Incorrect punctuation.
- **Standard problem-solving techniques:**
 1. Break down the sentence into its core components (subject, verb, object).
 2. Identify the part of speech for each word.
 3. Check for agreement (subject-verb, pronoun-antecedent).

4. Look for consistency in tense and structure.
5. Read the sentence aloud to catch awkward phrasing or obvious errors.

Grammatical Error

Definition and core idea: Grammatical errors are mistakes in the structure or usage of language that violate standard English grammar rules. Identifying and correcting them is key to effective communication.

- **Key properties and identities:**
 - Common error types include subject-verb agreement, pronoun agreement, tense consistency, parallelism, misplaced modifiers, comma splices, run-on sentences, fragment sentences, and incorrect word choice.
- **Common pitfalls:**
 - Overlooking subtle errors, especially in complex sentences.
 - Focusing only on obvious errors and missing nuanced ones.
 - Not knowing the specific rule being tested.
- **Standard problem-solving techniques:**
 1. Read the sentence carefully, looking for any awkward phrasing.
 2. Check for subject-verb agreement (singular subject, singular verb; plural subject, plural verb).
 3. Verify pronoun-antecedent agreement (pronoun matches the noun it refers to in number and gender).
 4. Ensure tense consistency throughout the sentence.
 5. Look for parallel structure in lists or comparisons.
 6. Identify misplaced or dangling modifiers.
 7. Check for correct use of articles, prepositions, and conjunctions.
 8. Consider punctuation errors.

Incorrect Sentence Part

Definition and core idea: This question type presents a sentence divided into several parts, one of which contains a grammatical error. The task is to identify the erroneous part.

- **Key properties and identities:**
 - Errors can span any grammatical rule: subject-verb agreement, tense, pronouns, articles, prepositions, parallelism, vocabulary, etc.
 - Often, the error is subtle and requires careful attention to detail.
- **Common pitfalls:**
 - Rushing through the sentence and missing the error.
 - Fixating on one type of error and ignoring others.
 - Assuming a part is correct because it "sounds" okay.
- **Standard problem-solving techniques:**
 1. Read the entire sentence first to understand its meaning and flow.
 2. Examine each part individually, checking for common grammatical errors:
 - **Subject-Verb Agreement:** Does the verb match the subject in number?
 - **Tense:** Is the tense consistent and appropriate for the context?
 - **Pronouns:** Do pronouns agree with their antecedents in number, gender, and case?
 - **Articles/Prepositions:** Are they used correctly?
 - **Parallelism:** Are items in a list or comparison structured similarly?
 - **Word Choice/Vocabulary:** Is the correct word used (e.g., affect vs. effect)?
 - **Modifiers:** Are adjectives and adverbs placed correctly?
 3. If no error is found, consider the "No Error" option if available.

Most Appropriate Word

Definition and core idea: This involves choosing the word that best fits the meaning and context of a given sentence or passage from a set of options.

- **Key properties and identities:**
 - Requires strong vocabulary and understanding of nuances in word meanings.
 - Context is paramount; a word might be grammatically correct but semantically inappropriate.
- **Common pitfalls:**
 - Choosing a word that is a near-synonym but doesn't fit the exact context.
 - Ignoring the tone or register of the sentence.
 - Misinterpreting the overall meaning of the sentence.

- **Standard problem-solving techniques:**

1. Read the sentence carefully to grasp its full meaning and tone.
2. Consider the part of speech required for the blank (noun, verb, adjective, adverb).
3. Substitute each option into the sentence.
4. Evaluate which option makes the most sense semantically and grammatically.
5. Look for collocations (words that naturally go together) or idiomatic expressions.
6. Eliminate options that are clearly incorrect in meaning or grammar.

Narrative Sequencing

Definition and core idea: This task involves arranging a set of jumbled sentences or paragraphs into a coherent and logical narrative sequence.

- **Key properties and identities:**

- A coherent narrative follows a logical progression, often chronological, cause-and-effect, or problem-solution.
- Transition words and pronouns are crucial clues.

- **Common pitfalls:**

- Ignoring transition words or pronoun references.
- Focusing only on the first sentence and not the overall flow.
- Assuming a chronological order when a different logical order is intended.

- **Standard problem-solving techniques:**

1. **Identify the opening sentence:** Look for sentences that introduce a topic, character, or setting, often without pronouns or conjunctions referring to previous sentences.
2. **Identify transition words/phrases:** Words like *however, therefore, meanwhile, subsequently, in addition, finally*, etc., indicate relationships between sentences.
3. **Look for pronoun references:** A pronoun (he, she, it, they, this, that) must refer back to a noun introduced in a previous sentence.
4. **Establish cause-and-effect or chronological order:** Determine if events follow a time sequence or if one event leads to another.
5. **Identify the concluding sentence:** Look for sentences that summarize or provide a final thought.
6. **Form pairs:** Try to identify two sentences that logically follow each other.
7. **Read the assembled sequence:** Once you have an order, read it through to ensure it flows logically and makes sense.

Noun Verb Adjective

Definition and core idea: This topic focuses on understanding the definitions, functions, and interrelationships of nouns (naming words), verbs (action/state words), and adjectives (describing words).

- **Important formulas, theorems, and results:**

1. **Noun:** Names a person, place, thing, or idea. Can be subject or object. E.g., *student, city, book, freedom*.
2. **Verb:** Expresses an action, occurrence, or state of being. E.g., *run, become, is*.
3. **Adjective:** Modifies or describes a noun or pronoun. Answers questions like "which one?", "what kind?", "how many?". E.g., *happy, blue, three*.
4. **Common Suffixes:**
 - Nouns: *-tion, -sion, -ment, -ness, -ity, -ance, -ence, -er, -or* (e.g., *information, development, happiness*).
 - Verbs: *-ate, -ify, -ize, -en* (e.g., *activate, simplify, modernize*).
 - Adjectives: *-ful, -less, -able, -ible, -ous, -ic, -al, -y* (e.g., *beautiful, hopeless, capable*).

- **Key properties and identities:**

- A word's function (noun, verb, adjective) can change based on context (e.g., "run" as a verb vs. "a run" as a noun).
- Understanding these parts of speech is fundamental to sentence construction and error identification.

- **Common pitfalls:**

- Confusing words that have similar forms but different parts of speech (e.g., "advice" (noun) vs. "advise" (verb)).
- Incorrectly identifying the function of a word in a sentence.

- **Standard problem-solving techniques:**

1. Identify the word's role in the sentence (e.g., is it naming something, performing an action, or describing something?).
2. Look for common suffixes that indicate a specific part of speech.
3. Test the word by replacing it with a clear example of a noun, verb, or adjective to see if the sentence structure

remains valid.

Passage Reading

Definition and core idea: This involves reading a given passage and answering comprehension questions based on its content, including main idea, specific details, inference, and author's tone.

- **Key properties and identities:**

- Questions can be direct (explicitly stated in the text) or inferential (requiring logical deduction from the text).
- Understanding the main idea and supporting details is crucial.

- **Common pitfalls:**

- Reading too slowly or too quickly without comprehension.
- Making assumptions not supported by the text.
- Getting bogged down by unfamiliar vocabulary.
- Misinterpreting the author's intent or tone.

- **Standard problem-solving techniques:**

1. **Skim the passage first:** Read quickly to get a general idea of the topic, main argument, and structure. Pay attention to the first and last sentences of paragraphs.
2. **Read the questions:** Understand what information you need to find. This helps you focus your reading.
3. **Scan for keywords:** For specific detail questions, scan the passage for keywords from the question.
4. **Read carefully for answers:** When you locate relevant sections, read them thoroughly to understand the context and find the exact answer.
5. **Inferential questions:** Look for clues, implications, and logical connections within the text. Avoid bringing in outside knowledge.
6. **Main idea questions:** Often found in the introduction or conclusion, or can be derived from summarizing the key points.
7. **Vocabulary in context:** Use surrounding words and sentences to deduce the meaning of unfamiliar words.

Phrasal Verb

Definition and core idea: A phrasal verb is a verb combined with a preposition or an adverb (or both) to create a new meaning that is often different from the original verb's meaning.

- **Important formulas, theorems, and results:**

1. *Verb + Preposition* (e.g., *look after* = care for)
2. *Verb + Adverb* (e.g., *break down* = stop functioning)
3. *Verb + Adverb + Preposition* (e.g., *look up to* = admire)

- **Key properties and identities:**

- Meanings are often idiomatic and cannot be deduced from the individual words.
- Can be separable (object can go between verb and particle) or inseparable.

- **Common pitfalls:**

- Assuming the literal meaning of the words.
- Confusing similar-looking phrasal verbs.
- Incorrectly using separable/inseparable phrasal verbs.

- **Standard problem-solving techniques:**

1. Memorize common phrasal verbs and their meanings.
2. Pay close attention to the context of the sentence to infer meaning.
3. If unsure, try to replace the phrasal verb with a single-word synonym to see if it fits.

Phrase Meaning

Definition and core idea: This involves understanding the meaning of idiomatic expressions, proverbs, or common phrases that often have a figurative meaning different from the literal interpretation of their individual words.

- **Key properties and identities:**

- Idioms are non-literal and culturally specific.
- Context is vital for interpreting the correct nuance of a phrase.

- **Common pitfalls:**

- Interpreting phrases literally.
- Confusing similar-sounding idioms.
- Not recognizing the figurative nature of the expression.

- **Standard problem-solving techniques:**

1. Read the sentence containing the phrase carefully to understand the overall context.
2. If the phrase is unfamiliar, try to infer its meaning from the surrounding words and the situation described.
3. Consider the options provided and see which one logically fits the context.
4. Eliminate options that represent a literal interpretation or are completely irrelevant.
5. Familiarity with common idioms and proverbs is the best preparation.

Prepositions

Definition and core idea: Prepositions are words that show the relationship between a noun or pronoun and other words in a sentence, indicating location, time, direction, or manner.

- **Important formulas, theorems, and results:**

1. **Prepositions of Place:**

- *in*: for enclosed spaces, cities, countries (e.g., *in the room*, *in India*).
- *on*: for surfaces, streets, specific days (e.g., *on the table*, *on Monday*).
- *at*: for specific points, addresses, events (e.g., *at the corner*, *at 5 PM*).

2. **Prepositions of Time:**

- *in*: for months, years, seasons, long periods (e.g., *in July*, *in 2023*).
- *on*: for specific days, dates (e.g., *on Friday*, *on August 15th*).
- *at*: for specific times (e.g., *at noon*, *at midnight*).

3. **Prepositions of Direction:** *to*, *into*, *onto*, *from*, *through*, *across*. E.g., *go to school*, *jump into the water*.

4. **Fixed Prepositional Phrases:** Many verbs, adjectives, and nouns are followed by specific prepositions (e.g., *afraid of*, *depend on*, *good at*).

- **Key properties and identities:**

- Prepositions always introduce a prepositional phrase, which includes a noun or pronoun (the object of the preposition).
- Their usage is often idiomatic and requires memorization.

- **Common pitfalls:**

- Confusing prepositions with similar meanings (e.g., "between" vs. "among").
- Incorrectly using fixed prepositional phrases.
- Omitting necessary prepositions or adding unnecessary ones.

- **Standard problem-solving techniques:**

1. Identify the relationship the preposition is trying to express (place, time, direction, etc.).
2. Check if the verb, adjective, or noun preceding the blank requires a specific preposition.
3. Consider the context and the object of the preposition.
4. Memorize common fixed prepositional phrases.

Pronouns

Definition and core idea: Pronouns are words that replace nouns to avoid repetition. They must agree with the nouns they refer to (antecedents) in number, gender, and case.

- **Important formulas, theorems, and results:**

1. **Types of Pronouns:**

- **Personal:** *I*, *you*, *he*, *she*, *it*, *we*, *they* (subjective); *me*, *you*, *him*, *her*, *it*, *us*, *them* (objective).
- **Possessive:** *mine*, *yours*, *his*, *hers*, *its*, *ours*, *theirs*.
- **Reflexive/Intensive:** *myself*, *yourself*, *himself*, *herself*, *itself*, *ourselves*, *yourselves*, *themselves*.
- **Relative:** *who*, *whom*, *whose*, *which*, *that*.
- **Demonstrative:** *this*, *that*, *these*, *those*.
- **Indefinite:** *everyone*, *somebody*, *nothing*, *all*, *many*, *few*.

2. **Pronoun-Antecedent Agreement:** A pronoun must agree with its antecedent in number (singular/plural) and gender (masculine/feminine/neuter). E.g., *The student submitted his assignment.*
The students submitted their assignments.

3. **Pronoun Case:**

- **Subjective Case:** Used when the pronoun is the subject of a verb (e.g., *He went*).
- **Objective Case:** Used when the pronoun is the object of a verb or preposition (e.g., *She saw him*, *Give it to me*).

- **Key properties and identities:**

- Pronouns clarify sentences by preventing redundancy.
- Correct pronoun usage is essential for grammatical accuracy.

- **Common pitfalls:**

- Incorrect agreement in number or gender with the antecedent.
- Confusing subjective and objective cases (e.g., "between you and I" instead of "between you and me").
- Ambiguous pronoun reference (unclear what noun the pronoun refers to).
- Using "its" (possessive) vs. "it's" (it is).

- **Standard problem-solving techniques:**

1. Identify the pronoun and its antecedent (the noun it replaces).
2. Check for agreement in number (singular/plural) and gender.
3. Determine if the pronoun is acting as a subject or an object to choose the correct case.
4. Ensure the pronoun's reference is clear and unambiguous.

Sentence Ordering

Definition and core idea: Similar to Narrative Sequencing, this involves arranging jumbled parts of a single sentence (or a short paragraph) into a grammatically correct and logically coherent order.

- **Key properties and identities:**

- Focuses on sentence structure, grammatical rules, and logical flow within a single sentence or a very short set of related sentences.
- Often involves identifying the subject, verb, and object, and then arranging modifiers correctly.

- **Common pitfalls:**

- Creating grammatically incorrect structures.
- Producing a sentence that is grammatically correct but makes no logical sense.
- Misplacing modifiers, leading to ambiguity.

- **Standard problem-solving techniques:**

1. **Identify the subject and main verb:** These usually form the core of the sentence.
2. **Look for introductory phrases/clauses:** These often come at the beginning.
3. **Identify conjunctions:** Words like *and*, *but*, *or*, *because*, *while* connect clauses or phrases.
4. **Check for pronoun references:** Ensure pronouns follow their antecedents.
5. **Consider logical flow:** What event or idea logically precedes another?
6. **Read the assembled sentence:** Ensure it is grammatically correct, makes sense, and flows naturally.

Statement Sufficiency

Definition and core idea: In this type of question, a question is given, followed by two or more statements. You need to determine whether the information in the statements is sufficient to answer the question.

- **Key properties and identities:**

- Focus is on sufficiency, not necessarily on finding the answer itself.
- Statements are evaluated independently first, then together if neither is sufficient alone.

- **Common pitfalls:**

- Trying to solve the question instead of just checking sufficiency.
- Assuming information not explicitly given in the statements.
- Incorrectly combining statements.

- **Standard problem-solving techniques:**

1. **Analyze the question:** Understand exactly what information is required to answer it.
2. **Evaluate Statement 1 alone:** Can the question be answered using only the information in Statement 1? Ignore Statement 2.
3. **Evaluate Statement 2 alone:** Can the question be answered using only the information in Statement 2? Ignore Statement 1.
4. **Evaluate Statements 1 and 2 together:** If neither statement alone is sufficient, check if combining them provides enough information.
5. **Select the appropriate option:**
 - Statement 1 alone is sufficient.
 - Statement 2 alone is sufficient.
 - Either Statement 1 alone or Statement 2 alone is sufficient.
 - Both Statement 1 and Statement 2 together are sufficient (but neither alone).
 - Neither Statement 1 nor Statement 2 nor both together are sufficient.

Statements Follow

Definition and core idea: This involves analyzing a set of given statements (premises) and determining which

conclusion(s) logically follow from them, often based on principles of deductive reasoning.

- **Important formulas, theorems, and results:**

1. **Deductive Reasoning:** If the premises are true, the conclusion must be true.
 - Example: *All A are B. All B are C. Therefore, All A are C.*
2. **Categorical Syllogisms:** Often involve "All," "No," "Some" statements.
 - *All X are Y* (Universal Affirmative)
 - *No X are Y* (Universal Negative)
 - *Some X are Y* (Particular Affirmative)
 - *Some X are not Y* (Particular Negative)
3. **Venn Diagrams:** A useful tool for visually representing relationships between categories and testing the validity of conclusions.

- **Key properties and identities:**

- Conclusions must be derived solely from the given statements, without external knowledge.
- Focus on logical necessity, not probability or common sense.

- **Common pitfalls:**

- Bringing in outside knowledge or personal opinions.
- Making assumptions not supported by the statements.
- Confusing "some" with "all" or "none."
- Errors in interpreting negative statements.

- **Standard problem-solving techniques:**

1. **Read statements carefully:** Understand the exact meaning of each premise.
2. **Use Venn Diagrams:** Draw circles to represent the categories and their relationships as described by the statements.
3. **Test each conclusion:** Check if the conclusion is necessarily true based on your diagram or logical deduction. If there's even one scenario where the conclusion is false, it does not follow.
4. **Avoid assumptions:** Stick strictly to the information provided in the statements.

Synonyms

Definition and core idea: Synonyms are words that have the same or very similar meanings. Understanding synonyms expands vocabulary and allows for varied expression.

- **Key properties and identities:**

- Few words are perfect synonyms; most have slightly different connotations or contexts of use.
- Can often be identified by root words, prefixes, and suffixes.

- **Common pitfalls:**

- Choosing a word that is related but not a true synonym.
- Ignoring the context in which the word is used.
- Misinterpreting the nuance of the given word.

- **Standard problem-solving techniques:**

1. Understand the precise meaning of the given word.
2. Consider the context if the word has multiple meanings.
3. Look for words with similar roots or derivations.
4. Substitute each option into a sentence where the original word is used to see which one fits best.
5. Eliminate options that are antonyms or unrelated.

Tenses

Definition and core idea: Tenses indicate the time an action occurs (past, present, future) and its aspect (simple, continuous, perfect, perfect continuous), showing whether an action is completed, ongoing, or habitual.

- **Important formulas, theorems, and results:**

1. **Simple Present:** $S + V1 + (s/es)$ (for 3rd person singular). Used for habits, facts, scheduled events. E.g., *He walks.*
2. **Present Continuous:** $S + is/am/are + V - ing$. Used for actions happening now or temporary actions. E.g., *He is walking.*
3. **Present Perfect:** $S + has/have + V3$. Used for actions completed in the past but with present relevance. E.g., *He has walked.*
4. **Present Perfect Continuous:** $S + has/have + been + V - ing$. Used for actions that started in the past and continue to the present. E.g., *He has been walking.*

5. **Simple Past:** $S + V2$. Used for completed actions in the past. E.g., *He walked*.
 6. **Past Continuous:** $S + was/were + V - ing$. Used for actions ongoing in the past. E.g., *He was walking*.
 7. **Past Perfect:** $S + had + V3$. Used for an action completed before another past action. E.g., *He had walked* before I arrived.
 8. **Past Perfect Continuous:** $S + had + been + V - ing$. Used for an action ongoing up to a point in the past. E.g., *He had been walking* for an hour when I saw him.
 9. **Simple Future:** $S + will/shall + V1$. Used for future actions. E.g., *He will walk*.
 10. **Future Continuous:** $S + will/shall + be + V - ing$. Used for actions ongoing at a specific time in the future. E.g., *He will be walking*.
 11. **Future Perfect:** $S + will/shall + have + V3$. Used for actions that will be completed by a certain time in the future. E.g., *He will have walked* 10 km by then.
 12. **Future Perfect Continuous:** $S + will/shall + have + been + V - ing$. Used for actions that will have been ongoing up to a point in the future. E.g., *He will have been walking* for two hours.
- **Key properties and identities:**
 - Tense consistency is crucial within a sentence or paragraph.
 - Time markers (e.g., yesterday, now, next week, since, for) often indicate the appropriate tense.
 - **Common pitfalls:**
 - Incorrectly mixing tenses within a sentence.
 - Confusing similar tenses (e.g., simple past vs. present perfect).
 - Incorrect verb forms (V1, V2, V3).
 - **Standard problem-solving techniques:**
 1. Identify the time frame of the action (past, present, future).
 2. Determine the aspect (simple, continuous, perfect, perfect continuous) based on whether the action is completed, ongoing, or has relevance to another time.
 3. Look for time expressions or other verbs in the sentence that provide clues about the correct tense.
 4. Ensure tense consistency, especially in complex sentences or reported speech.

Verbal Reasoning

Definition and core idea: Verbal Reasoning assesses the ability to understand and logically process information presented in words, including analogies, critical reasoning, and logical deduction from textual information.

- **Key properties and identities:**
 - Requires analytical thinking and the ability to identify patterns, relationships, and logical fallacies.
 - Often involves drawing conclusions, identifying assumptions, or evaluating arguments.
- **Common pitfalls:**
 - Making assumptions not explicitly stated.
 - Confusing correlation with causation.
 - Allowing personal bias to influence judgment.
 - Misinterpreting logical connectors (e.g., "if...then," "unless").
- **Standard problem-solving techniques:**
 1. **Read carefully:** Understand the exact meaning of all statements and the question.
 2. **Identify the core argument/relationship:** For analogies, determine the relationship between the first pair. For critical reasoning, identify the premise(s) and conclusion.
 3. **Break down complex statements:** Simplify sentences to their core logical components.
 4. **Look for keywords:** Words like *therefore*, *because*, *however*, *if*, *then*, *unless*, *all*, *some*, *none* are crucial for logical deduction.
 5. **Test options:** For multiple-choice questions, evaluate each option against the given information.
 6. **Avoid outside knowledge:** Base your reasoning strictly on the provided text.

Word Meaning

Definition and core idea: This involves understanding the precise definition of a word, often in a given context, and choosing the option that best reflects that meaning.

- **Key properties and identities:**
 - Words can have multiple meanings; context determines the relevant one.
 - Knowledge of root words, prefixes, and suffixes can help deduce meanings.
- **Common pitfalls:**
 - Choosing a common meaning of a word when the context implies a less common one.
 - Misinterpreting the nuance or connotation of a word.

- Ignoring contextual clues.
- **Standard problem-solving techniques:**
 1. Read the sentence or phrase containing the word carefully.
 2. Try to infer the meaning of the word from its context (surrounding words, overall sentence meaning).
 3. Look for synonyms or antonyms within the sentence that might provide clues.
 4. Break down the word into its root, prefix, and suffix if applicable.
 5. Substitute each option into the sentence to see which one makes the most sense.
 6. Eliminate options that are clearly incorrect or irrelevant.

Word Pairs

Definition and core idea: This question type presents a pair of words that share a specific relationship, and you must identify another pair of words from the options that shares the same relationship (analogies).

- **Important formulas, theorems, and results:**
 1. **Common Relationship Types:**
 - **Synonym:** *Happy : Joyful*
 - **Antonym:** *Hot : Cold*
 - **Part-Whole:** *Finger : Hand*
 - **Cause-Effect:** *Rain : Flood*
 - **Tool-User:** *Pen : Writer*
 - **Object-Function:** *Knife : Cut*
 - **Category-Member:** *Fruit : Apple*
 - **Intensity:** *Warm : Hot*
 - **Producer-Product:** *Baker : Bread*
 - **Action-Object:** *Read : Book*
- **Key properties and identities:**
 - The relationship must be consistent across both pairs.
 - The order of words in the pair often matters (e.g., cause-effect is different from effect-cause).
- **Common pitfalls:**
 - Identifying a weak or incorrect relationship in the original pair.
 - Finding a relationship in the options that is similar but not identical to the original.
 - Ignoring the order of the words in the pair.
- **Standard problem-solving techniques:**
 1. **Identify the precise relationship:** Clearly define how the two words in the given pair are related. Be specific (e.g., "is a type of," "is used to," "is the opposite of").
 2. **Formulate a sentence:** Create a simple sentence that expresses the relationship (e.g., "A finger is a part of a hand").
 3. **Apply the sentence to options:** Test each option by plugging its words into your sentence. The option that fits best is likely the answer.
 4. **Consider nuances:** If multiple options seem plausible, re-evaluate the strength and specificity of the relationship.
 5. **Check for order:** Ensure the order of the words in the option pair matches the order of the relationship in the original pair.

Quick Formula Reference

For Verbal Aptitude, "formulas" refer to core rules, structures, and patterns. Here's a consolidated list:

- **Articles:**
 - 'a': Singular, countable, consonant sound.
 - 'an': Singular, countable, vowel sound.
 - 'the': Specific, unique, or previously mentioned.
 - Zero Article: General plural/uncountable/abstract nouns, proper nouns.
- **Comparative Forms:**
 - 1-syllable: *Adj + -er / -est*
 - 2-syllable ending in -y: *Adj(-y → -i) + -er / -est*
 - 2+ syllables: *more / most + Adj*
 - Irregular: *good → better → best, bad → worse → worst*, etc.
- **English Grammar (Core):**
 - Sentence Structure: *S + V + O* (common).

- Subject-Verb Agreement: *SingularS* → *SingularV*, *PluralS* → *PluralV*.
- Tense Consistency: Maintain consistent tense.
- Parallelism: Similar items in same grammatical form.
- **Pronouns:**
 - Agreement: Pronoun matches antecedent in number, gender.
 - Case: Subjective (*I, he, she, we, they*) for subjects; Objective (*me, him, her, us, them*) for objects.
- **Tenses (Basic Structures):**
 - Simple Present: $S + V1 + (s/es)$
 - Present Continuous: $S + is/am/are + V - ing$
 - Present Perfect: $S + has/have + V3$
 - Simple Past: $S + V2$
 - Past Continuous: $S + was/were + V - ing$
 - Past Perfect: $S + had + V3$
 - Simple Future: $S + will/shall + V1$
- **Phrasal Verbs:** *Verb + Preposition/Adverb* (often idiomatic meaning).
- **Statement Sufficiency:** Evaluate Statement 1 alone, then Statement 2 alone, then both together.
- **Statements Follow (Deductive Logic):** If *Premise*₁ and *Premise*₂ are true, then *Conclusion* MUST be true. Use Venn Diagrams.
- **Word Pairs (Analogies):** Identify the specific relationship $A : B$ and find $C : D$ with the same relationship.

Important Tips for GATE

1. **Build a Strong Vocabulary:** Many questions (Synonyms, Antonyms, Word Meaning, Most Appropriate Word, Word Pairs, Phrasal Verbs, Phrase Meaning) directly test vocabulary. Read widely, use flashcards, and learn root words, prefixes, and suffixes.
2. **Master Grammar Rules:** Questions on Articles, Comparative Forms, English Grammar, Grammatical Error, Incorrect Sentence Part, Prepositions, Pronouns, and Tenses require a solid understanding of fundamental grammar. Practice identifying common errors.
3. **Read Actively and Critically:** For Passage Reading, Narrative Sequencing, and Verbal Reasoning, don't just read passively. Actively look for main ideas, supporting details, logical connections, transition words, and author's tone. Practice skimming and scanning.
4. **Context is King:** For vocabulary-based questions and even grammar, the context of the sentence or passage is crucial. A word's meaning or the correct grammatical form can change depending on its usage. Always read the full sentence.
5. **Practice Time Management:** Verbal Aptitude questions can be time-consuming, especially Passage Reading and Sentence Ordering. Practice solving questions under timed conditions to improve speed and accuracy. Don't get stuck on one question; move on and return if time permits.
6. **Eliminate Options Strategically:** For multiple-choice questions, if you're unsure of the answer, try to eliminate obviously incorrect options first. This increases your chances of selecting the correct answer, especially if negative marking is less severe.
7. **Don't Overthink Verbal Reasoning:** For Statement Sufficiency and Statements Follow, stick strictly to the information given in the statements. Do not bring in outside knowledge or make assumptions. Focus purely on logical deduction.
8. **Review Common Pitfalls:** Be aware of common errors like subject-verb agreement, pronoun-antecedent agreement, tense shifts, double comparatives, and literal interpretation of idioms. Regularly review these tricky areas.

11.1

Antonyms (7)

 **Practice Test:** Test 1 (7Q)

11.1.1 Antonyms: GATE CSE 2011 | Question: 60



Choose the word from the options given below that is most nearly opposite in the meaning to the given word

Amalgamate

- A. merge B. split C. collect D. separate

gatecse-2011 verbal-aptitude antonyms normal

Answer key 

11.1.2 Antonyms: GATE CSE 2014 | Set 3 | GA | Question: 3



Choose the word that is opposite in meaning to the word “coherent”.

- A. sticky B. well-connected C. rambling D. friendly

gatecse-2014-set3 verbal-aptitude antonyms easy

Answer key

11.1.3 Antonyms: GATE CSE 2026 | Set 1 | GA | Question: 1



The antonym of the word protagonist is _____.

- A. agnostic B. antagonist C. arsonist D. anarchist

gatecse-2026-set1 verbal-aptitude antonyms easy one-mark

Answer key

11.1.4 Antonyms: GATE DA 2026 | GA | Question: 1



Verbosity : Brevity :: Insolence : _____

Choose the word that best fills the blank.

- A. Innocence B. Respect C. Solace D. Wealth

gateda-2026 verbal-aptitude antonyms one-mark

Answer key

11.1.5 Antonyms: GATE2011 AG: GA-2



Choose the word from the questions given below that is most nearly opposite in meaning to the given word:

Frequency

- A. periodicity B. rarity C. gradualness D. persistency

general-aptitude verbal-aptitude gate2011-ag antonyms

Answer key

11.1.6 Antonyms: GATE2011 GG: GA-2



Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Polemical

- A. imitative B. conciliatory C. truthful D. ideological

gate2011-gg verbal-aptitude antonyms

Answer key

11.1.7 Antonyms: GATE2011 MN: GA-56



Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Deference

- A. aversion B. resignation C. suspicion D. contempt

verbal-aptitude gate2011-mn antonyms

Answer key

11.2

Articles (2)

11.2.1 Articles: GATE CSE 2016 | Set 2 | GA | Question: 01



The man who is now Municipal Commissioner worked as _____.

- A. the security guard at a university
C. a security guard at university

- B. a security guard at the university
D. the security guard at the university

gatecse-2016-set2 verbal-aptitude english-grammar normal articles

Answer key

11.2.2 Articles: GATE CSE 2025 | Set 1 | GA | Question: 1



Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man.

Select the option with the correct sequence of articles to fill in the blanks.

- A. a; a; an
C. a; an; a

- B. the; an; a
D. an; an; a

gatecse2025-set1 verbal-aptitude articles easy one-mark

Answer key

11.3 Comparative Forms (2)

Practice Test: Test 1 (7Q)

11.3.1 Comparative Forms: GATE CSE 2015 | Set 1 | GA | Question: 1



Didn't you buy _____ when you went shopping?

- A. any paper B. much paper C. no paper D. a few paper

gatecse-2015-set1 verbal-aptitude easy english-grammar comparative-forms

Answer key

11.3.2 Comparative Forms: GATE CSE 2021 | Set 2 | GA | Question: 1



Gauri said that she can play the keyboard _____ her sister.

- A. as well as B. as better as C. as nicest as D. as worse as

gatecse-2021-set2 verbal-aptitude comparative-forms one-mark

Answer key

11.4 English Grammar (5)

Practice Tests: Test 1 (15Q) Test 2 (11Q)

11.4.1 English Grammar: GATE CSE 2015 | Set 2 | GA | Question: 1



We _____ our friends's birthday and we _____ how to make it up to him.

- A. completely forgot --- don't just know
B. forgot completely --- don't just know
C. completely forgot --- just don't know
D. forgot completely --- just don't know

gatecse-2015-set2 verbal-aptitude easy english-grammar

Answer key

11.4.2 English Grammar: GATE CSE 2015 | Set 3 | GA | Question: 2



The Tamil version of _____ John Abraham-starrer *Madras Cafe* _____ cleared by the Censor Board with no cuts last week, but the film's distributor _____ no takers among the exhibitors for a release in Tamilnadu _____ this Friday.

- A. Mr., was, found, on
C. the, was, found, on
B. a, was, found, at
D. a, being, find, at

gatecse-2015-set3 verbal-aptitude normal english-grammar

Answer key

11.4.3 English Grammar: GATE CSE 2017 | Set 2 | GA | Question: 2



Saturn is _____ to be seen on a clear night with the naked eye.

- A. enough bright B. bright enough C. as enough bright D. bright as enough

gatecse-2017-set2 verbal-aptitude english-grammar

Answer key

11.4.4 English Grammar: GATE CSE 2020 | GA | Question: 1



Raman is confident of speaking English _____ six months as he has been practising regularly _____ the last three weeks

- A. during, for B. for, since C. for, in D. within, for

gatecse-2020 verbal-aptitude english-grammar one-mark

Answer key

11.4.5 English Grammar: GATE CSE 2021 | Set 1 | GA | Question: 3



Consider the following sentences:

- i. Everybody in the class is prepared for the exam.
- ii. Babu invited Danish to his home because he enjoys playing chess.

Which of the following is the **CORRECT** observation about the above two sentences?

- A. (i) is grammatically correct and (ii) is unambiguous
B. (i) is grammatically incorrect and (ii) is unambiguous
C. (i) is grammatically correct and (ii) is ambiguous
D. (i) is grammatically incorrect and (ii) is ambiguous

gatecse-2021-set1 verbal-aptitude english-grammar one-mark

Answer key

11.5

Grammatical Error (6)

Practice Test: Test 1 (11Q)

11.5.1 Grammatical Error: GATE CSE 2012 | Question: 59



Choose the grammatically **INCORRECT** sentence:

- A. They gave us the money back less the service charges of Three Hundred rupees.
B. This country's expenditure is not less than that of Bangladesh.
C. The committee initially asked for a funding of Fifty Lakh rupees, but later settled for a lesser sum.
D. This country's expenditure on educational reforms is very less.

gatecse-2012 verbal-aptitude grammatical-error normal

Answer key

11.5.2 Grammatical Error: GATE CSE 2013 | Question: 60



Choose the grammatically **INCORRECT** sentence:

- A. He is of Asian origin.
B. They belonged to Africa.
C. She is an European.
D. They migrated from India to Australia.

gatecse-2013 verbal-aptitude grammatical-error normal

Answer key

11.5.3 Grammatical Error: GATE CSE 2015 | Set 2 | GA | Question: 10



Out of the following 4 sentences, select the most suitable sentence with respect to grammar and usage:

- A. Since the report lacked needed information, it was of no use to them.
- B. The report was useless to them because there were no needed information in it.
- C. Since the report did not contain the needed information, it was not real useful to them.
- D. Since the report lacked needed information, it would not had been useful to them.

gatecse-2015-set2 verbal-aptitude normal english-grammar grammatical-error

Answer key

11.5.4 Grammatical Error: GATE CSE 2016 | Set 1 | GA | Question: 01



Out of the following four sentences, select the most suitable sentence with respect to grammar and usage.

- A. I will not leave the place until the minister does not meet me.
- B. I will not leave the place until the minister doesn't meet me.
- C. I will not leave the place until the minister meet me.
- D. I will not leave the place until the minister meets me.

gatecse-2016-set1 verbal-aptitude english-grammar easy grammatical-error

Answer key

11.5.5 Grammatical Error: GATE2012 AR: GA-3



Choose the grammatically CORRECT sentence:

- A. He laid in bed till 8 o'clock in the morning.
- B. He layed in bed till 8 o'clock in the morning.
- C. He lain in bed till 8 o'clock in the morning.
- D. He lay in bed till 8 o'clock in the morning.

gate2012-ar verbal-aptitude english-grammar easy grammatical-error

Answer key

11.5.6 Grammatical Error: GATE2013 EE: GA-4



Choose the grammatically **CORRECT** sentence:

- A. Two and two add four.
- B. Two and two become four.
- C. Two and two are four.
- D. Two and two make four.

gate2013-ee english-grammar verbal-aptitude grammatical-error

Answer key

11.6 Incorrect Sentence Part (5)

Practice Test: Test 1 (6Q)

11.6.1 Incorrect Sentence Part: GATE CSE 2014 | Set 3 | GA | Question: 1



While trying to collect I an envelope from under the table, Mr. X fell down and was losing consciousness.
IV

Which one of the above underlined parts of the sentence is NOT appropriate?

- A. I
- B. II
- C. III
- D. IV

gatecse-2014-set3 verbal-aptitude easy english-grammar incorrect-sentence-part

Answer key

11.6.2 Incorrect Sentence Part: GATE2012 AR: GA-4



Which one of the parts (A, B, C, D) in the sentence contains an **ERROR**?

No sooner had the doctor seen the results of the blood test, than he suggested the patient to see the specialist.

- A. no sooner had
- B. results of the blood test
- C. suggested the patient
- D. see the specialist

gate2012-ar verbal-aptitude incorrect-sentence-part

Answer key

11.6.3 Incorrect Sentence Part: GATE2012 CY: GA-2



One of the parts (A, B, C, D) in the sentence given below contains an ERROR. Which one of the following is **INCORRECT**?

I requested that he should be given the driving test today instead of tomorrow.

- A. requested that
- B. should be given
- C. the driving test
- D. instead of tomorrow

gate2012-cy verbal-aptitude incorrect-sentence-part

Answer key

11.6.4 Incorrect Sentence Part: GATE2013 AE: GA-4



All engineering students should learn mechanics, mathematics and how to do computation.

I II III IV

Which of the above underlined parts of the sentence is not appropriate?

- a. I
- b. II
- c. III
- d. IV

gate2013-ae verbal-aptitude incorrect-sentence-part

Answer key

11.6.5 Incorrect Sentence Part: GATE2013 CE: GA-2



The professor ordered to the students to go out of the class.

I II III IV

Which of the above underlined parts of the sentence is grammatically incorrect?

- A. I
- B. II
- C. III
- D. IV

gate2013-ce incorrect-sentence-part verbal-aptitude

Answer key

11.7 Most Appropriate Word (47)

Practice Tests: Test 1 (15Q) Test 10 (3Q) Test 2 (15Q) Test 3 (15Q) Test 4 (15Q) Test 5 (15Q) Test 6 (15Q) Test 7 (15Q)
Test 8 (15Q) Test 9 (15Q)

11.7.1 Most Appropriate Word: GATE CSE 2010 | Question: 56



Choose the most appropriate word from the options given below to complete the following sentence:

His rather casual remarks on politics _____ his lack of seriousness about the subject.

- A. masked
- B. belied
- C. betrayed
- D. suppressed

gatecse-2010 verbal-aptitude most-appropriate-word normal

Answer key

11.7.2 Most Appropriate Word: GATE CSE 2010 | Question: 58



Choose the most appropriate word from the options given below to complete the following sentence:

If we manage to _____ our natural resources, we would leave a better planet for our children.

- A. uphold B. restrain C. cherish D. conserve

gatecse-2010 verbal-aptitude most-appropriate-word easy

Answer key

11.7.3 Most Appropriate Word: GATE CSE 2011 | Question: 58

Choose the most appropriate word(s) from the options given below to complete the following sentence.

I contemplated _____ Singapore for my vacation but decided against it.

- A. to visit B. having to visit C. visiting D. for a visit

gatecse-2011 verbal-aptitude most-appropriate-word easy

Answer key

11.7.4 Most Appropriate Word: GATE CSE 2011 | Question: 59

Choose the most appropriate word from the options given below to complete the following sentence.

If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or _____.

- A. hyperbolic B. restrained C. argumentative D. indifferent

gatecse-2011 verbal-aptitude most-appropriate-word normal

Answer key

11.7.5 Most Appropriate Word: GATE CSE 2012 | Question: 57

Choose the most appropriate alternative from the options given below to complete the following sentence:

Despite several _____ the mission succeeded in its attempt to resolve the conflict.

- A. attempts B. setbacks C. meetings D. delegations

gatecse-2012 verbal-aptitude easy most-appropriate-word

Answer key

11.7.6 Most Appropriate Word: GATE CSE 2012 | Question: 60

Choose the most appropriate alternative from the options given below to complete the following sentence:

Suresh's dog is the one _____ was hurt in the stampede.

- A. that B. which C. who D. whom

gatecse-2012 verbal-aptitude most-appropriate-word normal

Answer key

11.7.7 Most Appropriate Word: GATE CSE 2014 | Set 1 | GA | Question: 2

Choose the most appropriate word from the options given below to complete the following sentence.

She could not understand the judges awarding her the first prize, because she thought that her performance was quite _____.

- A. superb B. medium C. mediocre D. exhilarating

gatecse-2014-set1 verbal-aptitude most-appropriate-word easy

Answer key

11.7.8 Most Appropriate Word: GATE CSE 2014 | Set 2 | GA | Question: 1

Choose the most appropriate phrase from the options given below to complete the following sentence.

India is a post-colonial country because

- A. it was a former British colony
- B. Indian Information Technology professionals have colonized the world
- C. India does not follow any colonial practices
- D. India has helped other countries gain freedom

gatecse-2014-set2 verbal-aptitude most-appropriate-word easy

Answer key

11.7.9 Most Appropriate Word: GATE CSE 2015 | Set 2 | GA | Question: 4



A generic term that includes various items of clothing such as a skirt, a pair of trousers and a shirt is

- A. fabric
- B. textile
- C. fiber
- D. apparel

gatecse-2015-set2 verbal-aptitude easy most-appropriate-word

Answer key

11.7.10 Most Appropriate Word: GATE CSE 2015 | Set 3 | GA | Question: 3



Extreme focus on syllabus and studying for tests has become such a dominant concern of Indian student that they close their minds to anything _____ to the requirements of the exam.

- A. related
- B. extraneous
- C. outside
- D. useful

gatecse-2015-set3 verbal-aptitude normal most-appropriate-word

Answer key

11.7.11 Most Appropriate Word: GATE CSE 2016 | Set 1 | GA | Question: 03



Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world."

The sentence above is an example of a _____ statement.

- A. figurative
- B. collateral
- C. literal
- D. figurine

gatecse-2016-set1 verbal-aptitude normal most-appropriate-word

Answer key

11.7.12 Most Appropriate Word: GATE CSE 2017 | Set 1 | GA | Question: 2



Research in the workplace reveals that people work for many reasons _____ .

- A. money beside
- B. beside money
- C. money besides
- D. besides money

gatecse-2017-set1 general-aptitude verbal-aptitude english-grammar most-appropriate-word

Answer key

11.7.13 Most Appropriate Word: GATE CSE 2018 | GA | Question: 1



"From where are they bringing their books? _____ bringing _____ books from _____"

The words that best fill the blanks in the above sentence are

- A. Their, they're, there
- B. They're, their, there
- C. There, their, they're
- D. They're, there, there

gatecse-2018 verbal-aptitude most-appropriate-word easy one-mark

Answer key

11.7.14 Most Appropriate Word: GATE CSE 2018 | GA | Question: 2



A _____ investigation can sometimes yield new facts, but typically organized ones are more successful.

The word that best fills the blank in the above sentence is

- A. meandering
- B. timely
- C. consistent
- D. systematic

Answer key 

11.7.15 Most Appropriate Word: GATE CSE 2019 | GA | Question: 1



The expenditure on the project _____ as follows: equipment Rs.20 lakhs, salaries Rs.12 lakhs, and contingency Rs.3 lakhs.

- A. break down B. break C. breaks down D. breaks

gatecse-2019 general-aptitude verbal-aptitude most-appropriate-word one-mark

Answer key 

11.7.16 Most Appropriate Word: GATE CSE 2019 | GA | Question: 2



The search engine's business model _____ around the fulcrum of trust.

- A. revolves B. plays C. sinks D. bursts

gatecse-2019 general-aptitude verbal-aptitude most-appropriate-word one-mark

Answer key 

11.7.17 Most Appropriate Word: GATE CSE 2019 | GA | Question: 5



A court is to a judge as _____ is to a teacher

- A. a student B. a punishment C. a syllabus D. a school

gatecse-2019 general-aptitude verbal-aptitude most-appropriate-word one-mark

Answer key 

11.7.18 Most Appropriate Word: GATE CSE 2022 | GA | Question: 1



The _____ is too high for it to be considered _____.

- A. fair / fare B. faer / fair C. fare / fare D. fare / fair

gatecse-2022 verbal-aptitude most-appropriate-word one-mark

Answer key 

11.7.19 Most Appropriate Word: GATE CSE 2023 | GA | Question: 1



We reached the station late, and _____ missed the train.

- A. near B. nearly C. utterly D. mostly

gatecse-2023 verbal-aptitude most-appropriate-word one-mark

Answer key 

11.7.20 Most Appropriate Word: GATE CSE 2024 | Set 1 | GA | Question: 1



If ' → ' denotes increasing order of intensity, then the meaning of the words [dry → arid → parched] is analogous to [diet → fast → _____].

Which one of the given options is appropriate to fill the blank?

- A. starve B. reject C. feast D. deny

gatecse-2024-set1 verbal-aptitude most-appropriate-word one-mark

Answer key 

11.7.21 Most Appropriate Word: GATE CSE 2024 | Set 1 | GA | Question: 6



In the given text, the blanks are numbered (i) - (iv). Select the best match for all the blanks.

Steve was advised to keep his head (i) before heading (ii) to bat; for, while he had a head (iii) batting, he could only do so with a cool head iv his shoulders.

- A. (i) down (ii) down (iii) on (iv) for
C. (i) down (ii) out (iii) for (iv) on

- B. (i) on (ii) down (iii) for (iv) on
D. (i) on (ii) out (iii) on (iv) for

gatecse-2024-set1 verbal-aptitude most-appropriate-word two-marks

Answer key

11.7.22 Most Appropriate Word: GATE CSE 2024 | Set 2 | GA | Question: 1



If '→' denotes increasing order of intensity, then the meaning of the words [walk → jog → sprint] is analogous to [bothered → _____ → daunted].

Which one of the given options is appropriate to fill the blank?

- A. phased B. phrased C. fazed D. fused

gatecse-2024-set2 verbal-aptitude most-appropriate-word one-mark

Answer key

11.7.23 Most Appropriate Word: GATE CSE 2025 | Set 1 | GA | Question: 2



The CEO's decision to downsize the workforce was considered myopic because it sacrificed long-term stability to accommodate short-term gains.

Select the most appropriate option that can replace the word "myopic" without changing the meaning of the sentence.

- A. visionary B. shortsighted C. progressive D. innovative

gatecse2025-set1 verbal-aptitude most-appropriate-word easy one-mark

Answer key

11.7.24 Most Appropriate Word: GATE CSE 2025 | Set 2 | GA | Question: 1



Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered.

Select the most appropriate option to complete the above sentence.

- A. ambivalence B. satisfaction C. resolve D. revolve

gatecse2025-set2 verbal-aptitude most-appropriate-word easy one-mark

Answer key

11.7.25 Most Appropriate Word: GATE DA 2025 | GA | Question: 1



Courage : Bravery :: Yearning : _____.

Select the most appropriate option to complete the analogy.

- A. Longing B. Yelling C. Yawning D. Glaring

gateda-2025 verbal-aptitude most-appropriate-word easy one-mark

Answer key

11.7.26 Most Appropriate Word: GATE DS&AI 2024 | GA | Question: 1



If '→' denotes increasing order of intensity, then the meaning of the words [sick → infirm → moribund] is analogous to [silly → _____ → daft].

Which one of the given options is appropriate to fill the blank?

- A. frown B. fawn C. vein D. vain

gate-ds-ai-2024 verbal-aptitude most-appropriate-word one-mark

Answer key

11.7.27 Most Appropriate Word: GATE2010 MN: GA-3



Choose the most appropriate word from the options given below to complete the following sentence:
The committee wrote a _____ report, extolling only the strengths of the proposal.

- A. reasonable B. supportive C. biased D. fragmented

general-aptitude verbal-aptitude gate2010-mn most-appropriate-word

Answer key

11.7.28 Most Appropriate Word: GATE2010 MN: GA-4



Choose the most appropriate word from the options given below to complete the following sentence:
If the country has to achieve real prosperity, it is _____ that the fruits of progress reach all, and in equal measure.

- A. inevitable B. contingent C. oblivious D. imperative

general-aptitude verbal-aptitude gate2010-mn most-appropriate-word

Answer key

11.7.29 Most Appropriate Word: GATE2010 TF: GA-3



Choose the most appropriate word from the options given below to complete the following sentence:
The two child norm with _____ for the violators will have significant implications for our demographic profile.

- A. disincentives B. incitements C. restrictions D. restraints

general-aptitude verbal-aptitude gate2010-tf most-appropriate-word

Answer key

11.7.30 Most Appropriate Word: GATE2010 TF: GA-4



Choose the most appropriate word from the options given below to complete the following sentence:
There is no fixed relation between food and famine; famines can occur with or without a substantial _____ in food output.

- A. aberration B. weakening C. decline D. deterioration

general-aptitude verbal-aptitude gate2010-tf most-appropriate-word

Answer key

11.7.31 Most Appropriate Word: GATE2011 AG: GA-1



Choose the most appropriate word from the options given below to complete the following sentence:
Under ethical guidelines recently adopted by the India Medical Association, human genes are to be manipulated only to correct diseases for which _____ treatments are unsatisfactory.

- A. similar B. most C. uncommon D. available

general-aptitude verbal-aptitude gate2011-ag most-appropriate-word

Answer key

11.7.32 Most Appropriate Word: GATE2011 AG: GA-3



Choose the most appropriate word from the options given below to complete the following sentence:
It was her view that the country's problem had been _____ by foreign technocrats, so that to invite them to come back would be counter-productive.

- A. identified B. ascertained
C. exacerbated D. analysed

general-aptitude verbal-aptitude gate2011-ag most-appropriate-word

Answer key

11.7.33 Most Appropriate Word: GATE2011 GG: GA-1



Choose the most appropriate word or phrase from the options given below to complete the following sentence.

The environmentalists hope _____ the lake to its pristine condition.

- A. in restoring
C. to restore
B. in the restoration of
D. restoring

gate2011-gg verbal-aptitude most-appropriate-word

Answer key

11.7.34 Most Appropriate Word: GATE2011 GG: GA-3



Choose the most appropriate word from the options given below to complete the following sentence.

Despite the mixture's _____ nature, we found that by lowering its temperature in the laboratory we could dramatically reduce its tendency to vaporize.

- A. acerbic
B. resilient
C. volatile
D. heterogeneous

gate2011-gg verbal-aptitude most-appropriate-word normal

Answer key

11.7.35 Most Appropriate Word: GATE2011 GG: GA-5



Choose the most appropriate words from the options given below to complete the following sentence.

Because she had a reputation for _____ we were surprised and pleased when she greeted us so _____.

- A. insolence irately
C. graciousness amiably
B. insouciance curtly
D. querulousness affably

gate2011-gg most-appropriate-word verbal-aptitude

Answer key

11.7.36 Most Appropriate Word: GATE2011 MN: GA-57



Choose the most appropriate word(s) from the options given below to complete the following sentence.

We lost confidence in him because he never _____ the grandiose promises he had made.

- A. delivered
B. delivered on
C. forgot
D. reneged on

gate2011-mn verbal-aptitude most-appropriate-word

Answer key

11.7.37 Most Appropriate Word: GATE2011 MN: GA-58



Choose the word or phrase that best completes the sentence below.

_____ in the frozen wastes of Arctic takes special equipment.

- A. To survive
B. Surviving
C. Survival
D. That survival

verbal-aptitude gate2011-mn most-appropriate-word

Answer key

11.7.38 Most Appropriate Word: GATE2011 MN: GA-60



Choose the most appropriate word from the options given below to complete the following sentence.

The _____ of evidence was on the side of the plaintiff since all but one witness testified that his story was correct.

- A. paucity B. propensity C. preponderance D. accuracy

verbal-aptitude gate2011-mn most-appropriate-word

Answer key

11.7.39 Most Appropriate Word: GATE2012 AE: GA-3

Choose the most appropriate alternative from the options given below to complete the following sentence:

The administrators went on to implement yet another unreasonable measure, arguing that the measures were already ____ and one more would hardly make a difference.

- A. reflective B. utopian C. luxuriant D. unpopular

gate2012-ae most-appropriate-word verbal-aptitude

Answer key

11.7.40 Most Appropriate Word: GATE2012 AE: GA-4

Choose the most appropriate alternative from the options given below to complete the following sentence:

To those of us who had always thought him timid, his ____ came as a surprise.

- A. intrepidity B. inevitability C. inability D. inertness

gate2012-ae verbal-aptitude most-appropriate-word

Answer key

11.7.41 Most Appropriate Word: GATE2012 AR: GA-2

Choose the most appropriate pair of words from the options given below to complete the following sentence:

The high level of ____ of the questions in the test was ____ by an increase in the period of time allotted for answering them.

- A. difficulty, compensated B. exactitude, magnified
C. aptitude, decreased D. attitude, mitigated

gate2012-ar most-appropriate-word verbal-aptitude normal

Answer key

11.7.42 Most Appropriate Word: GATE2012 CY: GA-4

Choose the most appropriate word from the options given below to complete the following sentence:

Given the seriousness of the situation that he had to face, his ____ was impressive.

- A. beggary B. nomenclature C. jealousy D. nonchalance

gate2012-cy most-appropriate-word

Answer key

11.7.43 Most Appropriate Word: GATE2013 CE: GA-4

Friendship, no matter how _____ it is, has its limitations.

- A. cordial B. intimate C. secret D. pleasant

gate2013-ce most-appropriate-word verbal-aptitude

Answer key

11.7.44 Most Appropriate Word: GATE2013 EE: GA-3

Complete the sentence:

Dare _____ mistakes.

- A. commit B. to commit C. committed D. committing

gate2013-ee most-appropriate-word easy verbal-aptitude

Answer key

11.7.45 Most Appropriate Word: GATE2014 AE: GA-2



Choose the most appropriate word from the options given below to complete the following sentence.

One of his biggest _____ was his ability to forgive.

- A. vice B. virtues C. choices D. strength

gate2014-ae most-appropriate-word

Answer key

11.7.46 Most Appropriate Word: GATE2014 AG: GA-1



Choose the most appropriate word from the options given below to complete the following sentence. A person suffering from Alzheimer's disease _____ short-term memory loss.

- A. Experienced B. Has experienced
C. Is experiencing D. Experiences

gate2014-ag verbal-aptitude most-appropriate-word normal

Answer key

11.7.47 Most Appropriate Word: GATE2014 AG: GA-2



Choose the most appropriate word from the options given below to complete the following sentence. _____ is the key to their happiness; they are satisfied with what they have.

- A. Contentment B. Ambition C. Perseverance D. Hunger

gate2014-ag verbal-aptitude most-appropriate-word easy

Answer key

11.8 Narrative Sequencing (1)

11.8.1 Narrative Sequencing: GATE DA 2025 | GA | Question: 6



Column-I has statements made by Shanthala; and, Column-II has responses given by Kanishk.

	Column-I		Column-II
P.	This house is in a mess.	1.	Alright, I won't bring it up during our conversations.
Q.	I am not happy with the marks given to me.	2.	Well, you can easily look it up.
R.	Politics is a subject I avoid talking about.	3.	No problem, let me clear it up for you.
S.	I don't know what this word means.	4.	Don't worry, I will take it up with your teacher.

Identify the option that has the correct match between Column-I and Column-II.

- A. P – 2; Q – 3; R – 1; S – 4
B. P – 3; Q – 4; R – 1; S – 2

- C. P – 4; Q – 1; R – 2; S – 3
D. P – 1; Q – 2; R – 4; S – 3

gateda-2025 verbal-aptitude narrative-sequencing two-marks

Answer key

11.9 Passage Reading (23)

 **Practice Tests:** Test 1 (6Q) Test 2 (15Q) Test 3 (15Q) Test 4 (15Q) Test 5 (5Q)

11.9.1 Passage Reading: GATE CSE 2010 | Question: 63



Modern warfare has changed from large scale clashes of armies to suppression of civilian populations. Chemical agents that do their work silently appear to be suited to such warfare; and regretfully, there exist people in military establishments who think that chemical agents are useful tools for their cause.

Which of the following statements best sums up the meaning of the above passage:

- A. Modern warfare has resulted in civil strife.
- B. Chemical agents are useful in modern warfare.
- C. Use of chemical agents in warfare would be undesirable.
- D. People in military establishments like to use chemical agents in war.

gatecse-2010 verbal-aptitude passage-reading normal

Answer key

11.9.2 Passage Reading: GATE CSE 2011 | Question: 61



Few school curricula include a unit on how to deal with bereavement and grief, and yet all students at some point in their lives suffer from losses through death and parting.

Based on the above passage which topic would not be included in a unit on bereavement?

- A. how to write a letter of condolence
- B. what emotional stages are passed through in the healing process
- C. what the leading causes of death are
- D. how to give support to a grieving friend

gatecse-2011 verbal-aptitude passage-reading normal

Answer key

11.9.3 Passage Reading: GATE CSE 2013 | Question: 63



After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again, the spider failed but that did not deter it to refrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

Which one of the following assertions is best supported by the above information?

- A. Failure is the pillar of success.
- B. Honesty is the best policy.
- C. Life begins and ends with adventures.
- D. No adversity justifies giving up hope.

gatecse-2013 verbal-aptitude passage-reading normal

Answer key

11.9.4 Passage Reading: GATE CSE 2014 | Set 1 | GA | Question: 6



The Palghat Gap (or Palakkad Gap), a region about 30 km wide in the southern part of the Western Ghats in India, is lower than the hilly terrain to its north and south. The exact reasons for the formation of this gap are not clear. It results in the neighbouring regions of Tamil Nadu getting more rainfall from the South West monsoon and the neighbouring regions of Kerala having higher summer temperatures.

What can be inferred from this passage?

Select one:

- A. The Palghat gap is caused by high rainfall and high temperatures in southern Tamil Nadu and Kerala
- B. The regions in Tamil Nadu and Kerala that are near the Palghat Gap are low-lying
- C. The low terrain of the Palghat Gap has a significant impact on weather patterns in neighbouring parts of Tamil Nadu and Kerala
- D. Higher summer temperatures result in higher rainfall near the Palghat Gap area

gatecse-2014-set1 verbal-aptitude passage-reading normal

Answer key

11.9.5 Passage Reading: GATE CSE 2014 | Set 2 | GA | Question: 6



The old city of Koenigsberg, which had a German majority population before World War 2, is now called Kaliningrad. After the events of the war, Kaliningrad is now a Russian territory and has a predominantly Russian population. It is bordered by the Baltic Sea on the north and the countries of Poland to the south and west and Lithuania to the east respectively. Which of the statements below can be inferred from this passage?

- A. Kaliningrad was historically Russian in its ethnic make up
- B. Kaliningrad is a part of Russia despite it not being contiguous with the rest of Russia
- C. Koenigsberg was renamed Kaliningrad, as that was its original Russian name
- D. Poland and Lithuania are on the route from Kaliningrad to the rest of Russia

gatecse-2014-set2 verbal-aptitude passage-reading normal

Answer key

11.9.6 Passage Reading: GATE CSE 2014 | Set 2 | GA | Question: 7



Number of people diagnosed with dengue fever (contracted from the bite of a mosquito) in North India is twice the number diagnosed last year. Municipal authorities have concluded that measures to control the mosquito population have failed in this region.

Which one of the following statements, if true, does not contradict this conclusion?

- A. A high proportion of the affected population has returned from neighbouring countries where dengue is prevalent
- B. More cases of dengue are now reported because of an increase in the Municipal Office's administrative efficiency
- C. Many more cases of dengue are being diagnosed this year since the introduction of a new and effective diagnostic test
- D. The number of people with malarial fever (also contracted from mosquito bites) has increased this year

gatecse-2014-set2 verbal-aptitude passage-reading normal

Answer key

11.9.7 Passage Reading: GATE CSE 2014 | Set 3 | GA | Question: 6



A dance programme is scheduled for 10.00 a.m. Some students are participating in the programme and they need to come an hour earlier than the start of the event. These students should be accompanied by a parent. Other students and parents should come in time for the programme. The instruction you think that is appropriate for this is

- A. Students should come at 9.00 a.m. and parents should come at 10.00 a.m.

- B. Participating students should come at 9.00 a.m. accompanied by a parent, and other parents and students should come by 10.00 a.m.
- C. Students who are not participating should come by 10.00 a.m. and they should not bring their parents. Participating students should come at 9.00 a.m.
- D. Participating students should come before 9.00 a.m. Parents who accompany them should come at 9.00 a.m. All others should come at 10.00 a.m.

gatecse-2014-set3 verbal-aptitude passage-reading easy

Answer key

11.9.8 Passage Reading: GATE CSE 2014 | Set 3 | GA | Question: 7



By the beginning of the 20th century, several hypotheses were being proposed, suggesting a paradigm shift in our understanding of the universe. However, the clinching evidence was provided by experimental measurements of the position of a star which was directly behind our sun.

Which of the following inference(s) may be drawn from the above passage?

- i. Our understanding of the universe changes based on the positions of stars
- ii. Paradigm shifts usually occur at the beginning of centuries
- iii. Stars are important objects in the universe
- iv. Experimental evidence was important in confirming this paradigm shift

- A. i, ii and iv B. iii only C. i and iv D. iv only

gatecse-2014-set3 verbal-aptitude passage-reading easy

Answer key

11.9.9 Passage Reading: GATE CSE 2015 | Set 3 | GA | Question: 9



Most experts feel that in spite of possessing all the technical skills required to be a batsman of the highest order, he is unlikely to be so due to lack of requisite temperament. He was guilty of throwing away his wicket several time after working hard to lay a strong foundation. His critics pointed out that until he addressed his problem, success at the highest level will continue to elude him.

Which of the statement(s) below is/are logically valid and can be inferred from the above passage?

- i. He was already a successful batsman at the highest level.
- ii. He was to improve his temperament in order to become a great batsman.
- iii. He failed to make many of his good starts count.
- iv. Improving his technical skills will guarantee success.

- A. iii and iv B. ii and iii C. i, ii and iii D. ii only

gatecse-2015-set3 verbal-aptitude normal passage-reading

Answer key

11.9.10 Passage Reading: GATE CSE 2016 | Set 2 | GA | Question: 07



Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required.

Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?

- (i) The author believes that computers are not good for us.
- (ii) Mobile computers and the internet are both intended inventions.

- A. (i) only B. (ii) only
C. Both (i) and (ii) D. Neither (i) nor (ii)

gatecse-2016-set2 verbal-aptitude passage-reading normal

Answer key

11.9.11 Passage Reading: GATE CSE 2017 | Set 1 | GA | Question: 6



"The hold of the nationalist imagination on our colonial past is such that anything inadequately or improperly nationalist is just not history."

Which of the following statements best reflects the author's opinion?

- A. Nationalists are highly imaginative.
- B. History is viewed through the filter of nationalism.
- C. Our colonial past never happened.
- D. Nationalism has to be both adequately and properly imagined.

gatecse-2017-set1 general-aptitude verbal-aptitude passage-reading

Answer key

11.9.12 Passage Reading: GATE CSE 2022 | GA | Question: 6



Some people believe that "what gets measured, improves". Some others believe that "what gets measured, gets gamed". One possible reason for the difference in the beliefs is the work culture in organizations. In organizations with good work culture, metrics help improve outcomes. However, the same metrics are counterproductive in organizations with poor work culture.

Which one of the following is the **CORRECT** logical inference based on the information in the above passage?

- A. Metrics are useful in organizations with poor work culture
- B. Metrics are useful in organizations with good work culture
- C. Metrics are always counterproductive in organizations with good work culture
- D. Metrics are never useful in organizations with good work culture

gatecse-2022 verbal-aptitude passage-reading two-marks

Answer key

11.9.13 Passage Reading: GATE CSE 2025 | Set 1 | GA | Question: 6



"I put the brown paper in my pocket along with the chinks, and possibly other things. I suppose every one must have reflected how primeval and how poetical are the things that one carries in one's pocket: the pocket-knife, for instance the type of all human tools, the infant of the sword. Once I planned to write a book of poems entirely about the things in my pocket. But I found it would be too long: and the age of the great epics is past."

(From G.K. Chesterton's "A Piece of Chalk")

Based only on the information provided in the above passage, which one of the following statements is true?

- A. The author of the passage carries a mirror in his pocket to reflect upon things.
- B. The author of the passage had decided to write a poem on epics.
- C. The pocket-knife is described as the infant of the sword.
- D. Epics are described as too inconvenient to write.

gatecse2025-set1 verbal-aptitude passage-reading two-marks

Answer key

11.9.14 Passage Reading: GATE2010 MN: GA-6



The ban on smoking in designated public places can save a large number of people from the well-known effects of environmental tobacco smoke. Passive smoking seriously impairs respiratory health. The ban rightly seeks to protect non-smokers from its ill effects.

Which of the following statements best sums up the meaning of the above passage:

- A. Effects of environmental tobacco are well known.
- B. The ban on smoking in public places protects the non-smokers.
- C. Passive smoking is bad for health.
- D. The ban on smoking in public places excludes passive smoking.

Answer key

11.9.15 Passage Reading: GATE2010 TF: GA-6



It has taken fifty-six long and frustrating, years to turn bronze, into gold for India's Olympics aspirations. Beijing 2008 marks a defining moment in India's Olympic history. From Delhi to Beijing is a long journey but one that our Olympians have undertaken with courage.

Which of the following statement best sums up the meaning of the above passage:

- A. India's participation in Olympics has been frustrating.
- B. Beijing Olympics was a landmark in India's Olympic history.
- C. Our Olympians have undertaken a long journey to Beijing.
- D. India's bronze medal turned into gold at Beijing.

Answer key

11.9.16 Passage Reading: GATE2011 AG: GA-10



The horse has played a little known but very important role in the field of medicine. Horses were injected with toxins of diseases until their blood built up immunities. Then a serum was made from their blood. Serums to fight with diphtheria and tetanus were developed this way.

It can be inferred from the passage, that horses were

- | | |
|------------------------------------|--|
| A. given immunity to diseases | B. generally quite immune to diseases |
| C. given medicines to fight toxins | D. given diphtheria and tetanus serums |

Answer key

11.9.17 Passage Reading: GATE2011 GG: GA-10



In order to develop to full potential, a baby needs to be physically able to respond to the environment.

It can be inferred from the passage that

- A. Full physical potential is needed in order for a baby to be able to respond to the environment.
- B. It is necessary for a baby to be able to physically respond to the environment for it to develop its full potential.
- C. Response to the environment of physically able babies needs to be developed to its full potential.
- D. A physically able baby needs to develop its full potential in order to respond to its environment.

Answer key

11.9.18 Passage Reading: GATE2011 MN: GA-65



Nimbus clouds are dark and ragged, stratus clouds appear dull in colour and cover the entire sky. Cirrus clouds are thin and delicate, whereas cumulus clouds look like cotton balls.

It can be inferred from the passage that

- A. A cumulus cloud on the ground is called fog
- B. It is easy to predict the weather by studying clouds
- C. Clouds are generally of very different shapes, sizes and mass
- D. There are four basic cloud types: stratus, nimbus, cumulus and cirrus

Answer key **11.9.19 Passage Reading: GATE2012 AE: GA-10**

In the early nineteenth century, theories of social evolution were inspired less by Biology than by the conviction of social scientists that there was a growing improvement in social institutions. Progress was taken for granted and social scientists attempted to discover its laws and phases.

Which one of the following inferences may be drawn with the greatest accuracy from the above passage?
Social scientists

- | | |
|---|---|
| A. did not question that progress was a fact. | B. did not approve of Biology. |
| C. framed the laws of progress. | D. emphasized Biology over Social Sciences. |

Answer key **11.9.20 Passage Reading: GATE2012 AR: GA-10**

The documents expose the cynicism of the government officials – and yet as the media website reflects, not a single newspaper has reported on their existence.

Which one of the following inferences may be drawn with the greatest accuracy from the above passage?

- A. Nobody other than the government officials knew about the existence of the documents.
B. Newspapers did report about the documents but nobody cared.
C. Media reports did not show the existence of the documents.
D. The documents reveal the attitude of the government officials.

Answer key **11.9.21 Passage Reading: GATE2012 CY: GA-6**

One of the legacies of the Roman legions was discipline. In the legions, military law prevailed and discipline was brutal. Discipline on the battlefield kept units obedient, intact and fighting, even when the odds and conditions were against them.

Which one of the following statements best sums up the meaning of the above passage?

- A. Thorough regimentation was the main reason for the efficiency of the Roman legions even in adverse circumstances.
B. The legions were treated inhumanly as if the men were animals.
C. Discipline was the armies' inheritance from their seniors.
D. The harsh discipline to which the legions were subjected to led to the odds and conditions being against them.

Answer key **11.9.22 Passage Reading: GATE2014 AE: GA-3**

Rajan was not happy that Sajan decided to do the project on his own. On observing his unhappiness, Sajan explained to Rajan that he preferred to work independently.

Which one of the statements below is logically valid and can be inferred from the above sentences?

- A. Rajan has decided to work only in a group.
B. Rajan and Sajan were formed into a group against their wishes.
C. Sajan had decided to give in to Rajan's request to work with him.
D. Rajan had believed that Sajan and he would be working together.

Answer key

11.9.23 Passage Reading: GATE2014 AG: GA-7



Moving into a world of big data will require us to change our thinking about the merits of exactitude. To apply the conventional mindset of measurement to the digital, connected world of the twenty-first century is to miss a crucial point. As mentioned earlier, the obsession with exactness is an artefact of the information-deprived analog era. When data was sparse, every data point was critical, and thus great care was taken to avoid letting any point bias the analysis. From "BIG DATA" Viktor Mayer-Schonberger and Kenneth Cukier. The main point of the paragraph is:

- A. The twenty-first century is a digital world
- B. Big data is obsessed with exactness
- C. Exactitude is not critical in dealing with big data
- D. Sparse data leads to a bias in the analysis

Answer key

11.10 Phrase Meaning (5)

Practice Test: Test 1 (8Q)

11.10.1 Phrase Meaning: GATE CSE 2014 | Set 1 | GA | Question: 1



Which of the following options is the closest in meaning to the phrase in bold in the sentence below?

It is fascinating to see life forms **cope with** varied environmental conditions.

- A. Adopt to
- B. Adapt to
- C. Adept in
- D. Accept with

Answer key

11.10.2 Phrase Meaning: GATE CSE 2014 | Set 1 | GA | Question: 3



In a press meet on the recent scam, the minister said, "The buck stops here". What did the minister convey by the statement?

- A. He wants all the money
- B. He will return the money
- C. He will assume final responsibility
- D. He will resist all enquiries

Answer key

11.10.3 Phrase Meaning: GATE CSE 2015 | Set 1 | GA | Question: 2



Which of the following options is the closest in meaning of the sentence below?

She enjoyed herself immensely at the party.

- A. She had a terrible time at the party
- B. She had a horrible time at the party
- C. She had a terrific time at the party
- D. She had a terrifying time at the party

Answer key

11.10.4 Phrase Meaning: GATE CSE 2015 | Set 1 | GA | Question: 7



Select the alternative meaning of the underlined part of the sentence.

The chain snatchers took to their heels when the police party arrived.

- A. Took shelter in a thick jungle
- B. Open indiscriminate fire

C. Took to flight

D. Unconditionally surrendered

gatecse-2015-set1 verbal-aptitude phrase-meaning easy

Answer key

11.10.5 Phrase Meaning: GATE CSE 2016 | Set 2 | GA | Question: 02



Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia.

Choose the option which is closest in meaning to the underlined phrase in the above sentence.

A. Put up with.

B. Put in with.

C. Put down to.

D. Put up against.

gatecse-2016-set2 verbal-aptitude phrase-meaning normal

Answer key

11.11

Prepositions (1)

Practice Test: Test 1 (10Q)

11.11.1 Prepositions: GATE DA 2026 | GA | Question: 6



'My friend and I parted ___ the door ___ the cabin that I had rented ___ the night.'

Choose the option with the correct sequence of words to fill the blanks.

A. at; of; for

B. for; at; of

C. of; for; in

D. in; of; for

gateda-2026 verbal-aptitude fill-in-the-blanks two-marks prepositions

Answer key

11.12

Sentence Ordering (2)

Practice Test: Test 1 (5Q)

11.12.1 Sentence Ordering: GATE CSE 2023 | GA | Question: 8



Which one of the following sentence sequences creates a coherent narrative?

- Once on the terrace, on her way to her small room in the corner, she notices the man right away.
- She begins to pant by the time she has climbed all the stairs.
- Mina has bought vegetables and rice at the market, so her bags are heavy.
- He was leaning against the parapet, watching the traffic below.

A. (i), (ii), (iv), (iii)

B. (ii), (iii), (i), (iv)

C. (iv), (ii), (i), (iii)

D. (iii), (ii), (i), (iv)

gatecse-2023 verbal-aptitude sentence-ordering two-marks

Answer key

11.12.2 Sentence Ordering: GATE CSE 2024 | Set 2 | GA | Question: 6



Sequence the following sentences in a coherent passage.

P: This fortuitous geological event generated a colossal amount of energy and heat that resulted in the rocks rising to an average height of 4 km across the contact zone.

Q: Thus, the geophysicists tend to think of the Himalayas as an active geological event rather than as a static geological feature.

R: The natural process of the cooling of this massive edifice absorbed large quantities of atmospheric carbon dioxide, altering the earth's atmosphere and making it better suited for life.

S: Many millennia ago, a breakaway chunk of bedrock from the Antarctic Plate collided with the massive Eurasian

Plate.

A. QPSR

B. QSPR

C. SPRQ

D. SRPQ

gatecse-2024-set2 verbal-aptitude sentence-ordering two-marks

Answer key

11.13

Statement Sufficiency (2)

11.13.1 Statement Sufficiency: GATE CSE 2015 | Set 1 | GA | Question: 4



Based on the given statements, select the most appropriate option to solve the given question.

If two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from the first floor to the second floor of the building?

Statements:

- I. Each step is $3/4$ foot high.
- II. Each step is 1 foot wide.

- A. Statements I alone is sufficient, but statement II alone is not sufficient.
- B. Statements II alone is sufficient, but statement I alone is not sufficient.
- C. Both statements together are sufficient, but neither statement alone is sufficient.
- D. Statements I and II together are not sufficient.

gatecse-2015-set1 verbal-aptitude easy statement-sufficiency

Answer key

11.13.2 Statement Sufficiency: GATE CSE 2015 | Set 2 | GA | Question: 5



Based on the given statements, select the most appropriate option to solve the given question.

What will be the total weight of 10 poles each of same weight?

Statements:

- I. One fourth of the weight of the pole is 5 Kg.
- II. The total weight of these poles is 160 Kg more than the total weight of two poles.

- A. Statement I alone is not sufficient.
- B. Statement II alone is not sufficient.
- C. Either I or II alone is sufficient.
- D. Both statements I and II together are not sufficient.

gatecse-2015-set2 normal logical-reasoning statement-sufficiency

Answer key

11.14

Statements Follow (1)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(7Q\)](#)

11.14.1 Statements Follow: GATE CSE 2015 | Set 1 | GA | Question: 8



The given statement is followed by some courses of action. Assuming the statement to be true, decide the correct option.

Statement:

There has been a significant drop in the water level in the lakes supplying water to the city.

Course of action:

- I. The water supply authority should impose a partial cut in supply to tackle the situation.
- II. The government should appeal to all the residents through mass media for minimal use of water.
- III. The government should ban the water supply in lower areas.

- A. Statements I and II follow.
C. Statements II and III follow.

- B. Statements I and III follow.
D. All the statements follow.

gatecse-2015-set1 verbal-aptitude normal statements-follow

Answer key

11.15

Synonyms (13)

 Practice Tests: Test 1 (15Q) Test 2 (6Q)

11.15.1 Synonyms: GATE CSE 2010 | Question: 57



Which of the following options is the closest in meaning to the word given below:

Circuitous

- A. cyclic B. indirect C. confusing D. crooked

gatecse-2010 verbal-aptitude word-meaning normal synonyms

Answer key

11.15.2 Synonyms: GATE CSE 2011 | Question: 56



Which of the following options is the closest in the meaning to the word below:

Inexplicable

- A. Incomprehensible B. Indelible
C. Inextricable D. Infallible

gatecse-2011 verbal-aptitude synonyms normal

Answer key

11.15.3 Synonyms: GATE CSE 2012 | Question: 58



Which one of the following options is the closest in meaning to the word given below?

Mitigate

- A. Diminish B. Divulge C. Dedicate D. Denote

gatecse-2012 verbal-aptitude synonyms easy

Answer key

11.15.4 Synonyms: GATE CSE 2013 | Question: 56



Which one of the following options is the closest in meaning to the word given below?

Nadir

- A. Highest B. Lowest C. Medium D. Integration

gatecse-2013 verbal-aptitude synonyms normal

Answer key

11.15.5 Synonyms: GATE CSE 2017 | Set 2 | GA | Question: 1



Choose the option with words that are not synonyms.

- A. aversion, dislike B. luminous, radiant
C. plunder, loot D. yielding, resistant

gatecse-2017-set2 verbal-aptitude synonyms

Answer key

11.15.6 Synonyms: GATE CSE 2026 | Set 2 | GA | Question: 1



Expedite, Hasten, Hurry, _____

Fill the blank by choosing a word with a meaning similar to that of the words given above.

- A. Accelerate B. Retard C. Provide D. Disable

gatecse-2026-set2 verbal-aptitude synonyms one-mark

Answer key 

11.15.7 Synonyms: GATE2010 MN: GA-1



Which of the following options is the closest in meaning to the word below:

Exhort

- A. urge B. condemn C. restrain D. scold

general-aptitude verbal-aptitude gate2010-mn synonyms

Answer key 

11.15.8 Synonyms: GATE2010 TF: GA-1



Which of the following options is the closest in meaning to the word below:

Ephemeral

- A. effeminate B. ghostlike C. soft D. short-lived

general-aptitude verbal-aptitude gate2010-tf synonyms

Answer key 

11.15.9 Synonyms: GATE2012 AR: GA-1



Which one of the following options is the closest in meaning to the word given below?

Pacify

- A. Excite B. Soothe C. Deplete D. Tire

gate2012-ar verbal-aptitude synonyms

Answer key 

11.15.10 Synonyms: GATE2012 CY: GA-3



Which one of the following options is the closest in meaning to the word given below?

Latitude

- A. Eligibility B. Freedom C. Coercion D. Meticulousness

gate2012-cy verbal-aptitude synonyms

Answer key 

11.15.11 Synonyms: GATE2013 CE: GA-3



Which of the following options is the closest in meaning to the word given below: **Primeval**

- A. Modern B. Historic C. Primitive D. Antique

gate2013-ce synonyms most-appropriate-word

Answer key 

11.15.12 Synonyms: GATE2013 EE: GA-1



They were requested not to **quarrel** with others.

Which one of the following options is the closest in meaning to the word **quarrel**?

- A. make out B. call out C. dig out D. fall out

gate2013-ee verbal-aptitude synonyms

Answer key 

11.15.13 Synonyms: GATE2014 AE: GA-1



A student is required to demonstrate a high level of comprehension of the subject, especially in the social sciences.

The word closest in meaning to **comprehension** is

- A. understanding B. meaning C. concentration D. stability

gate2014-ae synonyms verbal-aptitude

Answer key

11.16

Tenses (9)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(11Q\)](#)

11.16.1 Tenses: GATE CSE 2013 | Question: 59



Were you a bird, you _____ in the sky.

- A. would fly B. shall fly C. should fly D. shall have flown

gatecse-2013 verbal-aptitude tenses normal

Answer key

11.16.2 Tenses: GATE CSE 2014 | Set 2 | GA | Question: 2



Who _____ was coming to see us this evening?

- A. you said B. did you say C. did you say that D. had you said

gatecse-2014-set2 verbal-aptitude tenses normal

Answer key

11.16.3 Tenses: GATE CSE 2014 | Set 3 | GA | Question: 2



If she _____ how to calibrate the instrument, she _____ done the experiment.

- A. knows, will have B. knew, had
C. had known, could have D. should have known, would have

gatecse-2014-set3 verbal-aptitude easy english-grammar tenses

Answer key

11.16.4 Tenses: GATE CSE 2017 | Set 1 | GA | Question: 1



After Rajendra Chola returned from his voyage to Indonesia, he _____ to visit the temple in Thanjavur.

- A. was wishing B. is wishing C. wished D. had wished

gatecse-2017-set1 general-aptitude verbal-aptitude tenses english-grammar normal

Answer key

11.16.5 Tenses: GATE DA 2025 | GA | Question: 2



We _____ tennis in the lawn when it suddenly started to rain.

Select the most appropriate option to complete the above sentence.

- A. have been playing B. had been playing
C. would have been playing D. could be playing

gateda-2025 verbal-aptitude tenses easy one-mark

Answer key

11.16.6 Tenses: GATE2012 AE: GA-1



Choose the most appropriate alternative from the options given below to complete the following sentence:

I ____ to have bought a diamond ring.

- A. have a liking B. should have liked C. would like D. may like

gate2012-ae tenses verbal-aptitude

Answer key 

11.16.7 Tenses: GATE2012 AE: GA-2



Choose the most appropriate alternative from the options given below to complete the following sentence:

Food prices ____ again this month.

- A. have raised B. have been raising
C. have been rising D. have arose

gate2012-ae verbal-aptitude tenses

Answer key 

11.16.8 Tenses: GATE2012 CY: GA-5



Choose the most appropriate alternative from the options given below to complete the following sentence:

If the tired soldier wanted to lie down, he ____ the mattress out on the balcony.

- A. should take B. shall take
C. should have taken D. will have taken

gate2012-cy tenses verbal-aptitude

Answer key 

11.16.9 Tenses: GATE2013 AE: GA-2



The Headmaster _____ to speak to you. Which of the following options is incorrect to complete the above sentence?

- A. is wanting B. wants
C. want D. was wanting

gate2013-ae verbal-aptitude english-grammar tenses

Answer key 

11.17

Verbal Reasoning (15)

 **Practice Tests:** [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(15Q\)](#) [Test 4 \(2Q\)](#)

11.17.1 Verbal Reasoning: GATE CSE 2012 | Question: 61



Wanted Temporary, Part-time persons for the post of Field Interviewer to conduct personal interviews to collect and collate economic data. Requirements: High School-pass, must be available for Day, Evening and Saturday work. Transportation paid, expenses reimbursed.

Which one of the following is the best inference from the above advertisement?

- A. Gender-discriminatory B. Xenophobic
C. Not designed to make the post attractive D. Not gender-discriminatory

gatecse-2012 verbal-aptitude verbal-reasoning normal

Answer key 

11.17.2 Verbal Reasoning: GATE CSE 2014 | Set 1 | GA | Question: 7



Geneticists say that they are very close to confirming the genetic roots of psychiatric illnesses such as depression and schizophrenia, and consequently, that doctors will be able to eradicate these diseases through early identification and gene therapy.

On which of the following assumptions does the statement above rely?

Select one:

- A. Strategies are now available for eliminating psychiatric illnesses
- B. Certain psychiatric illnesses have a genetic basis
- C. All human diseases can be traced back to genes and how they are expressed
- D. In the future, genetics will become the only relevant field for identifying psychiatric illnesses

gatecse-2014-set1 verbal-aptitude verbal-reasoning normal

Answer key 

11.17.3 Verbal Reasoning: GATE CSE 2015 | Set 3 | GA | Question: 6



Alexander turned his attention towards India, since he had conquered Persia.

Which one of the statements below is logically valid and can be inferred from the above sentence?

- A. Alexander would not have turned his attention towards India had he not conquered Persia.
- B. Alexander was not ready to rest on his laurels, and wanted to march to India.
- C. Alexander was not completely in control of his army and could command it to move towards India.
- D. Since Alexander's kingdom extended to Indian borders after the conquest of Persia, he was keen to move further.

gatecse-2015-set3 verbal-aptitude normal verbal-reasoning

Answer key 

11.17.4 Verbal Reasoning: GATE CSE 2016 | Set 1 | GA | Question: 07



Indian currency notes show the denomination indicated in at least seventeen languages. If this is not an indication of the nation's diversity, nothing else is.

Which of the following can be logically inferred from the above sentences?

- A. India is a country of exactly seventeen languages.
- B. Linguistic pluralism is the only indicator of a nation's diversity.
- C. Indian currency notes have sufficient space for all the Indian languages.
- D. Linguistic pluralism is strong evidence of India's diversity.

gatecse-2016-set1 verbal-aptitude verbal-reasoning normal

Answer key 

11.17.5 Verbal Reasoning: GATE CSE 2017 | Set 2 | GA | Question: 6



"We lived in a culture that denied any merit to literary works, considering them important only when they were handmaidens to something seemingly more urgent – namely ideology. This was a country where all gestures, even the most private, were interpreted in political terms."

The author's belief that ideology is not as important as literature is revealed by the word:

- | | |
|--------------|----------------|
| A. 'culture' | B. 'seemingly' |
| C. 'urgent' | D. 'political' |

gatecse-2017-set2 passage-reading verbal-reasoning

Answer key 

11.17.6 Verbal Reasoning: GATE CSE 2019 | GA | Question: 6



The police arrested four criminals – P , Q , R and S . The criminals knew each other. They made the following statements:

- P says "Q committed the crime."
- Q says "S committed the crime."
- R says "I did not do it."
- S says "What Q said about me is false".

Assume only one of the arrested four committed the crime and only one of the statements made above is true. Who committed the crime?

A. *P*B. *R*C. *S*D. *Q*

gatecse-2019 verbal-aptitude verbal-reasoning two-marks

Answer key

11.17.7 Verbal Reasoning: GATE CSE 2019 | GA | Question: 8



“A recent High Court judgement has sought to dispel the idea of begging as a disease – which leads to its stigmatization and criminalization – and to regard it as a symptom. The underlying disease is the failure of the state to protect citizens who fall through the social security net.”

Which one of the following statements can be inferred from the given passage?

- A. Beggars are lazy people who beg because they are unwilling to work
- B. Beggars are created because of the lack of social welfare schemes
- C. Begging is an offence that has to be dealt with firmly
- D. Begging has to be banned because it adversely affects the welfare of the state

gatecse-2019 general-aptitude verbal-aptitude verbal-reasoning two-marks

Answer key

11.17.8 Verbal Reasoning: GATE CSE 2020 | GA | Question: 4



The dawn of the 21st century witnessed the melting glaciers oscillating between giving too much and too little to billions of people who depend on them for fresh water. The UN climate report estimates that without deep cuts to man-made emissions, at least 30% of the northern hemisphere's surface permafrost could melt by the end of the century. Given this situation of imminent global exodus of billions of people displaced by rising seas, nation-states need to rethink their carbon footprint for political concerns, if not for environmental ones.

Which one of the following statements can be inferred from the given passage?

- A. Nation-states do not have environmental concerns.
- B. Nation-states are responsible for providing fresh water to billions of people.
- C. Billions of people are responsible for man-made emissions.
- D. Billions of people are affected by melting glaciers.

gatecse-2020 verbal-aptitude verbal-reasoning passage-reading one-mark

Answer key

11.17.9 Verbal Reasoning: GATE CSE 2020 | GA | Question: 6



Goods and Services Tax (GST) is an indirect tax introduced in India in 2017 that is imposed on the supply of goods and services, and it subsumes all indirect taxes except few. It is a destination-based tax imposed on goods and services used, and it is not imposed at the point of origin from where goods come. GST also has a few components specific to state governments, central government and Union Territories (UTs).

Which one of the following statements can be inferred from the given passage?

- A. GST is imposed on the production of goods and services.
- B. GST includes all indirect taxes.
- C. GST does not have a component specific to UT.
- D. GST is imposed at the point of usage of goods and services.

gatecse-2020 verbal-aptitude verbal-reasoning passage-reading two-marks

Answer key

11.17.10 Verbal Reasoning: GATE CSE 2021 | Set 1 | GA | Question: 10



Some people suggest anti-obesity measures (AOM) such as displaying calorie information in restaurant menus. Such measures sidestep addressing the core problems that cause obesity: poverty and income inequality.

Which one of the following statements summarizes the passage?

- A. The proposed AOM addresses the core problems that cause obesity
- B. If obesity reduces, poverty will naturally reduce, since obesity causes poverty
- C. AOM are addressing the core problems and are likely to succeed
- D. AOM are addressing the problem superficially

gatecse-2021-set1 verbal-aptitude verbal-reasoning passage-reading two-marks

Answer key

11.17.11 Verbal Reasoning: GATE CSE 2021 | Set 2 | GA | Question: 6



Listening to music during exercise improves performance and reduces discomfort. Scientists researched whether listening to music while studying can help students learn better and the results were inconclusive. Students who needed external stimulation for studying fared worse while students who did not need any external stimulation benefited from music.

Which one of the following statements is the **CORRECT** inference of the above passage?

- A. Listening to music has no effect on learning and a positive effect on physical exercise
- B. Listening to music has a clear positive effect both in physical exercise and on learning
- C. Listening to music has a clear positive effect on physical exercise. Music has a positive effect on learning only in some students
- D. Listening to music has a clear positive effect on learning in all students. Music has a positive effect only in some students who exercise

gatecse-2021-set2 verbal-aptitude verbal-reasoning passage-reading two-marks

Answer key

11.17.12 Verbal Reasoning: GATE2013 AE: GA-3



Mahatma Gandhi was known for his humility as

- A. he played an important role in humiliating exit of British from India.
- B. he worked for humanitarian causes.
- C. he displayed modesty in his interactions.
- D. he was a fine human being

gate2013-ae verbal-aptitude verbal-reasoning

Answer key

11.17.13 Verbal Reasoning: GATE2013 EE: GA-5



Statement: You can always give me a ring whenever you need.

Which one of the following is the best inference from the above statement?

- A. Because I have a nice caller tune.
- B. Because I have a better telephone facility.
- C. Because a friend in need is a friend indeed.
- D. Because you need not pay towards the telephone bills when you give me a ring.

gate2013-ee verbal-reasoning verbal-aptitude

Answer key

11.17.14 Verbal Reasoning: GATE2013 EE: GA-7



Statement: There were different streams of freedom movements in colonial India carried out by the moderates, liberals, radicals, socialists, and so on.

Which one of the following is the best inference from the above statement?

- A. The emergence of nationalism in colonial India led to our Independence.
- B. Nationalism in India emerged in the context of colonialism.
- C. Nationalism in India is homogeneous.

D. Nationalism in India is heterogeneous

gate2013-ee passage-reading verbal-aptitude verbal-reasoning

Answer key

11.17.15 Verbal Reasoning: GATE2014 AG: GA-3



Which of the following options is the closest in meaning to the sentence below?

“As a woman, I have no country.”

- A. Women have no country.
- B. Women are not citizens of any country.
- C. Women’s solidarity knows no national boundaries.
- D. Women of all countries have equal legal rights.

gate2014-ag general-aptitude verbal-aptitude verbal-reasoning normal

Answer key

11.18

Word Meaning (5)

Practice Tests: Test 1 (15Q) Test 2 (4Q)

11.18.1 Word Meaning: GATE CSE 2014 | Set 2 | GA | Question: 3



Match the columns.

Column 1	Column 2
1. eradicate	P. misrepresent
2. distort	Q. soak completely
3. saturate	R. use
4. utilize	S. destroy utterly

- A. 1:S, 2:P, 3:Q, 4:R
- C. 1:Q, 2:R, 3:S, 4:P

- B. 1:P, 2:Q, 3:R, 4:S
- D. 1:S, 2:P, 3:R, 4:Q

gatecse-2014-set2 verbal-aptitude word-meaning normal match-the-following

Answer key

11.18.2 Word Meaning: GATE CSE 2015 | Set 2 | GA | Question: 2



Choose the statement where underlined word is used correctly.

- A. The industrialist had a personnel jet.
- B. I write my experience in my personnel diary.
- C. All personnel are being given the day off.
- D. Being religious is a personnel aspect.

gatecse-2015-set2 verbal-aptitude word-meaning normal

Answer key

11.18.3 Word Meaning: GATE CSE 2016 | Set 1 | GA | Question: 02



A rewording of something written or spoken is a _____.

- A. paraphrase
- B. paradox
- C. paradigm
- D. paraffin

gatecse-2016-set1 verbal-aptitude word-meaning normal

Answer key

11.18.4 Word Meaning: GATE CSE 2016 | Set 2 | GA | Question: 03



Find the odd one in the following group of words.

mock, deride, praise, jeer

- A. Mock B. Deride C. Praise D. Jeer

gatecse-2016-set2 verbal-aptitude word-meaning easy

Answer key

11.18.5 Word Meaning: GATE CSE 2020 | GA | Question: 2



His knowledge of the subject was excellent but his classroom performance was_____.

- A. extremely poor B. good C. desirable D. praiseworthy

gatecse-2020 verbal-aptitude english-grammar word-meaning one-mark

Answer key

11.19 Word Pairs (14)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(5Q\)](#)

11.19.1 Word Pairs: GATE CSE 2010 | Question: 60



The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.

Unemployed : Worker

- A. fallow : land B. unaware : sleeper
C. wit : jester D. renovated : house

gatecse-2010 verbal-aptitude word-pairs normal

Answer key

11.19.2 Word Pairs: GATE CSE 2013 | Question: 57



Complete the sentence:

Universalism is to particularism as diffuseness is to _____.

- A. specificity B. neutrality C. generality D. adaptation

gatecse-2013 verbal-aptitude normal word-pairs

Answer key

11.19.3 Word Pairs: GATE CSE 2015 | Set 1 | GA | Question: 5



Which one of the following combinations is incorrect?

- A. Acquiescence - Submission B. Wheedle - Roundabout
C. Flippancy - Lightness D. Profligate - Extravagant

gatecse-2015-set1 verbal-aptitude difficult word-pairs

Answer key

11.19.4 Word Pairs: GATE CSE 2015 | Set 3 | GA | Question: 4



Select the pair of best expresses a relationship similar to that expressed in the pair:

Children : Pediatrician

- A. Adult : Orthopaedist B. Females : Gynaecologist
C. Kidney : Nephrologist D. Skin : Dermatologist

gatecse-2015-set3 verbal-aptitude easy word-pairs

Answer key

11.19.5 Word Pairs: GATE CSE 2020 | GA | Question: 3



Select the word that fits the analogy:

Cook : Cook :: Fly : _____

- A. Flyer B. Flying C. Flew D. Flighter

gatecse-2020 verbal-aptitude word-pairs one-mark

Answer key

11.19.6 Word Pairs: GATE CSE 2021 | Set 1 | GA | Question: 5



_____ is to *surgery* as *writer* is to _____

Which one of the following options maintains a similar logical relation in the above sentence?

- A. Plan, outline B. Hospital, library
C. Doctor, book D. Medicine, grammar

gatecse-2021-set1 verbal-aptitude word-pairs one-mark

Answer key

11.19.7 Word Pairs: GATE CSE 2021 | Set 2 | GA | Question: 5



Pen : Write :: Knife : _____

Which one of the following options maintains a similar logical relation in the above?

- A. Vegetables B. Sharp C. Cut D. Blunt

gatecse-2021-set2 verbal-aptitude word-pairs one-mark easy

Answer key

11.19.8 Word Pairs: GATE CSE 2023 | GA | Question: 2



Kind : _____ : : Often : Frequently (By word meaning)

- A. Mean B. Type C. Cruel D. Kindly

gatecse-2023 verbal-aptitude word-pairs one-mark

Answer key

11.19.9 Word Pairs: GATE CSE 2025 | Set 2 | GA | Question: 2



Bird : Nest :: Bee : _____

- A. Kennel B. Hammock C. Hive D. Lair

gatecse2025-set2 verbal-aptitude word-pairs easy one-mark

Answer key

11.19.10 Word Pairs: GATE2010 MN: GA-2



The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.

Preamble : Constitution

- A. amendment : law B. prologue : play
C. episode : serial D. plot : story

general-aptitude verbal-aptitude gate2010-mn word-pairs

Answer key

11.19.11 Word Pairs: GATE2010 TF: GA-2



The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.

Erudition : Scholar

- A. steadfast : mercurial
C. skill : craftsman

- B. competence : strict
D. nurse : doctor

general-aptitude verbal-aptitude gate2010-tf word-pairs

Answer key

11.19.12 Word Pairs: GATE2011 AG: GA-5



The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair:

Gladiator : Arena

- A. dancer : stage
C. teacher : classroom

- B. commuter : train
D. lawyer : courtroom

general-aptitude verbal-aptitude gate2011-ag word-pairs

Answer key

11.19.13 Word Pairs: GATE2013 AE: GA-5



Select the pair that best expresses a relationship similar to that expressed in the pair:

water: pipe::

- A. cart: road
C. sea: beach

- B. electricity: wire
D. music: instrument

gate2013-ae verbal-aptitude word-pairs

Answer key

11.19.14 Word Pairs: GATE2013 CE: GA-5



Select the pair that best expresses a relationship similar to that expressed in the pair:

Medicine: Health

- A. Science: Experiment
C. Education: Knowledge

- B. Wealth: Peace
D. Money: Happiness

gate2013-ce word-pairs verbal-aptitude

Answer key

Answer Keys

11.1.1	D	11.1.2	C	11.1.3	B	11.1.4	B	11.1.5	B
11.1.6	B	11.1.7	D	11.2.1	B	11.2.2	A	11.3.1	A
11.3.2	A	11.4.1	C	11.4.2	C	11.4.3	B	11.4.4	D
11.4.5	C	11.5.1	D	11.5.2	C	11.5.3	A	11.5.4	D
11.5.5	D	11.5.6	D	11.6.1	D	11.6.2	D	11.6.3	B
11.6.4	D	11.6.5	B	11.7.1	C	11.7.2	D	11.7.3	C
11.7.4	B	11.7.5	B	11.7.6	A	11.7.7	C	11.7.8	A
11.7.9	D	11.7.10	B	11.7.11	A	11.7.12	D	11.7.13	B
11.7.14	A	11.7.15	C	11.7.16	A	11.7.17	D	11.7.18	D
11.7.19	B	11.7.20	A	11.7.21	C	11.7.22	C	11.7.23	B
11.7.24	C	11.7.25	A	11.7.26	D	11.7.27	C	11.7.28	D
11.7.29	A	11.7.30	C	11.7.31	D	11.7.32	C	11.7.33	C
11.7.34	C	11.7.35	D	11.7.36	B	11.7.37	C	11.7.38	C

11.7.39	D
11.7.44	A
11.9.1	D
11.9.6	D
11.9.11	B
11.9.16	A
11.9.21	A
11.10.3	C
11.12.2	C
11.15.2	A
11.15.7	A
11.15.12	D
11.16.4	C
11.16.9	C
11.17.5	B
11.17.10	D
11.17.15	C
11.18.5	A
11.19.5	A
11.19.10	B

11.7.40	A
11.7.45	B
11.9.2	C
11.9.7	B
11.9.12	B
11.9.17	B
11.9.22	D
11.10.4	C
11.13.1	A
11.15.3	A
11.15.8	D
11.15.13	A
11.16.5	B
11.17.1	D
11.17.6	B
11.17.11	C
11.18.1	A
11.19.1	A
11.19.6	C
11.19.11	C

11.7.41	A
11.7.46	D
11.9.3	D
11.9.8	D
11.9.13	C
11.9.18	D
11.9.23	C
11.10.5	A
11.13.2	C
11.15.4	B
11.15.9	B
11.16.1	A
11.16.6	C
11.17.2	B
11.17.7	B
11.17.12	C
11.18.2	C
11.19.2	A
11.19.7	C
11.19.12	D

11.7.42	D
11.7.47	A
11.9.4	C
11.9.9	B
11.9.14	B
11.9.19	A
11.10.1	B
11.11.1	A
11.14.1	A
11.15.5	D
11.15.10	B
11.16.2	B
11.16.7	C
11.17.3	A
11.17.8	D
11.17.13	C
11.18.3	A
11.19.3	B
11.19.8	B,D
11.19.13	B

11.7.43	B
11.8.1	B
11.9.5	B
11.9.10	D
11.9.15	B
11.9.20	D
11.10.2	C
11.12.1	D
11.15.1	B
11.15.6	A
11.15.11	C
11.16.3	C
11.16.8	A
11.17.4	D
11.17.9	D
11.17.14	D
11.18.4	C
11.19.4	B
11.19.9	C
11.19.14	C